

Semi-autonomous Driving Control of Multi-trailer Articulated Heavy Vehicles for Increasing Highway Traffic Safety

by

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A thesis submitted to the
School of Graduate and Postdoctoral Studies in partial
fulfillment of the requirements for the degree of

Doctor of Philosophy in Mechanical Engineering

Faculty of Engineering and Applied Science

University of Ontario Institute of Technology (Ontario Tech University)

Oshawa, Ontario, Canada

December 2025

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THESIS EXAMINATION INFORMATION

Submitted by: **Abbas Ajorkar**

Doctor of Philosophy in Mechanical Engineering

Thesis title: Semi-autonomous Driving Control of Multi Trailer Articulated Heavy
Vehicles (MTAHVs) for Increasing Highway Driving Safety

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ABSTRACT

This PhD research introduces innovative semi-autonomous driving strategies for Multi-trailer Articulated Heavy Vehicles (MTAHVs) to enhance highway safety and transportation efficiency. While autonomous driving has progressed, concerns about safety, reliability, and ethical issues remain, making the transition to full autonomy gradual. Unlike the extensive focus on semi-autonomous technologies for single-unit vehicles, limited attention has been given to Articulated Heavy Vehicles (AHVs), particularly MTAHVs, which have unique dynamics and a significantly higher accident risk. This study addresses the critical need for advancing semi-autonomous systems for AHVs and MTAHVs.

To fill the research gap, this study proposes an innovative and cost-effective methodology for exploring semi-autonomous driving systems for MTAHVs. To this end, an MTAHV with the configuration of A-train double is selected as the subject vehicle. To design the tracking-controller and the driver, the A-train double is modeled as a single-track yaw-plane model using the Lagrange formalism and symbolically manipulated in Maple software. A Nonlinear Model Predictive Controller (NLMPCC) is designed as the tracking-controller via multi-objective optimization technique; a Modified Visual Driver Model (MVDM) incorporating driver time delay and Arm Neuromuscular System (NMS) dynamics is devised to represent the MTAHV driver. A comprehensive framework is introduced to bridge the gap between high-level decision-making and low-level control/driver systems for MTAHVs in highway scenarios. The framework features a multi-layer decision-maker developed using deep reinforcement learning (DRL), designed to coordinate with the NLMPCC-based tracking controller or the MVDM-based driver. This

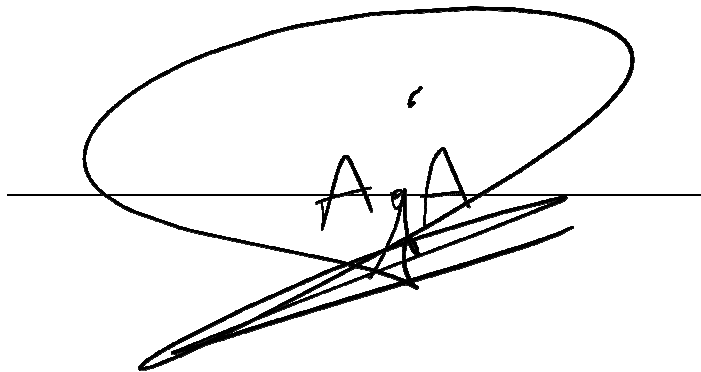
collaboration enables the vehicle to navigate effectively by integrating the decision-maker's commands with the controller/driver's outputs, ensuring seamless interaction between decision-making and control systems for safe and efficient highway operation. Finally, a novel formulation for a semi-autonomous controller tailored to the unique challenges of MTAHVs is introduced. The proposed semi-autonomous system is evaluated using a benchmark for non-cooperative maneuvers, demonstrating the system's capability to manage the complex, nonlinear dynamics of MTAHVs and effectively balance human and automated control objectives. This study provides an effective modeling-simulation based method for exploring semi-autonomous systems for MTAHV, and achieves insightful original findings, which may be used as the design guidelines for developing semi-autonomous MTAHVs.

Keywords: decision making; deep reinforcement learning; driver modeling; model predictive control; multi-trailer articulated heavy vehicle; semi-autonomous controller for highway driving

AUTHOR'S DECLARATION

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Abbas Ajorkar

STATEMENT OF CONTRIBUTIONS

Part of this dissertation have been published or to be published as:

- 1- A. Ajorkar and Y. He, “Tuning MPC Path-following Controllers Using Multi-objective Optimization”, Proceedings of the Canadian Society for Mechanical Engineering International Congress 2022, June 5-8, Edmonton, AB, Canada, <https://doi.org/10.7939/r3-0fqr-m015>
- 2- A. Ajorkar and Y. He, “Deep Reinforcement Learning Decision Maker and MPC Path-following Controller for the Autonomous Highway Driving”, Proceedings of the Canadian Society for Mechanical Engineering International Congress 2024, May 26-29, 2024, Toronto, Ontario, Canada.
- 3- A. Ajorkar and Y. He, “Design of autonomous driving controls for multi-trailer articulated heavy vehicles”, Journal of Vibration and Control. First published online January 16, 2025, <https://doi.org/10.1177/10775463251313657>
- 4- A. Ajorkar and Y. He, “Highway Decision Making Using Deep Reinforcement Learning for a Multi Trailer Articulated Heavy Vehicle”, Advanced Intelligent Equipment Design, Manufacturing, and Maintenance Conference 2025, September 19-21, Beijing, China.

ACKNOWLEDGEMENTS

I would like to express my deepest gratitude to my supervisor, Professor Yuping He, whose guidance, encouragement, and unwavering support have been instrumental in the completion of this journey. Your mentorship has been a source of inspiration, and I am profoundly thankful for the opportunities to grow and learn under your guidance.

This thesis is dedicated to my beloved parents and parents-in-law (Ali, Nosrat, Mehdi, and Mina), whose endless love, sacrifices, and unwavering belief in me have been the foundation of all my achievements. Your strength and support have been my guiding light.

To my dearest wife, Golnar, my partner and pillar of strength, thank you for your patience, love, and encouragement throughout this journey. Your unwavering faith in me has been my greatest motivation, and this work would not have been possible without you.

With heartfelt appreciation, I humbly dedicate this thesis to all of you.

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LIST OF ABBREVIATIONS AND SYMBOLS

ABS	Anti-lock Braking System
ACC	Adaptive Cruise Control
ADAS	Advanced Driver Assistance System
AGV	Autonomous Ground Vehicle
AHV	Articulated Heavy Vehicle
ATS	Active Trailer Steering
AV	Autonomous Vehicle
AVSS	Active Vehicle Safety System
CCOR	Control Coupled Output Regulation
CG	Centers of Gravity
CNN	Convolutional Neural Network
DDQN	Double Deep Q-network
DE	Differential Evolution
DOF	Degrees of Freedom
DQN	Deep Q-network
DRL	Deep reinforcement learning
DYC	Direct Yaw Control
ESC	Electronic Stability Control
EV	Ego Vehicle
FHWA	Federal Highway Administration
HGV	Heavy Goods Vehicle
HIL	Hardware in the Loop
HSLC	High-speed Lane Change
HSSO	High-speed Steady State Off-tracking
HSTO	High-speed Transient Off-tracking
IDM	Intelligent Driver Model
ISE	Integral of the Square Error
LCV	Long Combination Vehicle
LLC	Lower-Level Controller
LQR	Linear Quadratic Regulator

LV	Leading Vehicle
MDP	Markov Decision Process
MOBIL	Overall Braking Induced by Lane Changes
MVDM	Modified Visual Driver Model
MPC	Model Predictive Control
MTAHV	Multi-Trailer Articulated Heavy Vehicle
NLMPC	Nonlinear Model Predictive Control
NMS	Neuromuscular System
NSDA	Non-Dominated sorting Differential Evolution
NSC	National Safety Council
ODE	Ordinary Differential Equation
PER	Prioritized Experienced Replay
PI	Proportional Integral
PPO	Proximal Policy Optimization
PSO	Particle Swarm Optimization
PSS	Predictive Safety System
RAC	Rear Axle Center
RL	Reinforcement Learning
RMS	Root Mean Square
RPF	Randomized Prior Functions
RSOS	Residual Sum of Squares
RSS	Reactive Safety System
RWA	Rearward Amplification
SCSLA	Single-cycle Sinewave Lateral Acceleration
SIL	Software in the Loop
SLC	Single Lane Change
SQP	Sequential Quadratic Programming
STR	Static Rollover Threshold
SUMO	Simulation of Urban Mobility
SUV	Single Unit Vehicle
TD	Temporal Difference
TFAC	Tractor Front Axle Center

TJA	Traffic Jam Assistants
TSOS	Total Sum of Squares
ULC	Upper-level Speed Controller

Chapter 1

Introduction

1.1 Background

Nowadays, the growing demands of road freight transport necessitate the deployment of long and heavy vehicles to meet increasing safety and efficiency needs[1]. These vehicles, often referred to as Long Combination Vehicles (LCVs), offer numerous advantages over traditional tractor-semitrailers. Long combination vehicles are designed to maximize cargo capacity while minimizing their environmental and economic impact. Specifically, LCVs occupy less space on roads, resulting in improved traffic flow and reduced congestion. They also contribute to less road wear[2], contributing to lower maintenance costs for transportation infrastructure. Additionally, LCVs consume approximately 17% less fuel on average, making them a more sustainable option compared to their shorter counterparts. This fuel efficiency leads to a 30% reduction in the total cost of ownership on average, making them an economically attractive choice for freight operators[3].

However, despite these benefits, the safety challenges associated with heavy vehicles remain a significant concern. Based on the National Safety Council (NSC) injury report of 2020, heavy vehicles were involved in approximately 10% of road accident fatalities in North America[4]. This statistic underscores the pressing need to address safety issues associated with these large vehicles. As depicted in Figure 1-1, 4,842 heavy vehicles

were involved in fatal crashes in 2020. While this represents a 4% decrease compared to 2019, it is still alarming when considering the 33% increase since 2011.

The risks are further amplified by the specific characteristics of LCVs. These vehicles are longer and can be significantly heavier than tractor-semitrailers. Their increased size and weight contribute to unique operational challenges, such as reduced maneuverability, longer braking distances, and increased susceptibility to rollovers and jackknifing. Consequently, accidents involving LCVs have the potential to be more severe, leading to higher fatality rates and more extensive property damage[5]. As the use of LCVs continues to grow, addressing their safety becomes imperative to balance their economic and environmental benefits with public safety concerns.

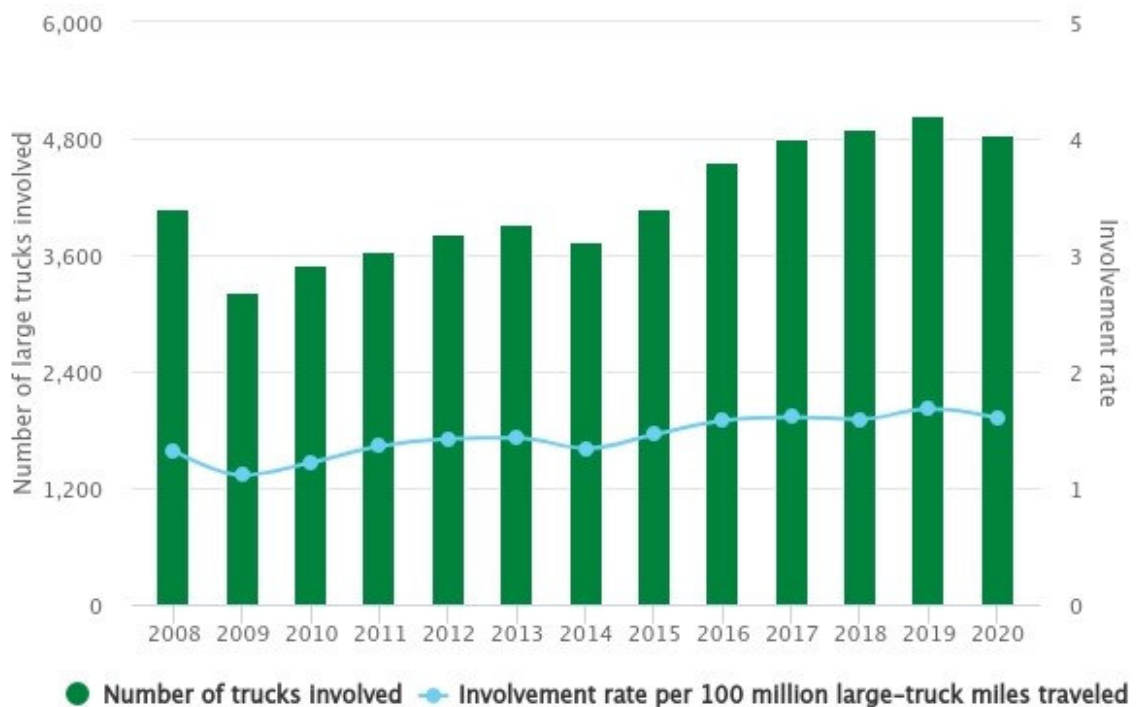


Figure 1-1: Heavy vehicle involvement in fatal crashes[4].

The automotive engineering industry is undergoing a transformative shift towards the era of Autonomous Ground Vehicles (AGVs). These vehicles represent a technological breakthrough in transportation, capable of navigating through highways, urban traffic, or even challenging off-road terrains without human intervention. The development of AGVs is made possible by the integration of cutting-edge technologies, including sophisticated software algorithms, powerful onboard computers, advanced sensor arrays, and reliable wireless communication systems. This integration is revolutionizing vehicle design, enabling AGVs to make complex decisions and respond to dynamic environments in real time.

One of the most compelling advantages of AGVs is their potential to significantly reduce accidents caused by human error, which accounts for many road mishaps. By leveraging automation, AGVs enhance safety for passengers, drivers, and pedestrians alike while reducing the financial burden associated with vehicle collisions. Advanced Driver Assistance Systems (ADAS) play a crucial role in this transformation. Systems like Adaptive Cruise Control (ACC), Traffic Jam Assistants (TJA), and collision avoidance systems have demonstrated their effectiveness in improving road safety. Between 2000 and 2019, the implementation of these technologies contributed to a substantial decline in fatal car crashes in North America, with reductions of 36% in the United States and an impressive 50% in Canada[6]. Figure 1-2 presents the evolution of driving assistance functions application and their potential future evolution.

Despite this positive trend, there is a notable divergence when it comes to heavy vehicle involvement in fatal crashes. While general automotive fatalities have been on a downward trajectory, incidents involving heavy vehicles have shown an upward trend. This

disparity highlights a critical gap in current safety technologies and emphasizes the need for focused research and innovation in heavy vehicle safety systems. Heavy vehicles, such as LCVs, present unique challenges due to their size, weight, and complex dynamics. Addressing these challenges requires the development of specialized systems tailored to mitigate risks in heavy vehicle operations.

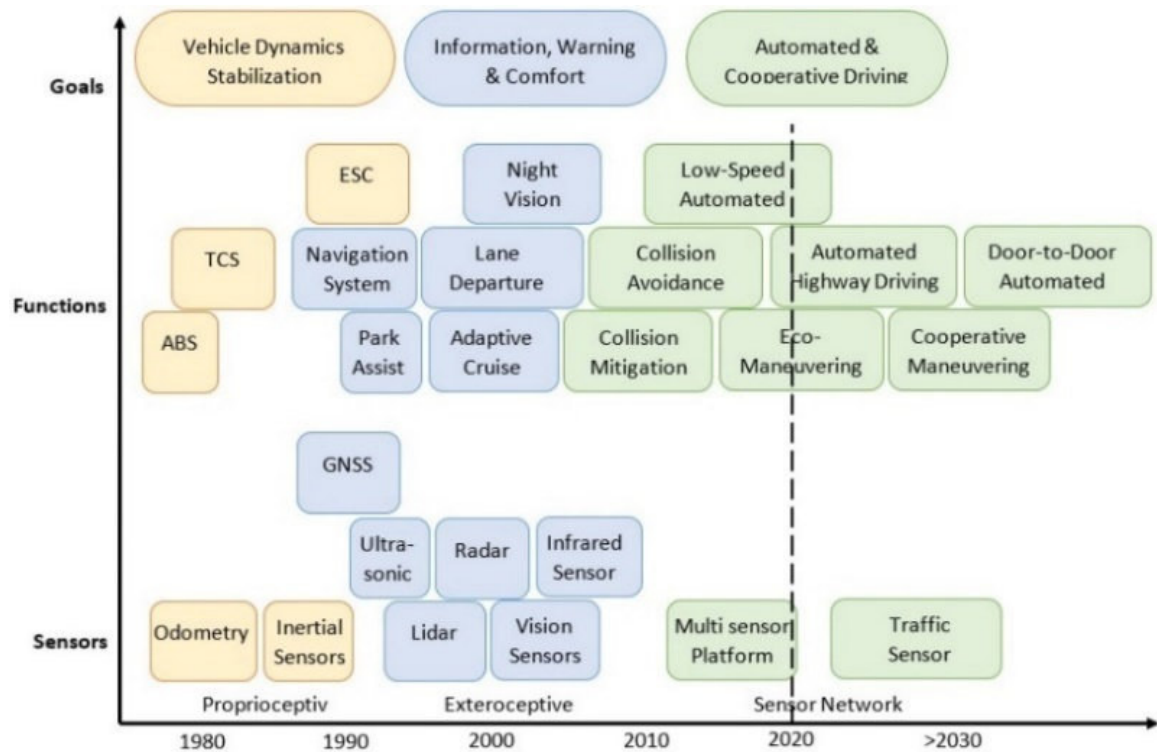


Figure 1-2: Past and future evolution of driving assistance functions from market perspective [4].

To enhance vehicle safety, Active Vehicle Safety Systems (AVSSs) are extensively researched, developed, and commercialized over recent decades. These systems, such as vehicle stability control, aim to prevent accidents by actively intervening in critical situations[7, 8]. Active vehicle safety systems can be broadly categorized into Reactive Safety Systems (RSSs), which are designed to respond to the current state of the vehicle and immediate threats[9]. For example, systems like Anti-lock Braking System (ABS) and

Electronic Stability Control (ESC) are highly effective at mitigating risks associated with skidding or loss of control. However, while RSSs have proven their value in enhancing vehicle safety, they operate within the limitations of responding to existing conditions and do not account for driver behavior or errors. This is a significant gap, as human errors are a leading cause of traffic collisions, contributing to many accidents[10]. Addressing this issue requires a paradigm shift, where the human factor is removed from the control loop through the deployment of autonomous driving systems[11]. Autonomous vehicles eliminate the variability and unpredictability associated with human drivers, offering the potential for a much safer transportation ecosystem. Despite their promise, the widespread deployment of autonomous driving systems faces significant challenges. One of the primary obstacles is the lack of formal, rigorous approaches for verifying and validating the safety of these systems in diverse and unpredictable scenarios[5]. Autonomous vehicles must be able to handle a wide array of driving conditions, ranging from adverse weather to complex urban environments, making comprehensive testing and certification critical.

Automated driving technologies have advanced significantly over the past few decades. However, the transition to fully autonomous driving will span a prolonged period during which most vehicles on the road will operate with partial autonomy. Table 1-1 outlines the timeline of developments in automated driving and safety functions. Since the late 1990s, ADASs, such as lane departure prevention, have emerged as vital tools for enhancing road safety. These systems are classified as Predictive Safety Systems (PSSs), which extend beyond reactive systems by considering not only the vehicle's current state but also predicting potential hazards and future states. Predictive safety systems enable proactive intervention, reducing the likelihood of accidents caused by anticipated threats.

In the last two decades, extensive research has focused on the development of semi-autonomous vehicles. These vehicles, categorized as Level 2 or Level 3 automation according to the SAE International standard[12], combine human driving with autonomous capabilities. Examples include adaptive cruise control with lane-keeping assist and traffic jam assist, where the vehicle can manage certain driving tasks under specific conditions. Semi-autonomous vehicles serve as a transitional technology, bridging the gap between traditional human-operated vehicles and fully autonomous systems. Their adoption paves the way for a gradual shift in public acceptance, technological advancement, and regulatory readiness for higher levels of autonomy.

Table 1-1: Vehicles automated driving timeline.

1926	● Radio-controlled car-following vehicle
1948	● Invention of modern cruise control
1960	● Firebird moved over a series of detector circuits buried in the highway
1980	● Vision-guided driver-less Mercedes-Benz van
1987	● Electronic Stability System was introduced by BMW
1995	● Mitsubishi Diamante revealed leaser-based adaptive cruise control technology
1999	● Autonomous shuttle service started working in Netherlands
2001	● Nissan Cima introduced lane departure warning system
2003	● Toyota Harrier came with pre-crash mitigation
2009	● Google waymo was introduced
2010	● VisLab international Autonomous Vehicle (AV) challenge
2012	● Nevada offered licenses for AGVs
2013	● Ford launched fleet of 10 Fusion hybrid autonomous research vehicles
2014	● Autonomous driving levels introduced by SAE
2014	● Mercedes-Benz unveiled FT2025, the world's first autonomous truck
2017	● Tesla unveiled semi-autonomous vehicle, the first electric Articulated Heavy Vehicle (AHV)

- 2018 • Ford F-Vision, a novel level 4 autonomous AHV
- 2018 • Volvo Trucks introduced Vera, the first self-driving fully electric truck
- 2019 • AXL, the latest autonomous heavy vehicle of Scania was introduced

As mentioned above, research activities in semi-autonomous and autonomous driving have predominantly focused on passenger cars, given their prevalence and the relatively straightforward dynamics of single-unit vehicles. Heavy Goods Vehicles (HGVs), on the other hand, represent a significantly higher risk in highway operations—approximately 7.5 times greater than passenger cars[13]. This elevated risk stems from their larger mass, extended braking distances, and greater potential for catastrophic outcomes in accidents. Despite these challenges, research efforts have largely overlooked the application of PSSs and ADAS for heavy goods vehicles, particularly articulated vehicles such as MTAHVs.

Articulated vehicles pose unique challenges due to their complex structure, which includes multiple pivoted points and dynamic interactions between the tractor and trailers. These complexities become even more pronounced in MTAHVs, where additional trailers further amplify instability risks. While some recent studies have addressed autonomous driving for specific types of articulated vehicles, such as construction vehicles, automated reverse-parking systems, and autonomous construction trucks[14, 15], these systems have primarily been designed for low-speed operations. For instance, low-speed trajectory planning and tracking have often relied on kinematic control models that fail to capture the intricate dynamic behaviors exhibited by articulated vehicles during high-speed operations. Such high-speed dynamics include critical issues like trailer sway, jackknifing, and rollover—phenomena that are not only dangerous but also difficult to predict and mitigate.

Addressing these dynamics requires a shift from purely kinematic control strategies to more sophisticated dynamic control approaches that can adapt to a wider range of operating conditions.

Interactions between human operators and machines are studied to some extent for improving the ADAS capabilities of articulated vehicles[16, 17]. These studies typically focus on enhancing driver awareness and control, such as through lane-keeping assistance, adaptive cruise control, and automated braking systems. While these efforts have contributed to safety improvements, they fall short of exploring the potential for semi-autonomous and fully autonomous operation in articulated heavy vehicles.

Multi trailer articulated heavy vehicles remain significantly underexplored in this regard. The added complexity of multiple trailers and their coupling dynamics introduces unique challenges in control, sensing, and decision-making that are not encountered in single-unit vehicles or even standard articulated vehicles. Despite their critical role in long-haul freight transport and their potential for improving logistics efficiency, MTAHVs have received limited attention in research on autonomous and semi-autonomous technologies.

1.2 Motivations

The primary goals of designing autonomous/semi-autonomous driving systems for road vehicles are to enhance operational safety and increase transportation efficiency. MTAHVs are becoming more common on highways due to their benefits in freight transport. Compared to single-trailer AHVs, MTAHVs with an A-train double configuration can cut travel mileage by 44%, lower fuel consumption and greenhouse gas emissions by 32%, and reduce tire-road wear by 40%[15]. Despite these advantages,

MTAHVs face challenges like poor maneuverability at low speeds and reduced lateral stability at high speeds due to their multi-unit structures, large dimensions, and high Centers of Gravity (CGs).

During high-speed turns or evasive maneuvers, MTAHVs often experience a phenomenon called Rearward Amplification (RWA), where the trailing units exhibit exaggerated lateral and yaw motions[15]. The rear trailer, in particular, experiences higher lateral acceleration than the leading tractor, making it more prone to rollover. This trailer is often the first to tip over in such situations, and drivers may not recognize the instability in time to take corrective action[16]. Drivers primarily rely on the motion cues of the leading unit since the cues from trailing units are dampened by the tractor cabin suspension, articulation joints, and other factors[17]. Consequently, operating MTAHVs demands high levels of training, skill, and experience from drivers.

Multi trailer articulated heavy vehicles play a vital role in freight transportation, especially for moving goods across long distances. Enhancing the safety and performance of MTAHVs has the potential to positively impact the economy and the lives of people who depend on transported goods. Autonomous and semi-autonomous technologies offer transformative opportunities by making freight transportation safer, more efficient, and sustainable. Automating the driving process, either partially or fully, can lead to smoother operations and fewer accidents, benefiting both drivers and other road users. As the demand for MTAHVs grows, the development of advanced technologies to improve their safety and performance is crucial. Active safety systems like Active Trailer Steering (ATS), which combine sensors, control algorithms, and cutting-edge technologies, can help prevent accidents and enhance the safety of MTAHVs. However, the current state of

MTAHV safety technology is limited, necessitating further research to address safety challenges and develop effective solutions.

Research in autonomous/semi-autonomous driving for MTAHVs is critical to overcoming these challenges and achieving more advanced, safer vehicles. These technologies can yield significant economic benefits by reducing transportation costs, enhancing operational efficiency, and cutting expenses related to fuel consumption and tire wear. As a result, they can make the transportation industry more competitive and sustainable. Moreover, autonomous/semi-autonomous driving can alleviate the issue of driver fatigue, a common concern for MTAHV operators. By automating the driving process, these systems allow drivers to focus on other tasks, reducing the risk of fatigue-related accidents and improving their overall well-being.

To bridge this gap, further investigation is needed to develop advanced control systems tailored specifically for MTAHVs. This includes the integration of high-fidelity dynamic models into control strategies, the use of machine learning algorithms to predict and mitigate high-risk scenarios, and the design of robust sensing and perception systems capable of handling the extended length and blind spots of MTAHVs. Moreover, the transition from human-driven to autonomous/semi-autonomous MTAHVs must also account for regulatory frameworks, infrastructure compatibility, and public acceptance, all of which are critical for successful deployment. By addressing these challenges, research can pave the way for safer and more efficient heavy vehicle operations, reducing the risks associated with highway transportation while leveraging the benefits of automation for the freight industry.

1.3 Objectives

The primary objective of this thesis is to contribute to the advancement of semi-autonomous driving control systems for MTAHVs, addressing the unique challenges that these vehicles pose in terms of stability, control, and decision-making in highway driving environments. Given the complexity of MTAHVs, which involve intricate dynamics and require robust control mechanisms, this research aims to develop a comprehensive framework encompassing accurate modeling, advanced control design, human-like driver integration, and intelligent decision-making. The expanded objectives are as follows:

1- Generation and validation of MTAHV models

Develop an effective dynamic model for an MTAHV with the configuration of A-train double, focusing on capturing the intricate interactions between the towing and trailing vehicle units. The model will account for various real-world conditions, including variable longitudinal speed, nonlinear lateral and longitudinal forces, trailer loads, and aerodynamic effects. Validation of this model will be carried out through simulation and comparison with the highly nonlinear similar vehicle model that is build in TruckSim to ensure its reliability in representing the system's behavior under different highway scenarios. The validated MTAHV model will be used to design the tracking-controller and the driver for the semi-autonomous A-train double.

2- Design, tuning and implementation of an NLMPC-based tracking-controller

Design a Nonlinear Model Predictive Control (NLMPC) based tracking-controller for the semi-autonomous MTAHV. This controller will be responsible for ensuring accurate path-following, maintaining system stability, and mitigating oscillatory behaviors

such as trailer sway or jackknifing. The controller will incorporate different driving conditions and payload variations to ensure robust and stability operation even under challenging conditions.

3- An integrated modified visual driver model for human-like interaction

Introduce an integrated driver model based on a visual flow approach, modified to account for human-like perceptual and cognitive responses during highway driving. This model aims to simulate driver behavior realistically, enabling better interaction between the NLMPC-based tracking-controller and the human driver. The integration focuses on optimizing the driving experience, ensuring that the MTAHV can operate semi-autonomously while allowing seamless human intervention when necessary.

4- Development of a highway decision-making framework using deep reinforcement learning

Leverage the state-of-the-art deep reinforcement learning (DRL) techniques to enable the MTAHV to make intelligent, real-time decisions in complex highway traffic environments. This includes tasks such as lane changing, longitudinal constant acceleration and deceleration, as well as obstacle avoidance. The DRL framework will be trained on a wide range of scenarios to generalize its decision-making capabilities across diverse conditions, including varying traffic densities and surrounding vehicles speed.

5- Semi-autonomous driving framework for MTAHVs

Combine the MTAHV, NLMPC-based tracking-controller, and the human into a cohesive system to form a novel simulation platform to explore semi-autonomous driving systems for MTAHVs. The platform will be rigorously tested to ensure its effectiveness in

a non-cooperative maneuver when the reference paths for the human driver and the NLMPC-based tracking-controller conflict each other.

1.4 Thesis organization

The rest of the thesis is organized as follows.

Chapter 2: Literature review

This chapter will conduct an extensive literature review in semi-autonomous vehicles and identify the research gap in semi-autonomous MTAHVs.

Chapter 3: Vehicle dynamic model

Chapter 3 employs a systematic approach to model and validate the dynamic behavior of a MTAHV with an A-train double configuration. A simplified single-track model is developed to represent the lateral dynamics of the vehicle, capturing critical factors such as trailer lengths, coupling mechanisms, and load distribution. The model serves as a foundation for analyzing vehicle stability, maneuverability, and safety under various driving conditions. The equations of motion are derived systematically in Maple which can be easily modified for other types of a MTAHV configuration. To validate the single-track model, a high-fidelity nonlinear dynamic model of the A-train double configuration is created in TruckSim. Co-simulations using Matlab and TruckSim are conducted to validate the model based on an open-loop standardized lane-change maneuver. This integrated methodology ensures that the developed model is both computationally efficient and accurate, laying the groundwork for advanced control strategy design in subsequent chapters.

Chapter 4: NLMPC-based tracking-controller design and tuning

This chapter presents the design of the NLMPC-based tracking-controller using Matlab/Simulink for autonomous driving of the A-train double. The controller combines a real-time optimizer, which calculates control inputs, and the single-track yaw-plane prediction model with 11-degrees-of-freedom (DOF), enabling accurate trajectory tracking over a finite prediction horizon. The control inputs include the tractor's front wheel steering angle and driving/braking torque. By utilizing a novel upper-level multi-objective optimization method, which is solved by the NSDA (Non-Dominated Sorting Differential Evolution), the weighting matrices in the cost function of NLMPC are selected to minimize the vehicle lateral acceleration and accumulating tracking error.

To validate the controller's performance, a co-simulation platform integrates the NLMPC with a high-fidelity 3D TruckSim model, forming a closed-loop feedback system. The platform tests the controller's performance under single lane-change maneuver. Then, the controller's stability and robustness are investigated based on the varying parameters and different driving conditions. This methodology provides a robust framework for autonomous control of MTAHVs, combining predictive modeling with realistic simulation to ensure safety and operational effectiveness.

Chapter 5: An integrated modified visual driver model

This chapter presents the generation of an Integrated Modified Visual Driver Model (Integrated MVDM) for highway driving of MTAHVs. The model simulates human driver behavior by incorporating visual and cognitive perception principles to execute lane-keeping and lane-change maneuvers. The Integrated MVDM evaluates the road path ahead

and determines a desired steering angle based on visual information. Key elements include signal delays that emulate the driver's response time and neuromuscular system (NMS) dynamics, which account for the physical actions of the driver's arm in executing steering commands. The model operates in a discrete time framework, updating vehicle states based on the driver's control inputs. The Integrated MVDM parameters are optimized based on the two different highway scenarios which are simulated within a co-simulation platform, integrating the driver model with a high-fidelity MTAHV simulator in TruckSim. Its performance is evaluated based on maneuver accuracy to replicate realistic driver behavior. This methodology ensures the Integrated MVDM provides a robust, human-like interface for simulating and refining autonomous control systems in MTAHVs.

Chapter 6: Deep-reinforcement-learning based decision making for MTAHVs

This chapter outlines the methodology for designing a DRL-based decision-making module for autonomous driving of MTAHVs on a three-lane highway using Python. The DRL framework is customized to account for two decision-making paradigms: autonomous control and human-like driving. The model aims to safely navigate the MTAHV through complex highway scenarios by making optimal decisions for acceleration, deceleration, and overtaking. By simulating dynamic traffic conditions with randomly distributed vehicles, the environment challenges the DRL algorithm to handle realistic situations, ensuring safety and efficiency.

To train and validate the DRL model a complete co-simulation environment is developed. Simulation of Urban Mobility (SUMO), an open-source traffic simulation platform is employed to create a high-fidelity highway simulation. The simulation

environment provides a setting for the DRL agent to learn and test its decision-making capabilities. Then, the controller/driver is simulated in python to perform the commands from the decision-making level and the MTAHV is simulated in the TruckSim as the high-fidelity nonlinear model. The methodology focuses on safety, efficiency, and adaptability, enabling the DRL model to handle diverse highway scenarios. This approach ensures that the autonomous system can make robust and reliable decisions while navigating multi-lane traffic conditions.

Chapter 7: Semi-autonomous driving control

This chapter presents the methodology for developing a semi-autonomous control system for MTAHVs, where both the human driver and the NLMPC-based tracking-controller share the control of the vehicle. The system models the dynamic interaction between the two controllers as a non-cooperative game, with each contributing to the vehicle's state evolution. The human driver and autonomous controller may have differing objectives, such as following separate target paths. To address this, the autonomous controller is designed to adapt its strategy by accounting for the driver's actions, mitigating potentially unsafe behaviors like inattentiveness or errors. Similarly, the driver may attempt to counteract the autonomous controller's inputs to maintain their desired trajectory, leading to a strategic interplay of control.

Because of the highly nonlinear dynamics of the MTAHV, the conventional game theory methods that are based on the linear system dynamics are not applicable for this case. A novel method is developed to address this difficulty and complexity in the design of the shared control. The non-cooperative game framework is conceptualized to capture

this interaction, emphasizing the mutual influence of steering actions. The model enables the design of an autonomous controller on the extended model (combination of vehicle model and Integrated MVDM as the driver) to response to the different and conflicting commands. By analyzing these dynamics, the system can achieve a balance between autonomous intervention and driver autonomy.

Chapter 8: Conclusions

This chapter highlights the insightful findings achieved in this study, emphasizes the effectiveness of the proposed modeling and simulation framework for exploring the semi-autonomous systems for MTAHVs, predicts the impacts of the achieved findings on the design and development of semi-autonomous technologies for MTAHVs, and lists a few promising research directions for developing these technologies.

Chapter 2

Literature review

2.1 Introduction

This chapter begins by reviewing the various types of articulated heavy vehicles (AHVs), including multi-trailer configurations, and introduces their common layouts. Next, Section 2.3 highlights the primary instability behaviors of articulated vehicles in both the yaw and roll planes. These include issues such as trailer sway, tractor/trailer jackknifing, and rollover occurrences. Furthermore, the fundamentals of MPC, one of the most widely used control methods in autonomous driving, are presented along with its application in vehicle autonomous control.

2.2 Articulated heavy vehicles configuration

The vast majority of articulated vehicles are commercial heavy vehicles designed primarily for the efficient and cost-effective transportation of goods and materials. To accommodate various transportation needs, a wide range of truck combinations is available. The Federal Highway Administration (FHWA) classifies vehicles into 13 categories, as illustrated in Figure 2-1[18]. Categories 8-13 are considered as the articulated heavy vehicles which are marked by the red lines. Truck combinations display considerable diversity in their axle arrangements, articulation points, load capacities, and auxiliary features. Additionally, when more than two units are connected, a fifth-wheel or a dolly is commonly used between the units.

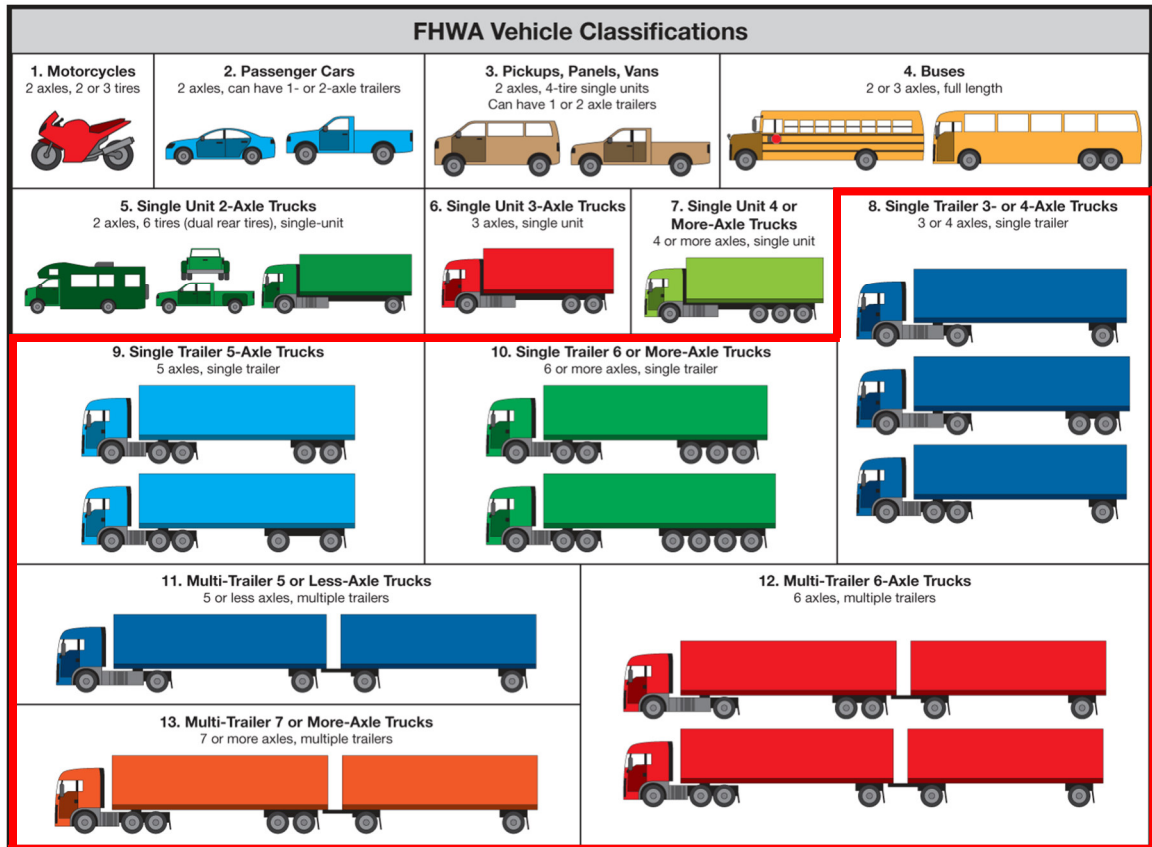


Figure 2-1: Federal highway administration 13-category scheme for vehicle classifications[167].

2.2.1 Multi-trailer articulated heavy vehicles

A MTAHV is defined as a combination of a tractor and two or more trailing units that exceed specific length or weight limits, which typically vary by country. For instance, in Canada, an MTAHV is defined by a minimum total length of 27.5 meters[19], whereas in the United States, a minimum gross vehicle weight of 80,000 pounds is used as the criterion[20]. On another hand, while many countries limit the maximum length of 18.75 meters and a weight of 40 tons for an AHV, certain nations allow longer and heavier configurations. For example, Sweden and Finland permit LCVs with lengths of up to 25.25 meters and weights of up to 60 tons[21].

From another point of view, a MTAHV can be classified as A-, B-, or C-trains based on the type of coupling used between the trailers[22].

- 1- A-train double: In this configuration, trailers are connected using a turntable dolly or an A-dolly mounted on the rear of the preceding trailer.
- 2- B-train double: In this configuration, the trailers are connected to each other by a fifth-wheel installed on the extended frame from the preceding trailer.
- 3- C-train double: In this configuration, the trailers are connected using a C-dolly mounted on the rear of the preceding trailer.

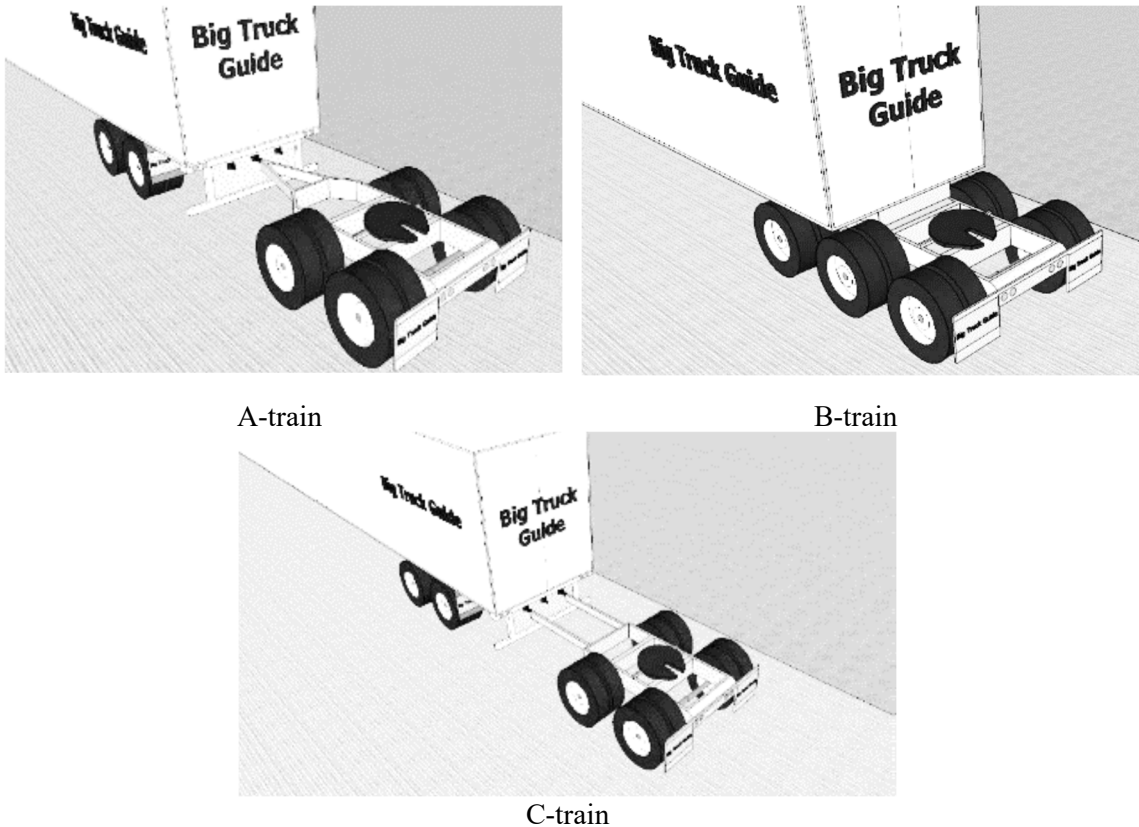


Figure 2-2: Various methods of connecting two trailers in North America[23].

2.3 Dynamics performance and instability

There are some major differences between articulated heavy vehicles and passenger vehicles because of their different designs and applications. Articulated heavy vehicles often incorporate one or more articulation joints along with multiple axles, significantly complicating their dynamics. For example, various maneuvers can induce oscillatory motion modes (yaw instability)[24]. Another key distinction is the susceptibility of multi-axle vehicles to rollover, driven by a high wheelbase-to-track ratio, elevated center of gravity, and uneven or variable weight distribution across axles. This implies that such trucks may display instability at lateral accelerations of 0.3g to 0.4g during steering and braking maneuvers[25, 26], while the passenger cars can typically withstand lateral acceleration ranging from 0.8g to 1.2g before experiencing a rollover[27].

2.3.1 Yaw-plane instability.

Yaw-plane instability refers to the unstable behavior of a vehicle's lateral or yaw motion during sudden lane changes, high-speed curve entry, or straight-line driving. As illustrated in Figure 2-3, articulated vehicles can exhibit three distinct types of yaw-plane instability, as identified in studies[28-32].

- 1) **Trailer Sway:** This phenomenon involves periodic trailer yaw oscillations, a dynamic instability that can result in increasing oscillatory amplitude, commonly referred to as snaking or sway, as depicted in Figure 2-3(a). Trailer sways typically occur at high speeds and are often triggered by external disturbances, such as wind gusts or the passing of large vehicles. Internal factors, including sharp steering inputs, unstable

trailer configurations (e.g., high payload or poorly positioned center of gravity), can also induce sway [33].

- 2) **Trailer Jackknife:** Jackknifing occurs when the units of an articulated vehicle fold into a "V" shape rather than maintaining alignment. As illustrated in Figure 2-3(b), a trailer jackknife represents a divergent instability where the tractor remains stable on the road, but the trailer rotates uncontrollably around the articulation joint. This phenomenon often arises when the trailer axle wheels lock during braking on slippery surfaces, such as in combined braking and steering scenarios.
- 3) **Tractor Jackknife:** In tractor jackknife, the trailer remains stable on the road while the tractor slides sideways, as illustrated in Figure 2-3(c). This instability typically occurs during excessive braking, such as when the rear wheels lock up, or during acceleration, such as when the rear wheels spin on a low-friction surface. The loss of grip or traction in the tractor's rear wheels is a primary cause of this phenomenon[31, 27].

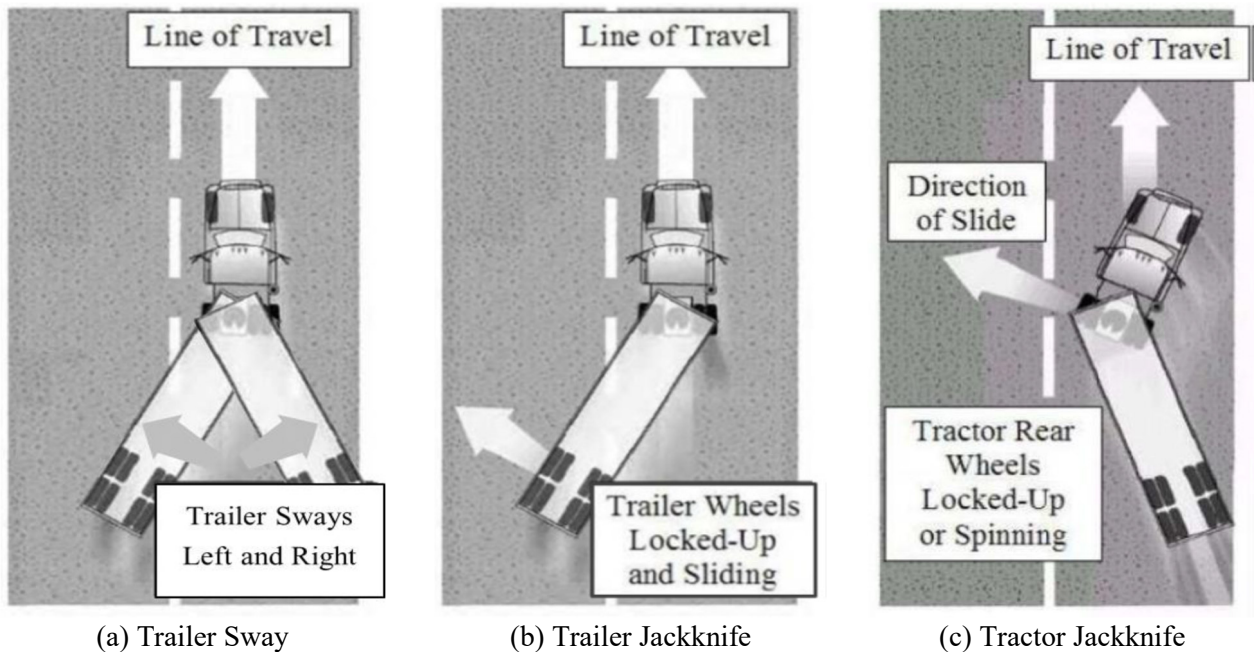


Figure 2-3: Unstable motion modes for an AHV in the yaw-plane[34].

Both sway and jackknife are highly dangerous unstable motion modes in AHVs, with jackknife being a leading cause of fatal accidents. Additionally, the trailer off-tracking phenomenon, while is not an instability issue, poses significant risks. For instance, during a low-speed 90-degree turn, off-tracking refers to the deviation of the semi-trailer's axles or articulation hitch from the path followed by the lead tractor's steering axle[35]. In non-articulated vehicles, the front and rear wheels take slightly different paths while cornering, though the difference is usually minimal. However, in AHVs, the trailer deviates significantly from the lead tractor's path during a turn or lane change. This deviation can be hazardous, as it may cause the semi-trailer to cross the lane boundary or collide with an adjacent vehicle, even when both the AHV and the other car are within safe speed limits. To ensure safety and stability, it becomes necessary for the driver or active control systems to compensate for this off-tracking behavior.

2.3.2 Roll-plane instability

Rollover accidents in AHVs constitute a significant portion of crashes. According to statistics from NHTSA/NCSA, approximately half of heavy truck crash fatalities in the United States involve rollovers, with similar findings reported in European studies[36]. Roll instability refers to the likelihood of a vehicle tipping over, as illustrated in Figure 2-4. Various factors, including vehicle characteristics (such as the height of the center of gravity and track width), driver behavior, and environmental conditions (such as road conditions), play a critical role in rollover incidents[37]. Additionally, rollover risk is influenced by factors like suspension compliance, lash, and the distribution of roll stiffness across the

axles[24]. Vehicle rollovers can be categorized into two types: tripped and untripped rollovers.

A tripped rollover typically occurs in two scenarios: when a vehicle's tires skid and sink into soft soil, or when the vehicle collides with a tripping mechanism (e.g., a curb) at high lateral velocity[38]. Static rollover indicators, such as the tilt table ratio, critical sliding velocity, static stability factor, and track width ratio[39], are used to identify tripped rollovers. These indicators rely solely on the vehicle's parameters to assess the static rollover risk. Untripped rollovers primarily occur due to significant lateral accelerations caused by extreme operating conditions, such as abrupt directional changes at high speeds, excessive velocity while navigating curves, and lateral load transfer[40]. The unique dynamic characteristics of RWA cause the rollover in AHVs usually initiates with the semi-trailer. The static rollover threshold, measured as lateral acceleration in terms of gravitational acceleration (g), serves as the fundamental indicator of roll stability[41]. This threshold represents the maximum steady lateral acceleration a vehicle can withstand without experiencing a divergent roll response, such as rollover.



Figure 2-4: Truck rollover due to excessive speed in a turn[42].

2.4 Measures of directional performance

Directional performance measures for AHVs encompass a set of criteria used to assess their dynamic behavior in navigating various turns and maneuvers while minimizing risks to other road users and infrastructure. These measures are essential for ensuring the safe operation of AHVs, which are large and complex vehicles presenting unique challenges for drivers and operators. As outlined in Section 2.3, examples of such instability and unsafe operations include trailer sway, jackknife, and rollover, among others. These criteria help evaluate the design and operation of AHVs, highlighting potential areas for improvement to enhance safety and performance, particularly when control systems are integrated. Common measures for high-speed maneuvers useful in highway driving are discussed below.

2.4.1 High-speed performance measures

High-speed directional performance poses additional safety challenges that are harder to manage compared to the low-speed performance measures. The most critical safety concern for AHVs is their roll instability, which can lead to severe accidents resulting in significant injuries or fatalities. The Static Rollover Threshold (SRT) is a key measure of roll stability, defined as the lateral acceleration at the vehicle's CG during a steady turn, without rolling over, expressed in g's. A higher SRT value generally indicates greater roll stability for heavy vehicles. SRT testing can be performed using either a tilt table or a high-speed circular turn with a constant radius of at least 100 meters, conducted over a specific time until rollover occurs. For longer vehicles like AHVs, the test requires

maintaining an initial speed until the vehicle reaches a steady-turn condition, after which the speed is gradually increased until rollover happens. While SRT effectively evaluates the roll stability of AHVs under near-static conditions, it does not consider the lateral acceleration response of AHVs during sudden maneuvers. As a result, an AHV may roll over at a lateral acceleration significantly different from its SRT when performing dynamic maneuvers such as High-speed Lane Changes (HSLC)[43]. Calculating the SRT for an AHV differs from that of a single-unit heavy vehicle because the lateral acceleration of each unit must be accounted for, as these values vary. According to the National Transport Commission of Australia[44], the resultant lateral acceleration of an AHV comprising N roll-coupled units can be determined by:

$$a_y = \frac{\sum_{n=1}^N m_n h_n a_{y_n}}{\sum_{n=1}^N m_n h_n} \quad (2.1)$$

where m_n represents the n^{th} sprung mass, h_n is the height of the n^{th} sprung mass CG, and a_{y_n} is the lateral acceleration at the CG of each vehicle unit.

Reward amplification is a critical high-speed dynamic performance measure for MTAHVs. It is defined as the ratio of the maximum motion variable (such as yaw rate or lateral acceleration) of the rearmost roll-coupled towed unit to that of the hauling unit during a high-speed obstacle-avoidance maneuver. This maneuver uses a Single-cycle Sinewave Lateral Acceleration (SCSLA) steering input at a specified road friction level. Typically, RWA values based on lateral acceleration and yaw rate are similar. However, under specific conditions—such as reduced tire-road friction, the use of self-steered trailer axles, or challenges in placing acceleration sensors precisely at the CG of vehicle units—RWA results for these two variables can differ significantly[45]. Therefore, the choice of

variable should be made carefully. The closer the RWA value is to unity, the better the directional performance of an AHV during HSLC maneuvers, making it less susceptible to dynamic rollover[46]. It should be noted that at a forward speed of 100 km/h, the RWA values for an A-train and B-train double reach 2.08 and 1.55, respectively[47]. Hence, in these MTAHVs, such elevated RWA levels during high-speed rapid obstacle-avoidance maneuvers can lead to an early rollover of the rear trailer.

Articulated heavy vehicles also exhibit High-speed Transient Off-tracking (HSTO) during high-speed obstacle avoidance maneuvers. High-speed transient off-tracking refers to the lateral displacement between the center of the last axle of the rearmost towed unit and the path of the steer axle's center of the lead unit during a sudden lane-change maneuver at a specific road friction level and predetermined high speed. This displacement can be either positive or negative, indicating overshoot or undershoot. The primary goal of the HSTO metric is to prevent the last trailers of the AHV from encroaching into adjacent lanes and to reduce the risk of traffic accidents. While both RWA and HSTO are measured under similar maneuvers, RWA governs lateral stability, while HSTO reflects the vehicle's ability to follow its path or maneuver. Therefore, an AHV with poor HSTO performance might have a lower rollover risk due to a lower CG[48]. Figure 2-5 offers a schematic representation of HSTO.

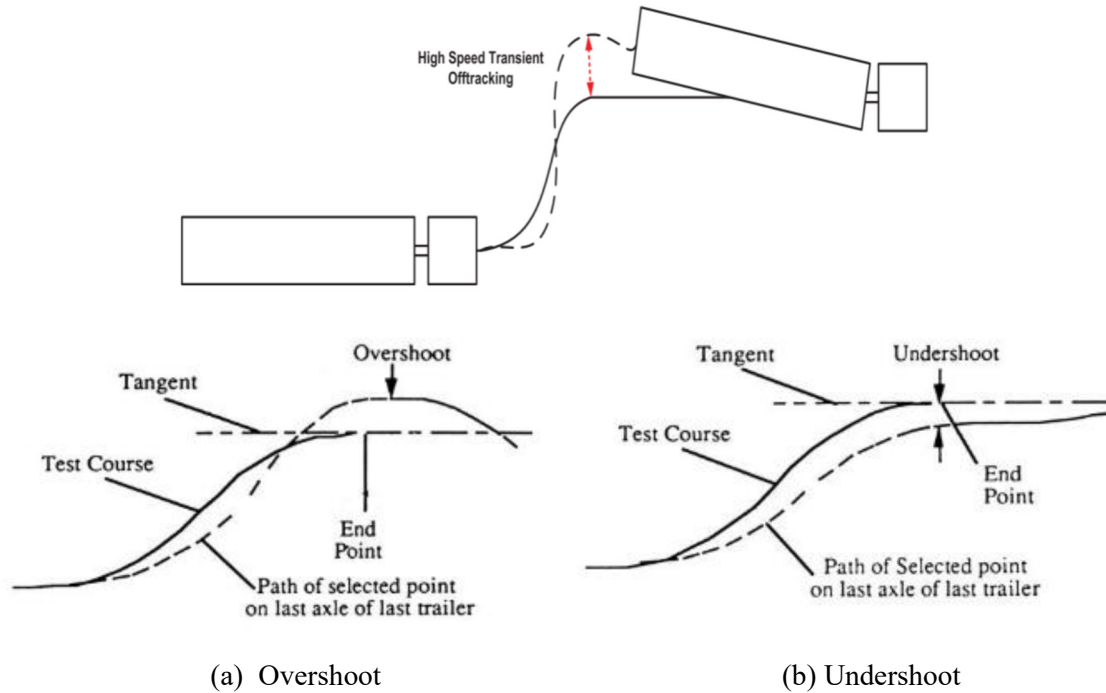


Figure 2-5: High-speed transient off-tracking performance measure along with overshoot and undershoot scenario[49, 50].

When navigating a steady curve with a large radius, the rearmost trailer of an AHV creates a lateral displacement between its Rear Axle Center (RAC) and the path of the Tractor Front Axle Center (TFAC). This displacement is referred to as High-speed Steady State Off-tracking (HSSO). High-speed steady state off-tracking serves as a measure of steady-state path-following performance and is evaluated when the AHV travels at a constant high speed on a road with a specified friction level. Higher HSSO values increase the risk of collisions with adjacent traffic or roadside obstacles. Figure 2-6 illustrates the concept of HSSO.

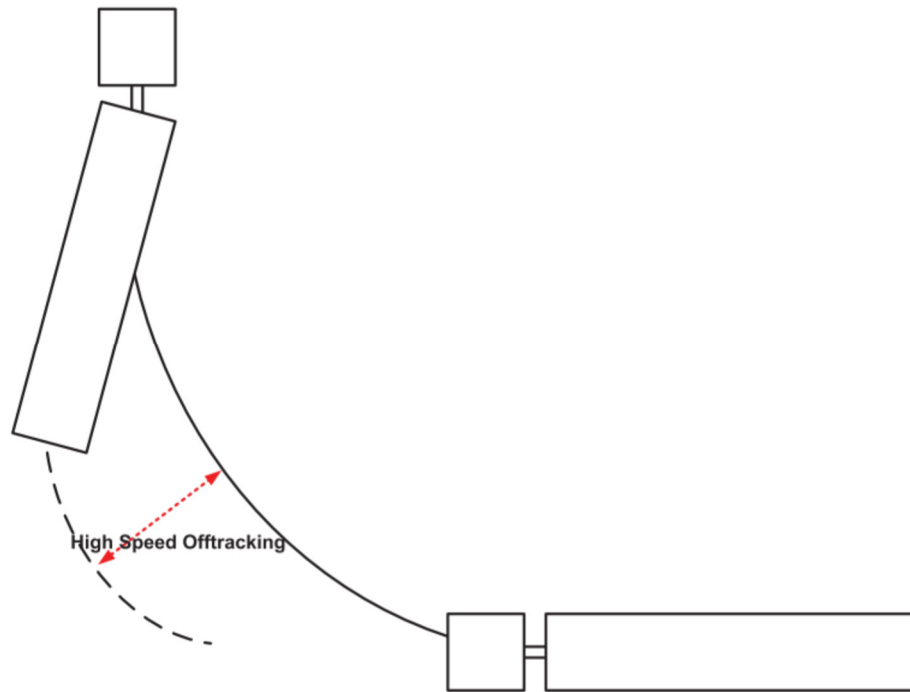


Figure 2-6: HSSO performance measure[50].

2.5 Model predictive control

Over the past decade, MPC has gained significant attention from researchers for its application in vehicle control systems in a real-time manner due to its predictive nature and ability to handle constraints systematically. It is characterized by its ability to predict future system behavior and optimize control inputs over a finite time horizon while respecting constraints. These constraints are addressed and satisfied in real time, resulting in a locally optimal control solution. The essence of MPC is based on three key components: (1) a prediction model, (2) optimization over a temporal window, and (3) feedback correction (Figure 2-7). These components are typically executed continuously by online algorithms.

- 1) The model of the system (or "plant") is utilized to predict its future behavior. At each time step, the current system state is sampled, and an optimal control strategy is

computed for a short future horizon $[t, t + T]$ using numerical minimization. Only the first control action is applied, after which the state is re-sampled, and the optimization is repeated from the updated state.

- 2) MPC is also an optimal control algorithm, which calculates future control actions based on a penalty or cost function. At each time-step, an open-loop optimal control problem is formulated, incorporating a performance index (or cost function) and operational constraints over a finite horizon. The solution to this problem provides a sequence of control actions, but only the first action is applied to the system. This process repeats at each subsequent time step, using updated state measurements and shifting the prediction horizon forward. Additionally, external factors, such as slip history, may be included as inputs to the model. This approach is effective for complex systems with dynamic changes and uncertainties such as MTAHVs, and there is no need to assess optimization performance over the entire time range[51].
- 3) Feedback correction plays a crucial role in ensuring the control system remains effective in real-world applications where uncertainties, disturbances, or modeling inaccuracies are present. Feedback correction involves updating the control actions based on the latest measurements of the system's state, enabling the controller to adapt dynamically to changes. In another word, feedback correction in MPC ensures that the control system remains effective, adaptive, and robust by continuously updating the control actions based on the system's real-time behavior. It bridges the gap between model predictions and actual system dynamics, making MPC suitable for practical applications.

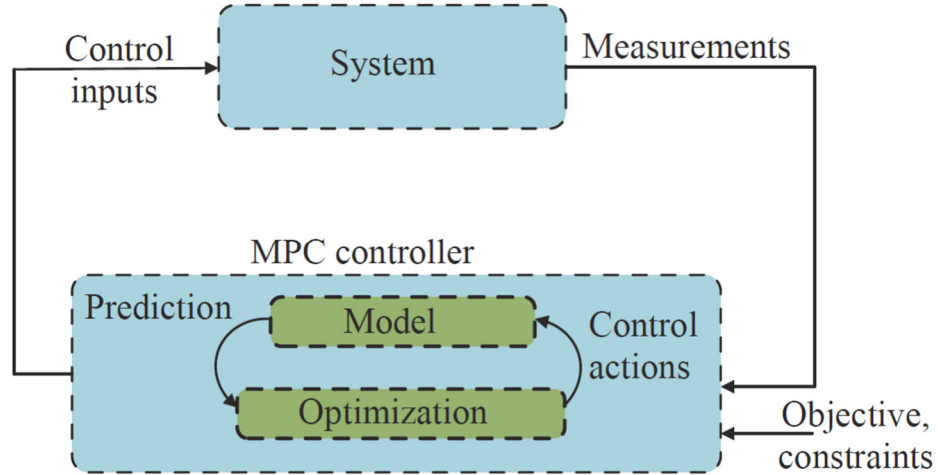


Figure 2-7: Illustration of model predictive control[52].

In the context of vehicle stability control, the primary objectives are to maintain longitudinal, yaw, and roll stability by tracking desired states or reducing deviations to acceptable thresholds. Safety constraints often limit the output, and the physical capabilities of actuators, such as tire limits, must also be considered. These requirements align well with the MPC formulation, as the approach solves finite-time optimization problems online, balancing objectives and constraints effectively.

Numerous studies and publications have explored MPC's applications to vehicle stability control. For example, Falcone[53] employed an MPC-based method to control yaw and lateral dynamics through active front steering and braking. Using both a detailed tenth-order vehicle model and a simplified bicycle model, the study found that while the simplified model struggled at high speeds, the detailed model provided robust performance across a range of speeds, albeit with significant computational demands. Beal[51] introduced a model predictive envelope controller designed to enforce safe handling boundaries in real time. This controller only activates to stabilize the vehicle when it approaches or exceeds the safe handling envelope. Additionally, Jalaliyazdi[54] proposed

an integrated MPC system addressing vehicle stability, traction control, and power distribution simultaneously. This controller was designed to be reconfigurable, accommodating different drivelines and braking actuator configurations. These examples demonstrate MPC's versatility and effectiveness in vehicle stability control, offering a powerful method for managing complex dynamics and constraints in real-time scenarios.

2.6 MTAHV modeling

In[55], Brock et al. evaluate various reduced-order lateral dynamic models used to represent tractor-trailer systems. The study identifies several existing models from research and assesses their performance in estimating the lateral dynamics of these vehicles. In[56] Jindra presents a simplified linear analysis of the lateral dynamics of tractor-trailer systems based on the Newton's law. The study models the vehicle combination as a dynamic system with five degrees of freedom, accounting for the sideslip and yaw motions of the tractor, as well as the yaw motions of the semitrailer, converter dolly, and full trailer. In[57], a kinematic motion planning controller for the general n-trailer system is proposed. This approach utilizes a kinematic model of the tractor-trailer system, intentionally omitting the modeling of tire forces and the inertia of the truck and trailer. Although Rouchon demonstrates the controller's path-planning capabilities through simulations, he does not validate the model against experimental data or high-fidelity simulations. In[58], the researchers create a reduced-order lateral dynamic model based on the Newton-Euler force balance approach. They develop two versions of the model: one incorporating second-order steering tire dynamics (resulting in additional states) and another without it. The dynamic parameters of their truck were determined experimentally, and the model's simulation results were compared with experimental data for speeds ranging from 15 to 30 miles per

hour. However, the researchers do not provide any quantitative metrics to assess the accuracy of either model in their report. In[31], the author employs linear single-track models to examine the stability of various heavy vehicle combinations, including tractor-trailer systems. Using the Euler-Lagrange approach, he develops reduced-order models for both a five-axle configuration and configurations with additional trailers. But he does not validate the models using real-world data. Wolfe develops his dynamic model using the Euler-Lagrange method, adopting a five-axle bicycle model. He also investigates how different assumptions influenced the model's accuracy. To validate the model, Wolfe compares its performance with high-fidelity TruckSim simulations and experimental data. His findings indicate that the model accurately predicts the articulation angle during common steady-state maneuvers but does not evaluate its effectiveness in highly transient scenarios, such as emergency lane changes[59].

In[60], authors conducted an in-depth analysis of dynamic models for MTAHVs. The study aims to provide guidelines for selecting appropriate vehicle models for control algorithm development, design optimization, and linear stability analysis. The researchers developed and validated both a linear four DOF yaw-plane model and a linear seven DOF yaw-roll model of a tractor with two semitrailers. These models were compared against nonlinear simulations using the TruckSim software to assess their accuracy. The validation process demonstrated that the linear models could effectively predict the dynamic behavior of MTAHVs under various conditions. Additionally, the paper includes eigenvalue and frequency-response analyses of the linear models to identify unstable motion modes and predict unique dynamic characteristics of MTAHVs in the frequency domain. In[11], authors introduced nonlinear mathematical models (single and double track) aimed at

enhancing the computational efficiency of simulating multi-trailer road vehicles. The authors derive equations of motion using Lagrangian mechanics, resulting in a system of Ordinary Differential Equations (ODEs) that maintain essential nonlinear vehicle dynamics while minimizing computational demands. In double-track model, they provide a more detailed representation by considering individual wheels on each axle, allowing for the analysis of roll dynamics and load transfer effects. To validate these models, the authors compare simulation results with experimental data, demonstrating that the proposed models can accurately predict vehicle behavior in various driving scenarios. The balance between model fidelity and computational efficiency makes these models suitable for applications such as real-time vehicle dynamics simulations, control system development, and optimization studies. In[61], authors developed a nonlinear dynamic model to investigate the stability of articulated considering nonlinear tire model and a detailed hydraulic steering system. The model is formulated using Newtonian mechanical equilibrium equations to derive the necessary state equations for vehicle dynamics and stability analysis.

2.7 Autonomous driving control

The focus of autonomous driving research has primarily been on Single Unit Vehicles (SUVs), such as passenger cars, while articulated vehicles—particularly MTAHVs—have received significantly less attention. Despite their growing prevalence in highway operations, MTAHVs present substantial safety risks, with heavy goods vehicles being 7.5 times more likely to cause accidents than passenger cars[12]. Research efforts on PSSs for articulated vehicles remain limited, especially for MTAHVs[62]. Recent studies have explored autonomous systems for articulated construction vehicles and automated

reverse parking, but these systems primarily address low-speed motion planning and trajectory tracking through kinematic control models. They neglect high-speed dynamic behaviors, such as rollover, jackknifing, and trailer sway, which are critical for the safe operation of MTAHVs[23, 50, 63, 64].

Autonomous driving technologies are developed to improve both transportation safety and operational efficiency. MTAHVs offer significant advantages over single-trailer AHVs, such as reduced mileage, decreased fuel consumption, lower greenhouse gas emissions, and minimized tire-road wear[65, 66]. However, their multi-unit structures, large sizes, and high centers of gravity introduce critical challenges, such as poor low-speed maneuverability and reduced high-speed lateral stability. A prominent issue associated with MTAHVs is rearward amplification[67], where trailing units experience exaggerated lateral or yaw motions during high-speed maneuvers. This effect causes the rearmost trailer to encounter the highest lateral acceleration, making it more susceptible to rollover. In such scenarios, drivers often fail to detect and react to the imminent rollover risk in time to prevent accidents[37]. Human drivers rely primarily on motion cues from the leading units of MTAHVs because the motion signals from trailing units are significantly attenuated by the suspension system of the tractor cabin and the articulation joints. This dependency requires MTAHV drivers to possess substantial training, skill, and experience. The introduction of autonomous driving systems is therefore crucial to enhance the safety and operational reliability of MTAHVs[68].

In[69], the authors present the development of an ATS system for MTAHVs. The study focuses on improving the lateral stability and maneuverability of MTAHVs during low-speed and high-speed operations. The authors propose an optimal control strategy for

the ATS system based on a real-time simulation framework. The framework integrates dynamic models of MTAHVs with active steering algorithms to reduce issues such as rearward amplification, trailer sway, and jackknifing. Islam et al.[60] develop an LQR-based lateral control strategy for articulated vehicles, emphasizing trajectory tracking and yaw stability. The controller used a linearized model for computational efficiency and demonstrated improvements in maneuverability and stability. Ghandriz et al.[11] propose a computationally efficient nonlinear MPC approach for MTAHVs. Their model captures critical nonlinearities in trailer dynamics while maintaining real-time feasibility. This approach shows enhanced performance in reducing rearward amplification and mitigating trailer sway. In[70], a simplified single-unit vehicle models and kinematic models are employed for autonomous systems in tractor/semitrailer configurations. However, the model disregards critical dynamic behaviors of the trailing units, and little effort is made to develop automated driving systems that address the unique dynamics of MTAHVs. In[62], a predictive model is generated for path planning and trajectory following for a tractor/semitrailer. However, the autonomous driving control for a small-scale tractor/semitrailer prototype is based on the kinematic model of the vehicle and a vision-based system. Furthermore, these studies lack a comprehensive approach to simultaneously improving maneuverability and lateral stability across multiple vehicle units under dynamic conditions.

Ghazali et al.[71] develop a multi-layer controller to prevent high-CG vehicles from rolling over. This controller consists of an Upper-level Speed Controller (ULC) and a Lower-level Controller (LLC). The LLC in turn comprises a brake and a steering controller. In the LLC, the steering controller is designed using MPC to predict the vehicle's condition

over an upcoming time-horizon; the MPC control can intervene in the driver's steering control to prevent an imminent rollover, which may result in a potential deviation from the driver's intended path as a penalty. Meanwhile, the ULC calculates the desired speed to keep the error within a controllable range. The speed determined by the ULC is tracked by the lower-level braking controller. Consequently, the steering controller can simultaneously limit tracking errors while preventing rollovers. Sharma and He[63] design a tracking-controller using NLMPC for a tractor/semi-trailer during a lane-change maneuver. They consider collision avoidance in the maneuver. The vehicle is able to successfully track the predefined path and speed profile. It should be mentioned that the dynamics of the tractor/semi-trailer are much simpler than that of a MTAHV, and the NLMPC controller parameters are tuned manually.

The current literature review identifies the following issues on autonomous driving control designs for AHVs:

- 1) Extensive efforts are being made to explore new techniques for trajectory-tracking controls for passenger cars, but little attention is paid to developing associated techniques for MTAHVs[12].
- 2) The MPC technique is widely applied to the trajectory-tracking control designs. Although reliable guidelines for tuning the control parameters, for example, prediction and control horizons, are established, tuning output and control input weighting is still open for discussions[72].
- 3) Linear single-track MTAHV models have been employed for lateral stability control[65]. However, their applicability to trajectory-tracking control is limited, as

MTAHVs can reach overall lengths of up to 40 m and exhibit strongly nonlinear dynamics[3].

2.8 Driver modeling

Driver modeling is a cornerstone in understanding and replicating human behavior within traffic systems. It plays a pivotal role in improving road safety, optimizing traffic flow, and advancing the development of both ADAS and AVs. By creating accurate and reliable models of driver behavior, engineers can predict how drivers are likely to respond in various driving scenarios. This predictive capability enhances vehicle control systems, facilitates safer human-machine interaction, and aids in the design of systems that seamlessly integrate with human-driven traffic[54].

The challenge of driver modeling lies in the significant uncertainties present under diverse operating conditions. Human drivers exhibit a wide range of responses to identical scenarios, influenced by factors such as individual driving styles, situational awareness, and environmental conditions[73]. This variability makes it difficult to develop predictive models that account for every possible outcome. Consequently, extensive research is devoted to analyzing and categorizing human driving behaviors. Studies have focused on evaluating driving skills, understanding reaction times, decision-making processes, and stress responses to various road conditions. These efforts aim to capture the complex dynamics of driver-vehicle interaction, which is essential for designing systems that can adapt to unpredictable human inputs and improve overall traffic system efficiency and safety[74, 75].

Numerous driver models for single-unit vehicles are developed for use in closed-loop simulations. Many of these models have undergone thorough validation and examination in various studies, either through theoretical simulations or experimental road tests[76-79]. However, a MTAHV significantly differs from a passenger car in terms of dynamics, particularly during high-speed maneuvers such as negotiating curves and lane changes, due to substantial variations in dimensions, weight, and configuration[37, 48, 80]. Therefore, driver models designed for single-unit vehicles may not be compatible with MTAHVs, highlighting the need for the creation of specialized driver models tailored to these larger vehicles.

Classical driver models represent a mathematical framework for simulating driver behavior as a control system. These models describe the relationship between a driver's input and the vehicle's response in the frequency domain to consider only the compensatory action[23]. In[62], the authors propose a multi-loop framework that considers factors such as the tractor's lateral deviation and the articulation angle rates to determine steering demand. However, the study primarily focuses on analyzing the vehicle's response to driver behavior. As the result, it does not explore the maneuverability, lateral stability, or potential improvements of the driver model in various driving scenarios for AHVs. In[81], a driver model for tractor/semi-trailers is developed, utilizing the feedback of the vehicle units' real-time states, including a time delay, to compute the optimal steering angle. A cost function with feedback gains is defined, and sensitivity analysis is conducted. However, the model does not account for distinct motion cues perceived from the towing and towed units, limiting its accuracy in representing driver perception. Ding and He investigates the effects of motion cues from the tractor, the trailer, and the combination of both. Their findings

indicate that incorporating motion cues from all vehicle units enhances the vehicle's directional performance[82]. Zhu et al. create a driver model for a B-train double using a sliding mode controller, incorporating motion cues from all vehicle units. They evaluate it with a high-speed single-lane-change maneuver, where it outperforms other models. However, the model is not tested for other types of maneuvers, including low-speed or various high-speed scenarios[83]. Although classical driver models are simple and computationally efficient, they use simplified linear vehicle dynamics which may reduce accuracy in highly dynamic or nonlinear scenarios.

An optimal predictive/preview driver modeling method incorporates future information about the desired trajectory or road conditions into the vehicle driver command using combined pursuit/compensatory approach. Among the predictive/preview driver models, McAdam driver model is highly regarded and well-validated, which is based on optimal preview control for linear systems[84]. In[76], the authors present a driver model that adapts based on lateral preview information to improve vehicle path-following performance. The model accounts for driver behavior and adjusts steering commands to enhance stability and responsiveness, particularly in dynamic driving situations. In[23], the author introduces a new predictive/preview driver model which used the cue information from all units of a MTAHV. The author considers the driver controller in two different modes. 1) a low-speed path-following mode minimizes lateral deviation for all vehicle units along the target path; 2) the high-speed lateral stability mode ensures the leading unit follows the desired path. Simultaneously, the trailing units track the TFAC trajectory to maintain the RWA close to 1.0.

A driver model is designed considering two preview points from the road. For the first time in[75], the authors propose a two-point visual control model of steering, where drivers use both near and far visual points to guide their steering behavior. The near point, positioned at a fixed distance ahead of the driver, helps maintain lane position, while the far point, representing a notable point farther away aids in anticipating road curvature. The model integrates these visual cues to simulate human-like steering and is validated through experiments and simulations, demonstrating its accuracy in replicating real-world driving behavior. In[85], Sentouh et al. develop a sensorimotor driver model that simulates human steering behavior using visual and kinesthetic feedback for a passenger vehicle. The model is validated and showed promise for applications in vehicle design and driver assistance. Nilsson et al. develop a driver model that leverages visual information for both longitudinal and lateral control of long vehicle combinations[86]. The model simulates human driving behavior, focusing on stability and path-following accuracy, and is validated through simulation studies, showing its effectiveness in improving vehicle control. In[87], the authors conduct a simulator study comparing manual (two-point model) and automated driving (optimal control) during lane changes for LCVs. The study evaluates differences in driver performance, vehicle stability, and safety between the two modes. Results show that automated driving improves lane-change precision and stability, reducing the risk of instability commonly observed in manual control. Sentouh et al. propose a shared control (two-point visual + optimal controller) approach for lane-keeping assist systems that balances driver input and automation[88]. The method enhances driver-automation cooperation by adapting control based on driver behavior and system feedback. Evaluations demonstrate improved lane-keeping performance and reduced driver

workload, promoting safer and more intuitive interactions. Although the two-point visual method is a relatively new approach for modeling driver behavior during vehicle steering, it suffers from the following shortcomings in the literature.

- 1) Due to the nonlinear and complex dynamics of MTAHVs, most of the research has focused on passenger vehicles.
- 2) Among those who have utilized the two-point visual model for the MTAHVs, none have incorporated the position and/or orientation information of the trailers/dollies.
- 3) Most of the research has considered the MTAHV as a linear model and has not evaluated the results based on a nonlinear model in the software/hardware-in-the loop.
- 4) Nearly all studies in this field consider the driver as an ideal model and do not account for any dynamics and/or delays in the driver's response.

2.9 Deep-reinforcement-learning based decision making

An autonomous driving system for ground vehicles generally consists of three modules[89]: 1) perception; 2) motion-planning and decision-making; and 3) tracking-control. Decision-making in autonomous vehicles is a critical sub-module, which processes sensor data, interprets surrounding environment, and makes real-time decisions[90]. The decision-making task for an autonomous vehicle is complex, as the system must navigate a variety of environments, engage with other road users, and account for uncertainties in sensor data[91]. Therefore, predicting all potential scenarios and coding appropriate responses manually is not feasible. Consequently, leveraging artificial intelligence methods to train decision-making sub-modules for autonomous vehicles becomes a more effective approach[90, 92]. Reinforcement Learning (RL) is a branch of artificial intelligence where

an agent learns the most effective control strategy through interactions with its environment. Compared to other approaches, deep reinforcement learning (DRL) is distinct in that it does not rely on an action sequence from an expert driver[93]. A DRL algorithm has the following benefits:

- 1) **Adaptability:** Deep reinforcement learning allows autonomous vehicles to adapt to different environments and unforeseen scenarios. By training in a wide range of situations, the vehicle can generalize its learned behaviors and adapt it to new, unseen conditions[94].
- 2) **Handling Uncertainty:** Real-world driving involves uncertainties due to factors like unpredictable traffic and environmental changes. Deep reinforcement learning techniques are well-suited for handling these uncertainties by continuously adjusting actions based on real-time feedback[95].
- 3) **Learning from Experience:** Unlike rule-based systems, DRL enables autonomous vehicles to learn from their experiences. Through trial and error, AVs can discover strategies that work across different driving tasks, leading to more robust decision-making capabilities[96].

Thus, in recent years, research efforts have been made in this field to address this issue in a safe and reliable manner.

In[97], to solve the problem of decision making for passenger vehicles, the authors introduce a Double Deep Q-network (DDQN) for autonomous highway driving, which is simulated in Simulation of Urban Mobility (SUMO)[168]. The proposed DDQN agent could successfully learn the optimal policy, thereby preventing collisions in all situations. Although it looks great, it needs more consideration to approach 100% success. Liao et al.

propose a framework for the autonomous highway driving and compare the result for a reference model, Deep Q-network (DQN), and DDQN. The results show that DDQN outperforms the others in terms of collision avoidance, average reward, and vehicle speed[98]. In[92], the authors propose a safe controller with the DRL module to check the DDQN agent decisions to prevent collision. By doing so, they could successfully enhance the vehicle safety in a highway driving scenario. In[99], the authors concentrate on scenarios where an AV is approached by an emergency vehicle. By making correct decisions in such situations, the vehicle could enhance both autonomous driving and emergency response efficiency in a highway environment. Yuan et al. consider Prioritized Experienced Replay (PER) in combination with DQN to consider the effect of PER in performance. They find that PER DQN enhances the training efficiency and interpretability of the end-to-end driving decision-making framework. This technique surpasses the standard DQN by achieving a higher average speed and increasing overtakes, while minimizing lane-change time. This indicates that the PER DQN enables driving decisions that emphasize both safety and time efficiency[100]. In[101], the authors explore the use of a Bayesian reinforcement learning technique, which relies on an ensemble of neural networks combined with Randomized Prior Functions (RPF), to estimate the uncertainty of decision-making in autonomous driving. The results show more efficient decisions and more stable performance in comparison with DQN method. In[102], the authors explore the use of DRL algorithms, such as DQN and Proximal Policy Optimization (PPO), to enhance decision-making for autonomous vehicles. It focuses on improving reward functions to increase model adaptability and robustness in complex traffic scenarios.

Comparative analysis shows that DRL-based methods outperform traditional approaches, demonstrating superior adaptability and decision-making in dynamic environments.

In summary, there is a lot of research that addresses the open problem of decision-making in the field of autonomous driving. Making intelligent real-time decisions in a diverse, dynamic, and often unpredictable environment is a very challenging task. Deep reinforcement learning has recently emerged as a powerful approach to learn optimal driving behaviors through experience and interaction with the environment. In this regard, DQN has significantly advanced the field of reinforcement learning. Subsequent improvements, such as Double DQN, PER DQN, and Rainbow DQN, have further enhanced stability, efficiency, and performance. Despite significant advancements in this field, most of the research suffers from the following shortcomings:

- 1) Due to the nonlinear and complex dynamics of MTAHVs, most of this research has focused on passenger vehicles.
- 2) They consider a very simple model (point mass model) for the vehicle in the traffic simulation. They do not investigate the effect of vehicles dimensions in the decision-making and path-following control.
- 3) They use simple controller in the lateral and longitudinal direction to change the lane and avoid collision such as Intelligent Driver Model (IDM) for the longitudinal direction and Minimizing Overall Braking Induced by Lane Changes (MOBIL) for the lateral direction.
- 4) It is necessary for the results to be thoroughly evaluated in a Hardware-in-the-Loop (HIL) real-time simulation or, at least, in a Software-in-the-Loop (SIL) real-time simulation.

2.10 Semi-autonomous driving

Semi-autonomous vehicles, which integrate advanced interactions between human drivers and automated driving control systems, have experienced significant advancements in their designs, functionalities, and performance measures in recent years. These developments have enhanced their ability to support drivers by seamlessly blending manual and automated operations, making them more efficient, reliable, and adaptive to diverse driving conditions[103]. The advancement and integration of shared driving control of vehicles has the potential to revolutionize the transportation industry, providing a significant technological leap forward. These vehicles can enhance operational efficiency, reduce congestion, and minimize environmental impact by optimizing shared mobility solutions. Furthermore, their deployment could lead to safer roads by reducing human errors, lower transportation costs through ride-sharing models, and increased accessibility for underserved populations. As a transformative innovation, semi-autonomous vehicles represent a critical step toward a more sustainable, inclusive, and technologically advanced transportation ecosystem[104-106].

The interactions between human drivers and automated systems in semi-autonomous vehicles represent a highly intricate and challenging issue. This complexity stems from the difficulty in accurately modeling human behavior, which is often unpredictable, and anticipating drivers' intentions in various real-world scenarios. Human factors, such as reaction time, decision-making processes, and situational awareness, add further layers of uncertainty, making the seamless integration of human inputs with automated controls a daunting task. Ethical and trust-related concerns further complicate this dynamics, as users may question the reliability and decision-making logic of automated

systems[107]. It is emphasized that advancements in semi-autonomous technology, including shared control schemes, are aimed at improving the interaction between human drivers and machines, fostering greater collaboration for tasks, such as lane-keeping, highway merging, and obstacle avoidance[108]. Addressing these challenges is vital for broader adoption of semi-autonomous systems, which have the potential to significantly improve safety and efficiency in modern transportation networks[109].

Game theory offers a robust and promising framework for modeling and managing the dynamic interactions between multiple agents operating in complex and unpredictable environments. Its strength lies in its ability to provide structured strategies for decision-making, enabling systems to address challenges that arise from the interplay between drivers and automation in semi-autonomous vehicles. Specifically, game theory excels in dealing with uncertainties related to driver behavior models and environmental variables, making it a valuable tool for enhancing the efficiency and safety of human-automation interaction.

It is reported that the integration of cooperative game-theoretic approaches in shared control systems has shown significant potential in optimizing task allocation and conflict resolution between drivers and automated systems[110]. This is particularly crucial in semi-autonomous contexts, where human inputs must be harmonized with automated controls to ensure seamless performance. An innovative game-theoretical framework is proposed to model the dynamic interaction between human drivers and active steering systems[111, 112]. Both the driver and the active steering controller are treated as players in a differential game, each aiming to minimize their specific cost functions within a path-following context. The researchers investigate multiple interaction paradigms, including

cooperative (Pareto), non-cooperative (Nash and Stackelberg), and decentralized paradigms, where the driver and collision avoidance controller act independently without mutual consideration. The study reveals that non-cooperative Nash and Stackelberg paradigms effectively capture driver behavior, with collision avoidance controllers actively adjusting to the driver's steering actions. Conversely, the Pareto paradigm demonstrates a cooperative approach, optimizing the interaction between the driver and the controller for mutual benefit. To analyze these paradigms, MPC and Linear Quadratic Dynamic Optimization techniques are used, illustrating their effectiveness in solving game-theoretical problems. It is further shown that variations in path-error cost function weights across paradigms could lead to different steering behaviors, highlighting the adaptability of the approach to various driving conditions. As one of the first studies to apply game theory in vehicle dynamics, this research primarily focused on the steering problem. Nonetheless, it lays the groundwork for future exploration into broader aspects of driver-automation interaction, including lane-keeping, obstacle avoidance, and traffic coordination. The findings underscore the promise of game-theoretical approaches in advancing the safety and efficiency of semi-autonomous systems, making a significant contribution to the field of vehicle dynamics and control.

The Nash game theory framework has been used to model a two-player cooperative differential game involving the driver and the Direct Yaw Control (DYC) system as key players[113-115]. It is demonstrated that employing optimal preview control could enhance vehicle stability significantly, particularly by mitigating lateral velocity, yaw rate, and roll angle. To address this, the researchers developed an innovative algorithm tailored for solving the two-player Nash game in discrete systems, providing a computationally

efficient solution for real-time applications. The research efforts advance the methodology by extending the Nash game framework to encompass both two-player and three-player games in continuous systems. This expanded approach not only refines the mathematical modeling and solution strategies but also broadens the scope of application to include more intricate and dynamic interactions in vehicular systems. Beyond the interaction between steering and DYC, this work explores scenarios such as the integration of multiple subsystems, cooperative maneuvers, and real-time decision-making in semi-autonomous and autonomous vehicles. These advancements pave the way for designing more sophisticated control architectures, enhancing vehicle performance, stability, and safety in diverse driving conditions.

Huang et al.[116] extended the Linear Quadratic Regulator (LQR) method by introducing Control Coupled Output Regulation (CCOR), a framework designed to enhance roll stability in heavy articulated vehicles with multiple axles. Their study introduced the Rollover Index as a metric to evaluate roll stability performance, focusing on the role of active anti-roll bars associated with different axles. These active anti-roll bars, treated as players, aim to minimize a performance index that incorporates multiple rollover indices. Additionally, Huang's work presented a design algorithm to analyze and compare various control configurations, facilitating the selection of an optimal design. However, the approach was limited to single-player games, leaving the potential for its extension to multi-player scenarios unexplored. Building upon Huang's foundational work, the current study investigates the application of the CCOR framework in multi-player game contexts. Specifically, it explores its utility in regulating sprung mass vertical acceleration, a critical factor in improving ride quality for heavy vehicles. By extending the method to

accommodate two-player and three-player games, the study broadens the applicability of CCOR to problems involving cooperative or competitive interactions among multiple control agents. This advancement not only refines the theoretical framework but also opens new avenues for designing advanced control systems, which enhance both stability and ride quality in complex vehicle dynamics scenarios.

In[117], the authors focus on leveraging game theory to address challenges in vehicle dynamics and control systems. The study models the interaction between human drivers and vehicle control systems, such as steering and yaw control, as a game-theoretic framework. It explores both cooperative and non-cooperative paradigms to enhance vehicle safety, stability, and performance under various driving scenarios. The investigators developed mathematical models for two-player and multi-player games, incorporating drivers and automated systems as players with distinct objectives. The study examines different game strategies, such as Nash equilibrium and Stackelberg solutions, to predict and improve driver-controller interactions. Applications are extended to scenarios like collision avoidance, lane-keeping, and roll stability, with optimization techniques used to achieve desired outcomes. This research demonstrates the potential of game theory as a robust tool for advancing vehicle control systems, contributing to the development of safer and more adaptive semi-autonomous vehicles.

The application of game theory in shared autonomous control is a burgeoning area of research within the field of autonomous driving. While promising, current implementations are constrained by simplifying assumptions that fail to capture the full complexity of semi-autonomous vehicle systems. Most existing studies focus on single-unit passenger vehicles, e.g., passenger cars and rigid trucks, operating under constant

velocities and using simplified linear vehicle model. These limitations hinder the ability of game-theoretical approaches to address the challenges inherent in MTAHVs and their semi-autonomous operation.

Chapter 3

Vehicle dynamic modeling

3.1 Introduction

Vehicle dynamic models play a vital role in the design, development, and control of MTAHVs. These vehicles feature complex configurations, with their dynamic behavior influenced by various factors, such as the number and length of trailers, the type of coupling mechanism between the tractor and trailers, and load distribution. Dynamic models are essential for understanding MTAHVs' behavior, predicting their responses to different inputs, and designing control systems that ensure safe and efficient operation. By developing these models, engineers can simulate vehicle behavior under various operating conditions and optimize control strategies to enhance safety, maneuverability, and stability.

In this study, a single-track model of a MTAHV with the configuration of A-train double is used[21]. An A-train utilizes a dolly equipped with a single drawbar to connect the second semi-trailer, introducing an additional articulation point that allows for independent yaw motion between the dolly and the lead trailer. In contrast, a B-train eliminates the dolly entirely by mounting a fifth wheel directly on the rear of the lead trailer, while a C-train employs a dolly with a double-drawbar design to rigidly constrain yaw rotation. Consequently, the A-train exhibits significantly higher degrees of freedom and is more susceptible to Rearward Amplification (RWA) compared to the inherent stability of B-train and C-train configurations, making it a more challenging candidate for stability control systems. As shown in Figure 3-1, the vehicle combination consists of four units: a

tractor, semi-trailer 1, a dolly, and semi-trailer 2. The dolly is hooked up to a pintle hitch at the rear of semi-trailer 1, and semi-trailer 2 is in turn connected to the dolly via a fifth wheel. It is one of several vehicle combinations denoted as prospective modular combinations. The prospective modular combinations have the potential to increase productivity but result in extended complexity unique vehicle dynamics.

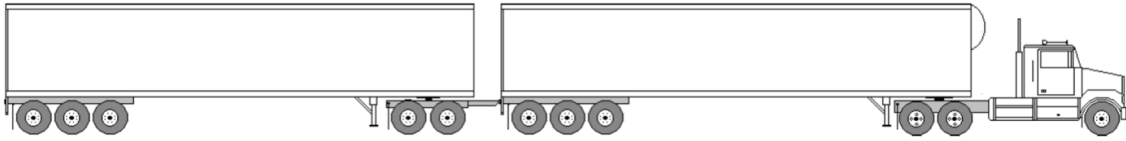


Figure 3-1: A MTAHV with the configuration of A-train double[118]

To validate the effectiveness of the single-track yaw-plane model, the highly nonlinear and high-fidelity A-train double model in TruckSim[119] is employed. Co-simulations using Matlab/TruckSim are conducted to test the models in various MTAHV maneuvers and driving scenarios.

3.2 Nonlinear 11-DOF single-track MTAHV model

Figure 3-2 depicts the nonlinear 11-DOF single-track yaw-plane model for representing the lateral dynamics of the A-train double. To derive this MTAHV model, the following assumptions are made: 1) the tractor rear tandem-axle group is simplified as a single axle, the semi-trailer 1 tridem-axle group as a single axle, the dolly tandem-axle group as a single axle, and the semi-trailer 2 tridem-axle group as a single axle; 2) the simplified axle arrangement leads to five axles in total, which are denoted as axles 1 to 5 from the tractor front-axle to the semi-trailer 2 axle; 3) all wheels on each axle are lumped as a single wheel located at the axle centre; 4) tractor, semi-trailer 1, dolly, and semi-trailer 2 are treated as four rigid-bodies, and adjacent units are connected by an equivalent

3.2.1 Equations of motion for rigid bodies of vehicle units

The mathematical modeling of the vehicle dynamics was established using the Lagrange formulation. A primary advantage of this approach is the elimination of the need for explicit constraint-force calculations, which significantly streamlines the derivation of the equations of motion. Furthermore, this formulation provides a modular framework that is easily extendable, allowing for the efficient adaptation of the model to simulate different vehicle configurations or additional trailers without fundamental restructuring[120]. The Lagrange equations for the four rigid-bodies with six motions are given by:

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_i} - \frac{\partial T}{\partial q_i} + \frac{\partial V}{\partial q_i} = Q_i, \quad i = 1, 2, \dots, 6 \quad (3.1)$$

where T denotes the kinetic energy, V the potential energy, Q_i the generalized forces, and q_i the generalized coordinates, which are defined as:

$$\mathbf{q} = [X, Y, \phi_1, \theta_1, \theta_2, \theta_3] \quad (3.2)$$

where X and Y are the global coordinates of the tractor CG, which correspond to the longitudinal and lateral velocities of the tractor, i.e., v_x and v_y , ϕ_1 is the heading angle of the tractor, θ_1 denote the articulation angle between the tractor and semi-trailer 1, θ_2 the articulation angle between semi-trailer 1 and the dolly, and θ_3 the articulation angle between the dolly and semi-trailer 2.

The notation of the vehicle model is illustrated in Figure 3-2. For simplicity, the tractor, semi-trailer 1, dolly, and semi-trailer 2 are termed as units 1 to 4. As shown in the figure, a_1 and b_1 denote the distances from the tractor CG to its front and rear axles, respectively, a_i and b_i ($i = 2, 3, 4$) the distances from the i^{th} vehicle unit's CG to its front

articulation joint and rear axle, accordingly, and c_i ($i = 1, 2, 3, 4$) represents the distance from the i^{th} vehicle unit to its rear articulation joint.

In Equation (3.1), the generalized force, Q_i , is given by:

$$Q_i = \sum_{k=1}^{N_{ax}} \left(F_{Xk} \frac{\partial P_{Xk}}{\partial q_i} + F_{Yk} \frac{\partial P_{Yk}}{\partial q_i} \right) \quad (3.3)$$

where N_{ax} represent the total number of axles for the vehicle mode (as shown in Figure 3-2, $N_{ax} = 5$), F_{Xk} and F_{Yk} denote the X and Y components of the total external force on the k^{th} wheel expressed in the global coordinate system, respectively, P_{Xk} and P_{Yk} the X and Y positions of the point (expressed in the global coordinate system), at which the total external force exerted on the k^{th} wheel, accordingly. The coordinates in the global reference frame for the contact point of each wheel ($P_k, i = 1, 2, 3, 4, 5$) are determined by:

$$P_1 = \begin{pmatrix} X + a_1 \cos \phi_1 \\ Y + a_1 \sin \phi_1 \end{pmatrix} \quad (3.4)$$

$$P_2 = \begin{pmatrix} X - b_1 \cos \phi_1 \\ Y - b_1 \sin \phi_1 \end{pmatrix} \quad (3.5)$$

$$P_3 = \begin{pmatrix} X - c_1 \cos \phi_1 - (a_2 + b_2) \cos \phi_2 \\ Y - c_1 \sin \phi_1 - (a_2 + b_2) \sin \phi_2 \end{pmatrix} \quad (3.6)$$

$$P_4 = \begin{pmatrix} X - c_1 \cos \phi_1 - (a_2 + c_2) \cos \phi_2 - (a_3 + b_3) \cos \phi_3 \\ Y - c_1 \sin \phi_1 - (a_2 + c_2) \sin \phi_2 - (a_3 + b_3) \sin \phi_3 \end{pmatrix} \quad (3.7)$$

$$P_5 = \begin{pmatrix} X - c_1 \cos \phi_1 - (a_2 + c_2) \cos \phi_2 - (a_3 + c_3) \cos \phi_3 - (a_4 + b_4) \cos \phi_4 \\ Y - c_1 \sin \phi_1 - (a_2 + c_2) \sin \phi_2 - (a_3 + c_3) \sin \phi_3 - (a_4 + b_4) \sin \phi_4 \end{pmatrix} \quad (3.8)$$

The MTAHV assumes travel on level road, the potential energy of the vehicle system is thus set to zero. The system's kinetic energy is determined by calculating the sum of the rotational and translational energy of all units:

$$T = \frac{1}{2} \sum_{i=1}^4 \left(m_i (v_{xi}^2 + v_{yi}^2) + J_i \dot{\phi}_i^2 \right) \quad (3.9)$$

where m_i is the mass for the i^{th} vehicle unit, v_{xi} and v_{yi} are the X and Y components of the velocity at the CG of the i^{th} vehicle unit, respectively, $\dot{\phi}_i$ and J_i the yaw rate and the yaw mass-moment of inertial around for the i^{th} vehicle unit, accordingly. To facilitate the derivation of the vehicle model, the velocity at the CG of the first unit is converted into its body-fixed coordinate system, as seen in Figure 3-2. The relationship between the velocities expressed in the global coordinate system $(\dot{X}, \dot{Y})$ and those described in the body-fixed coordinate system (v_x, v_y) is determined by[9],

$$\begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} \cos(\phi_1) & \sin(\phi_1) \\ -\sin(\phi_1) & \cos(\phi_1) \end{bmatrix} \begin{bmatrix} \dot{X} \\ \dot{Y} \end{bmatrix} \quad (3.10)$$

In order to derive the equations of motion in the direction of the quasi-coordinates, this method is followed:

$$\frac{\partial T}{\partial \dot{X}_i} = \frac{\partial T}{\partial u} \cdot \frac{\partial u}{\partial \dot{X}_i} + \frac{\partial T}{\partial v} \cdot \frac{\partial v}{\partial \dot{X}_i} = \frac{\partial T}{\partial u} \cdot \cos(\phi_1) - \frac{\partial T}{\partial v} \cdot \sin(\phi_1) \quad (3.11)$$

$$\frac{\partial T}{\partial \dot{Y}_i} = \frac{\partial T}{\partial u} \cdot \frac{\partial u}{\partial \dot{Y}_i} + \frac{\partial T}{\partial v} \cdot \frac{\partial v}{\partial \dot{Y}_i} = \frac{\partial T}{\partial u} \cdot \sin(\phi_1) + \frac{\partial T}{\partial v} \cdot \cos(\phi_1) \quad (3.12)$$

$$\frac{\partial T}{\partial X} = \frac{\partial T}{\partial Y} = 0 \quad (3.13)$$

The above equations of motions can be written as:

$$\begin{bmatrix} \cos(\phi_1) & -\sin(\phi_1) \\ \sin(\phi_1) & \cos(\phi_1) \end{bmatrix} \begin{bmatrix} \frac{d}{dt} \frac{\partial T}{\partial u} - \frac{\partial T}{\partial v} \cdot \dot{\phi}_1 \\ \frac{d}{dt} \frac{\partial T}{\partial v} + \frac{\partial T}{\partial u} \cdot \dot{\phi}_1 \end{bmatrix} = \begin{bmatrix} Q_X \\ Q_Y \end{bmatrix} \quad (3.14)$$

Pre-multiplying the above equation with the inverse of the yaw rotation matrix and introducing the generalized forces Q_x and Q_y defined in the local frame, it follows that the equations of motion in the quasi-coordinate' direction can be written as:

$$\begin{bmatrix} \frac{d}{dt} \frac{\partial T}{\partial u} - \frac{\partial T}{\partial v} \cdot \dot{\phi}_1 \\ \frac{d}{dt} \frac{\partial T}{\partial v} + \frac{\partial T}{\partial u} \cdot \dot{\phi}_1 \end{bmatrix} = \begin{bmatrix} Q_x \\ Q_y \end{bmatrix} \quad (3.15)$$

To be able to calculate Equation (3.9), all force elements, the positions at which they act, and the positions of the CGs and their velocities must be determined in terms of generalized coordinates, input variables and system parameters. By using system kinematics consistent with the coupling positions, the position of the CGs of the vehicle i^{th} unit can be found as:

$$\begin{bmatrix} X_i \\ Y_i \end{bmatrix} = \begin{bmatrix} X_{i-1} \\ Y_{i-1} \end{bmatrix} + \begin{bmatrix} -c_{i-1} \cos(\phi_{i-1}) - a_i \cos(\phi_i) \\ -c_{i-1} \sin(\phi_{i-1}) - a_i \sin(\phi_i) \end{bmatrix}, \quad i = 2, 3, 4 \quad (3.16)$$

where:

$$\phi_i = \phi_{i-1} + \theta_{i-1}, \quad i = 2, 3, 4 \quad (3.17)$$

The vehicle unit velocities in the global coordinate system can be calculated by taking the time derivative of the CG positions according to:

$$v_{xi} = \frac{dX_i}{dx}, \quad v_{yi} = \frac{dY_i}{dx} \quad (3.18)$$

Finally, the six Lagrangian equations of motion are expressed as:

$$\frac{d}{dt} \frac{\partial T}{\partial v_x} - \frac{\partial T}{\partial v_y} \cdot \dot{\phi}_1 = Q_x \quad (3.19)$$

$$\frac{d}{dt} \frac{\partial T}{\partial v_y} + \frac{\partial T}{\partial v_x} \cdot \dot{\phi}_1 = Q_y \quad (3.20)$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\phi}_1} - \frac{\partial T}{\partial v_x} \cdot v_y + \frac{\partial T}{\partial v_y} \cdot v_x = Q_{\phi_1} \quad (3.21)$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\theta}_1} - \frac{\partial T}{\partial \theta_1} = Q_{\theta_1} \quad (3.22)$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\theta}_2} - \frac{\partial T}{\partial \theta_2} = Q_{\theta_2} \quad (3.23)$$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\theta}_3} - \frac{\partial T}{\partial \theta_3} = Q_{\theta_3} \quad (3.24)$$

Note that in Equation (3.3), Q_i represents the generalized force; accordingly, in Equations (3.19)-(3.24), it denotes the combination of associated external forces: 1) lateral and longitudinal tire forces (to be determined in Section 3.2.2), and 2) the longitudinal aerodynamic drag on the first vehicle unit, which is determined by:

$$F_a = \frac{1}{2} C_D A_f \rho_a v_x^2 \quad (3.25)$$

where C_D is the aerodynamic drag coefficient, A_f is the tractor frontal area, and ρ_a is the air mass density. By using Maple software[121], the nonlinear rigid-body sub-model (6-DOF) of the single-track yaw-plane MTAHV model is derived and expressed in the compact format as:

$$\dot{\mathbf{x}}_0(t) = f_0[\mathbf{x}_0(t), \mathbf{u}_0(t)] \quad (3.26)$$

where $\mathbf{x}_0(t)$ and $\mathbf{u}(t)$ are the state and control variables defined as:

$$\mathbf{x}_0(t) = [v_y, \phi_1, \dot{\phi}_1, \theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2, \theta_3, \dot{\theta}_3, v_x]^T \quad (3.27)$$

$$\mathbf{u}_0(t) = [\delta_{1f}, F_{x1f,r}]^T \quad (3.28)$$

where δ_{1f} is the steering angle of tractor front axle wheels, and $F_{x1f,r}$ longitudinal forces due to tractor rear driving wheels or brake forces from all vehicle braking wheels. To

calculate the longitudinal and lateral tire forces, an analytical nonlinear tire model is used and introduced in the following subsection. The complete nonlinear equations of motion for the rigid body are detailed in Appendix A. The equations have been derived by assuming that the heading and articulation angles, as well as their rates of change, remain small.

3.2.2 Tire and wheel modeling

In this research, the nonlinear Dugoff tire model is used to calculate the longitudinal and lateral tire forces [122]. The tire model assumes a uniform vertical pressure distribution across the tire contact patch. Using this tire model, the longitudinal and cornering (lateral) tire forces are calculated as follows:

$$F_{li} = C_{\sigma i} \left[\frac{\sigma_{li}}{1 + \sigma_{li}} \right] f(\lambda_i), \quad i = 1, 2, 3, 4, 5 \quad (3.29)$$

$$F_{ci} = C_{\alpha i} \left[\frac{\tan(\alpha_i)}{1 + \sigma_{li}} \right] f(\lambda_i) \quad (3.30)$$

$$\lambda_i = \frac{\mu F_{zi}(1 + \sigma_{li})}{2\sqrt{(C_{\sigma i}\sigma_{li})^2 + (C_{\alpha i}\tan(\alpha_i))^2}} \quad (3.31)$$

$$f(\lambda_i) = \begin{cases} (2 - \lambda_i)\lambda_i & \text{for } \lambda_i < 1 \\ 1 & \text{for } \lambda_i \geq 1 \end{cases} \quad (3.32)$$

where F_{li} and F_{ci} represent the longitudinal and lateral tire forces, respectively, $C_{\sigma i}$ and $C_{\alpha i}$ denote tire longitudinal and lateral stiffness, respectively. Note that as shown in Figure 3-2, from the tractor front tire to the tire of semi-trailer 2, they are denoted as tire 1 to 5, i.e., $i=1, 2, 3, 4, 5$, respectively. The longitudinal and lateral slip ratios, σ_{li} and α_i , can be determined by:

$$\alpha_i = \left[\delta_i - \arctan\left(\frac{v_{ci}}{v_{li}}\right) \right], \quad i = 1, 2, 3, 4, 5 \quad (3.33)$$

$$\sigma_{li} = \begin{cases} \frac{R_{ei}\omega_i - v_{li}}{v_{li}}, & \text{if } v_{li} > R_{ei}\omega_i \text{ and } v_{li} \neq 0 \text{ (for braking), } i = 1, 2, \dots, 5 \\ \frac{R_{ei}\omega_i - v_{li}}{R_{ei}\omega_i}, & \text{if } v_{li} < R_{ei}\omega_i \text{ and } \omega_i \neq 0 \text{ (for driving), } i = 2 \end{cases} \quad (3.34)$$

where v_{ci} , v_{li} , and δ_i are lateral velocity, longitudinal velocity, and steering angle at the i^{th} wheel, respectively, R_{ei} denotes effective rolling radius, ω_i represents wheel angular velocity. The wheel equation of motion is governed by:

$$I_{ei}\dot{\omega}_i = T_i - F_{li}R_{ei} - F_{zi}a_i, \quad i = 1, 2, 3, 4, 5 \quad (3.35)$$

where I_{ei} is the equivalent wheel mass moment of inertia, T_i the applied torque, propulsion or braking, F_{li} longitudinal tire force, F_{zi} normal tire force, and a_i offset of the normal load from the centre of the wheel. Using the geometry depicted in Figure 3-2 for the MTAHV model, the normal loads on the tires of the tractor and semi-trailers are approximated by:

$$F_{z5} = \frac{m_4ga_4}{a_4 + b_4} \quad (3.36)$$

$$F_4 = \frac{m_4gb_4}{a_4 + b_4} \quad (3.37)$$

$$F_{z4} = \frac{m_3ga_3}{a_3 + b_3} + \frac{F_4(a_3 + c_3)}{a_3 + b_3} \quad (3.38)$$

$$F_3 = \frac{m_3gb_3}{a_3 + b_3} + \frac{F_4(b_3 - c_3)}{a_3 + b_3} \quad (3.39)$$

$$F_{z3} = \frac{m_2ga_2}{a_2 + b_2} + \frac{F_3(a_2 + c_2)}{a_2 + b_2} \quad (3.40)$$

$$F_2 = \frac{m_2gb_2}{a_2 + b_2} + \frac{F_3(b_2 - c_2)}{a_2 + b_2} \quad (3.41)$$

$$F_{z2} = \frac{m_1ga_1}{a_1 + b_1} + \frac{F_2(a_1 + c_1)}{a_1 + b_1} \quad (3.42)$$

$$F_{z1} = \frac{m_1gb_1}{a_1 + b_1} + \frac{F_2(b_1 - c_1)}{a_1 + b_1} \quad (3.43)$$

By combining Equations (3.1) and (3.35), the equations of motion for the rigid-bodies of vehicle units are combined with those governing the motions of the wheels, thereby leading to the augmented nonlinear equations of motion for the nonlinear 11-DOF single-track yaw-plane MTAHV model, which is expressed in the concise format as:

$$\dot{\mathbf{x}}(t) = f[\mathbf{x}(t), \mathbf{u}(t)] \quad (3.44)$$

where $\mathbf{x}(t)$ and $\mathbf{u}(t)$ are the state variables and control variables in the following format:

$$\mathbf{x}(t) = [v_y, \phi_1, \dot{\phi}_1, \theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2, \theta_3, \dot{\theta}_3, v_x, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5]^T \quad (3.45)$$

$$\mathbf{u}(t) = [\delta_{1f}, T_{x1f,r}]^T \quad (3.46)$$

where in the case of acceleration scenario, the traction torque, $T_{x1f,r}$, is determined by a simplified tractor powertrain model reported in [63]. Equation (3.45) can be written in the semi-matrix and continuous-time format as follows:

$$M_c \dot{\mathbf{x}}(t) = A_c(\mathbf{x}) + B_c(\mathbf{x})\mathbf{u}(t) \quad (3.47)$$

The above equation can be discretized simply by the discretization of the state transition function such as:

$$M_c \left(\frac{\mathbf{x}(k+1) - \mathbf{x}(k)}{T_s} \right) = A_c(\mathbf{x}(k)) + B_c(\mathbf{x}(k))\mathbf{u}(k) \quad (3.48)$$

$$M_c \mathbf{x}(k+1) = M_c \mathbf{x}(k) + T_s \left(A_c(\mathbf{x}(k)) + B_c(\mathbf{x}(k))\mathbf{u}(k) \right) = \quad (3.49)$$

$$M_c \mathbf{x}(k+1) = \left(M_c \mathbf{x}(k) + T_s A_c(\mathbf{x}(k)) \right) + T_s B_c(\mathbf{x}(k))\mathbf{u}(k)$$

or in the standard format (T_s is the sampling time):

$$M\mathbf{x}(k+1) = A(\mathbf{x}(k)) + B(\mathbf{x}(k))\mathbf{u}(k) \quad (3.50)$$

$$M = M_c, \quad A(\mathbf{x}(k)) = M_c \mathbf{x}(k) + T_s A_c(\mathbf{x}(k)), \quad B(\mathbf{x}(k)) = T_s B_c(\mathbf{x}(k))$$

In the derivation of the nonlinear 11-DOF single-track yaw-plane MTAHV model, the associated components/sub-models, i.e., the wheel model, Dugoff tire model, and rigid-body model for the vehicle units, are integrated according to the flowchart shown in Figure 3-3.

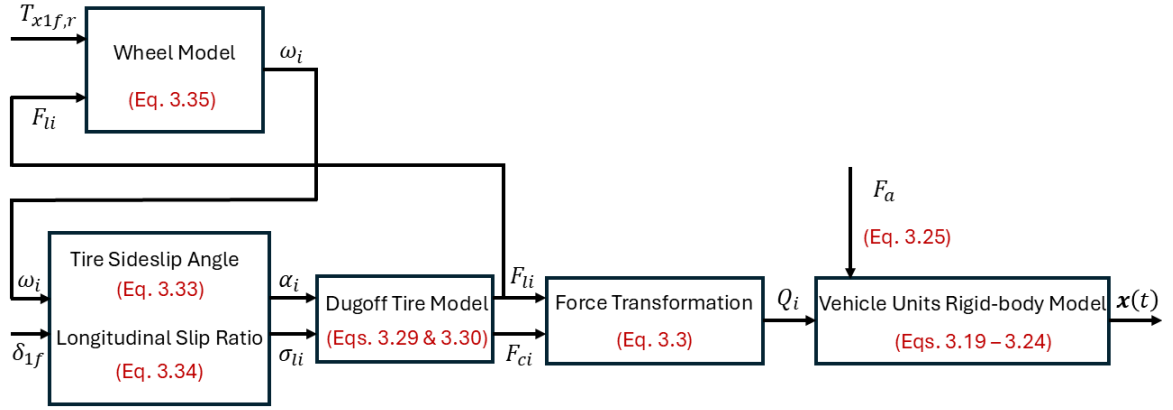


Figure 3-3: Flowchart for integrating the components/sub-models of the nonlinear 11-DOF single-track yaw-plane MTAHV model.

3.3 TruckSim model

TruckSim, developed by Mechanical Simulation Corporation, is a vehicle simulation software designed to model the behavior of heavy-duty commercial vehicles, including trucks, buses, and trailers. It is extensively utilized in the automotive industry for vehicle development, testing, and validation. TruckSim creates vehicle dynamics models by combining analytical and experimental data. The software is built using a symbolic multi-body program called VehicleSim Lisp, which produces equations of motion for 3D multi-body vehicle systems[119].

To conduct co-simulations, this software is used to generate a high fidelity nonlinear 3-D model for representing the A-train double with the configuration of

tractor/semi-trailer/dolly/semi-trailer, as shown in Figure 3-4. The nonlinear 3-D TruckSim model features the axle-arrangement scheme: S_SS + SSS + dSS + SSS, where "S" portrays a solid axle, an underscore "_" means a separation of axle groups, a "+" represents a fifth wheel connecting the two vehicle units, and "dSS" denotes a dolly with two solid axles. The TruckSim model assumes the nonlinear dynamics of suspension systems, pneumatic tires, and mechanical joints. The 3-D model comprises rigid-bodies representing the tractor, dolly, and semi-trailers, along with eleven axles. These rigid bodies may move longitudinally, laterally, and vertically, and they may also rotate in roll, pitch, and yaw directions. The fifth wheel is modelled as a ball joint, allowing roll, yaw, and pitch motions. Each axle is modelled as a beam capable of rolling and bouncing relative to its connected sprung mass. In addition, each wheel is permitted to spin around its axle. Using TruckSim software, the A-train double is modelled as a nonlinear 3-D model with 59-DOF.

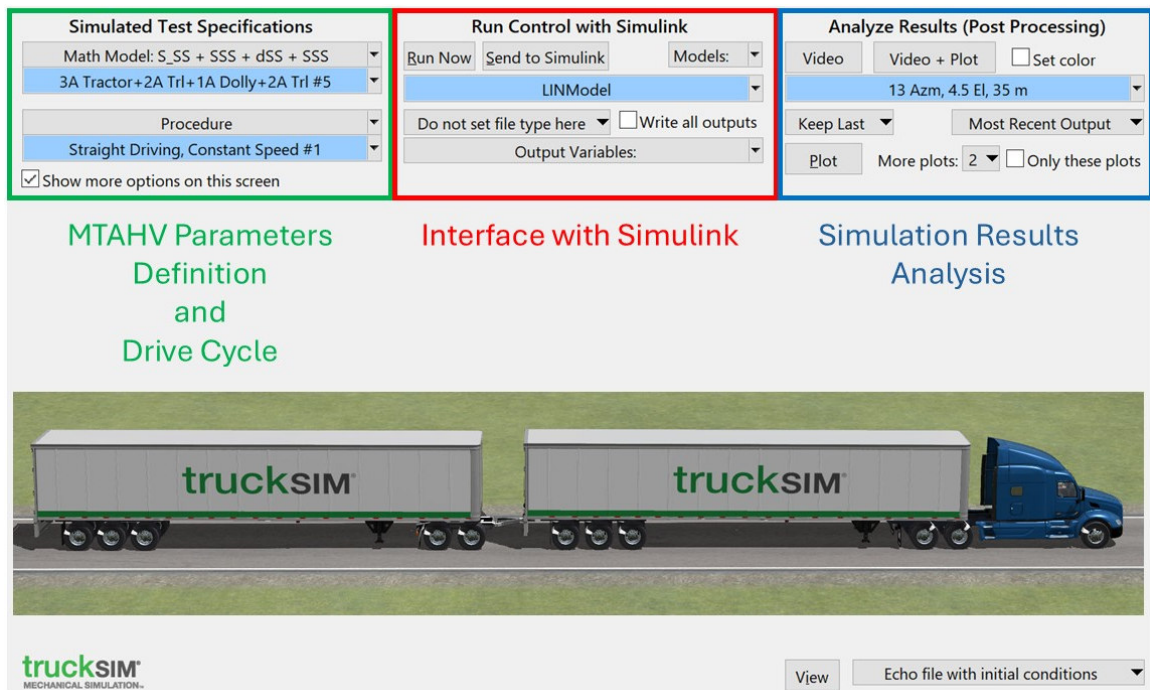


Figure 3-4: The A-train double model generated using TruckSim.

As shown in Figure 3-4, using TruckSim software, the MTAHV configuration and parameters can be specified. According to the evaluation requirements, the respective testing maneuver may be prescribed. To assess the NLMPC tracking controller designed with Simulink/Matlab (to be introduced in Section 4.4), co-simulation can be conducted by integrating the controller with the MTAHV model generated in TruckSim using the S-function as the interface between the two software packages. The attained simulation data may be post processed and the respective results can be used to evaluate the performance of the tracking controller.

It should be mentioned that the TruckSim environment utilizes the Pacejka Magic formula, which captures the transient decay of lateral force beyond the saturation point. In contrast, the 11-DOF model employs the nonlinear analytical Dugoff model. While the Dugoff formulation offers computational efficiency and differentiability suitable for control synthesis, it tends to exhibit monotonic saturation behavior. Consequently, the 11-DOF model may slightly overestimate the vehicle's lateral stability limit compared to the TruckSim benchmark, as it does not aggressively model the reduction in cornering force at extreme slip angles. However, in the linear operating region (lateral acceleration $< 4 \text{ m/s}^2$), the models show high correlation, confirming the validity of the 11-DOF model for the intended control strategy.

3.4 Comparing MTAHV models

To examine the derived nonlinear 11-DOF single-track yaw-plane model, numerical simulations are conducted under an open-loop Single Lane Change (SLC) maneuver specified by ISO-14791[123]. In the SLC testing maneuver, the MTAHV travels

at a constant speed of 88 km/h while the steering input to tractor front wheels is a single-cycle sine wave with a frequency 0.4 Hz and an amplitude of 0.0698 rad (4.00 degrees), as illustrated in Figure 3-5. The simulation results of the nonlinear 11-DOF MTAHV model acquired from the simulated SLC maneuver are compared with those of the TruckSim model.

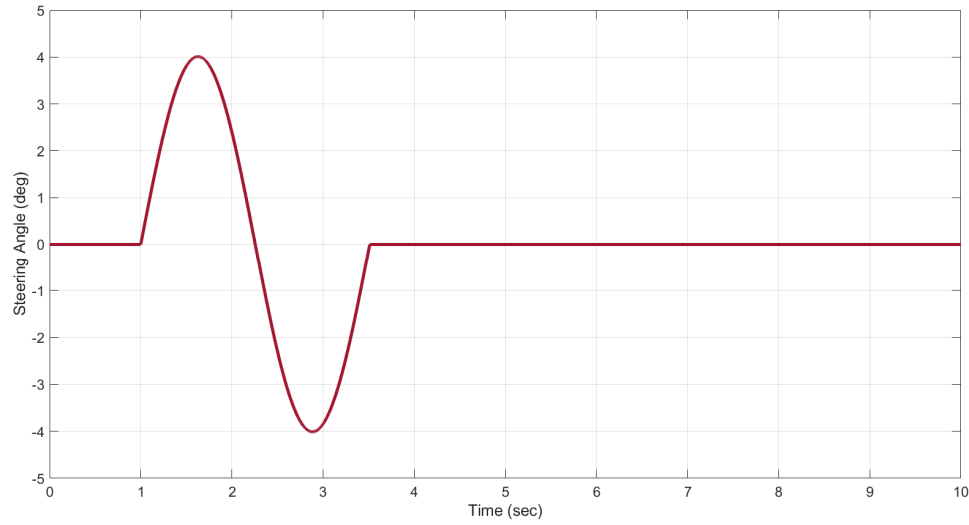


Figure 3-5: Steering angle input for the SLC testing maneuver.

Figure 3-6 shows the comparison between the lateral velocity curves of the nonlinear 11-DOF and TruckSim models. As expected, the simulation results shown in this figure achieve good agreement. Figure 3-7 illustrates a comparison of yaw rates for the nonlinear 11-DOF and TruckSim models. Once again, the time histories of yaw rate based on the two models match reasonably. Thus, in this study, the nonlinear 11-DOF model is used for the design of NLMPC controller for the autonomous A-train double.

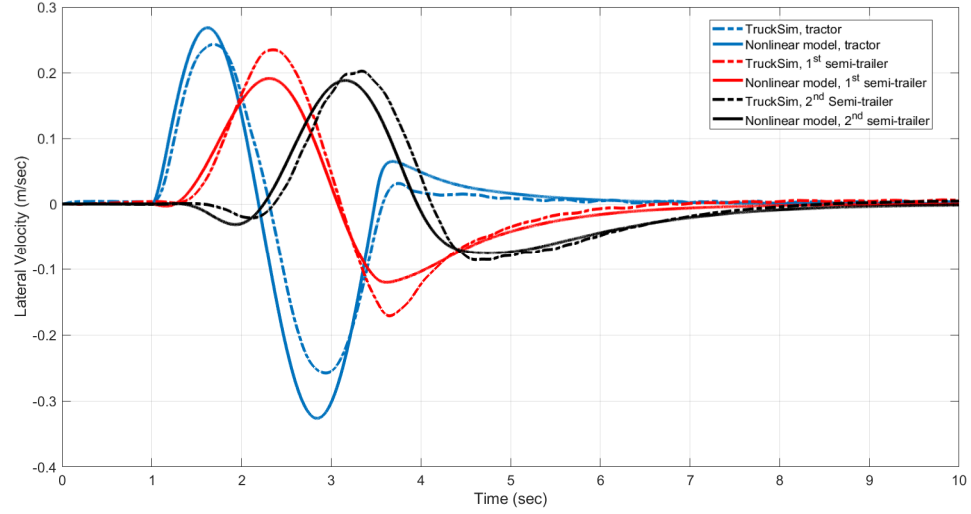


Figure 3-6: Time history of lateral velocity of the nonlinear 11-DOF and TruckSim models.

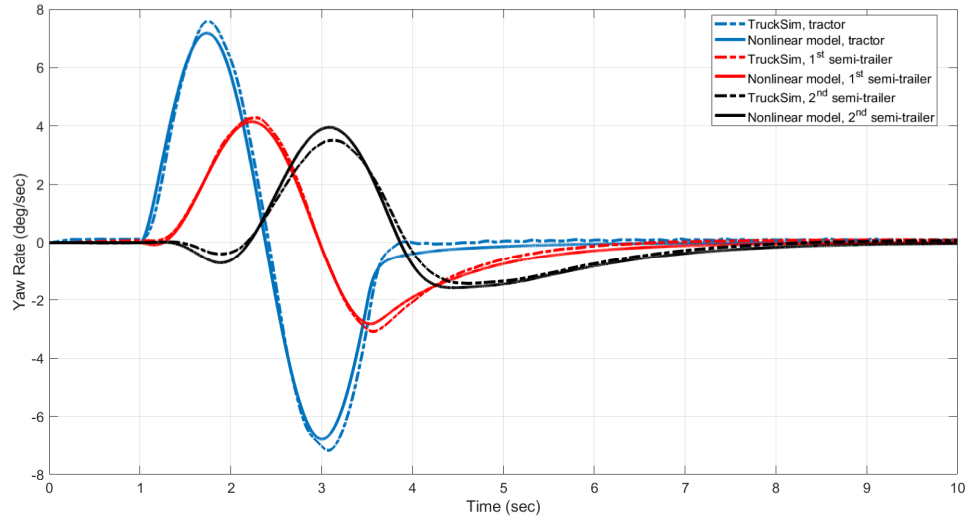


Figure 3-7: Time history of yaw rate of the nonlinear 11-DOF and TruckSim models.

3.5 Summary

This chapter details the development and validation of a nonlinear 11-DOF single-track yaw-plane model for an A-train double MTAHV, intended as a foundation for advanced control system design. The model was validated against high-fidelity TruckSim simulations using a standard ISO-14791 single lane-change maneuver at 88 km/h. Results demonstrated a strong correlation in lateral velocity and yaw rate profiles, confirming the

model's accuracy and establishing it as a robust framework for the subsequent development of a NLMPC. Future validation efforts should advance from qualitative visual inspections to rigorous quantitative assessments by calculating specific error metrics such as root mean square error or coefficient of determination.

Additionally, to address the limitations imposed by the assumption of static normal tire loads without compromising computational efficiency, it is recommended that a quasi-static load transfer approximation be integrated into the prediction plant. By estimating the vertical load redistribution as a function of instantaneous lateral acceleration, the model can account for the critical reduction in cornering stiffness on inner tires during aggressive maneuvers. This modification would enhance the model's predictive capability regarding limit-handling stability without necessitating the integration of fully coupled roll dynamics.

Finally, the generalization of the 11-DOF model warrants further investigation through comprehensive sensitivity analysis. Currently, validation is restricted to the specific conditions of the ISO-14791 maneuver at 88 km/h. Future validation protocols should extend the operational envelope to include variations in longitudinal velocity and payload distribution. Testing the model against these variable parameters is essential to ensure that the derived equations of motion remain robust and accurate across the diverse operational scenarios inherent to MTAHVs.

Chapter 4

NLMPC-based tracking-controller design and tuning

4.1 Introduction

This chapter introduces an innovative method for developing autonomous and semi-autonomous driving controls for MTAHVs, formulated as a multi-objective optimization problem aimed at enhancing ride quality and path-following performance. The proposed approach utilizes an NLMPC-based tracking-controller for an A-train double, employing an 11-DOF yaw-plane model to capture the MTAHV's dynamics. The nonlinear MTAHV model, developed in Chapter 3, serves as the prediction model in the tracking-controller design. Unlike conventional MPC designs, which rely on trial-and-error methods for adjusting weighting matrices—a tedious, time-consuming, and experience-dependent process[53]—the proposed method formulates the determination of weighting matrices as a multi-objective optimization problem. Design criteria and limitations are incorporated into the objective function and constraints, while the elements of the weighting matrices are treated as design variables. By solving the optimization problem using a stochastic search algorithm, multiple solutions for weighting matrices are generated, allowing designers to select the most suitable matrices based on their preferences to achieve desired performance measures. This NLMPC-based strategy effectively optimizes the tractor's directional and speed control while addressing the lateral

dynamics of the trailing units, ensuring improved trajectory tracking and dynamic stability for MTAHVs.

As mentioned before, the design process of an NLMPC-based tracking-controller for autonomous and semi-autonomous driving of a MTAHV is introduced. Additionally, to examine the performance of the designed controller, co-simulations are performed. Figure 4-1 shows the co-simulation platform, which consists of the NLMPC controller and the A-train double (i.e., the nonlinear 3-D TruckSim model). The NLMPC controller consists of an online real-time optimizer and a prediction vehicle model, i.e., the nonlinear 11-DOF single-track yaw-plane model. Given a reference trajectory determined from the upper level motion-planning module, within the finite time-window, the optimizer determines the desired control action $\mathbf{u}(t_k)$ at the sample time step k ; provided the control input in terms of tractor front wheel steering angle (δ_{1f}) and driving/braking torque ($T_{x1f,r}$), the A-train double tracks the prescribed reference trajectory. Then, the output of the A-train double ($\mathbf{y}$) is sent to the NLMPC controller to form a closed-loop feedback control system.

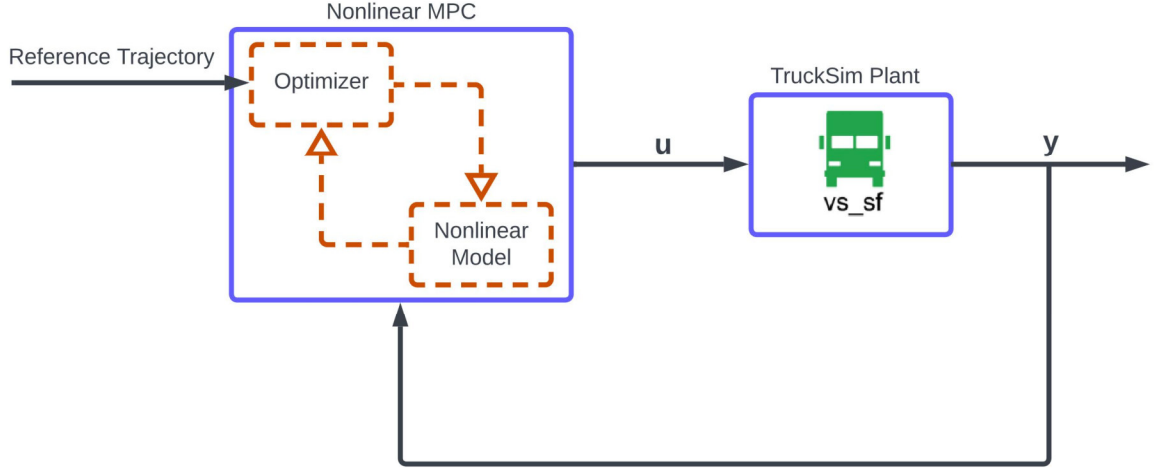


Figure 4-1: Co-simulation platform for devising NLMPC controller for autonomous driving of MTAHVs.

4.2 Reference motion trajectory

To assess the controller's path-tracking performance in highway operations, a SLC maneuver with a predefined testing course by ISO 14791 and a specified speed profile is selected. It is assumed that the vehicle tracks the specified path at a constant longitudinal acceleration. The reference motion trajectory is specified by[63]:

$$Y_{ref}(t) = \frac{L}{T} \left[t - \frac{T}{2\pi} \sin\left(\frac{2\pi t}{T}\right) \right] \quad (4.1)$$

$$X_{ref}(t) = v_{e,0}t + \frac{1}{2}\bar{a}t^2 \quad (4.2)$$

where $Y_{ref}(t)$ and $X_{ref}(t)$ represent the vertical and horizontal coordinates of the target path at time t , L denotes the maximum lateral displacement during the SLC maneuver, T the time period of the sinewave lateral acceleration input by the tractor, $v_{e,0}$ the initial horizontal vehicle velocity at the start of the maneuver, and $\bar{a}$ the constant horizontal acceleration throughout the maneuver.

4.3 Kinematics for the A-train double path-tracking

Figure 4-2 shows the geometry representation of the A-train double and the target path. At an arbitrary time instant t , the horizontal (X) and vertical (Y) components for the velocity of at tractor CG can be calculated by:

$$\dot{X} = v_x \cos \phi_1 - v_y \sin \phi_1 \quad (4.3)$$

$$\dot{Y} = v_x \sin \phi_1 + v_y \cos \phi_1 \quad (4.4)$$

where the output and reference vectors are defined as:

$$\mathbf{y} = [X, Y, \phi_1, \phi_2, \phi_3, \phi_4]^T \quad (4.5)$$

$$\mathbf{y}_{ref} = [X_{ref}(t), Y_{ref}(t), \phi_{ref}(t), \phi_{ref}(t - \tau_1), \phi_{ref}(t - \tau_2), \phi_{ref}(t - \tau_3)]^T \quad (4.6)$$

where the time delays, τ_1 , τ_2 , and τ_3 , are calculated by:

$$\tau_1 = \frac{v_X(t) - \sqrt{v_X(t)^2 - 2\bar{a}(c_1 + a_2)}}{\bar{a}} \quad (4.7)$$

$$\tau_2 = \frac{v_X(t) - \sqrt{v_X(t)^2 - 2\bar{a}(c_1 + a_2 + c_2 + a_3)}}{\bar{a}} \quad (4.8)$$

$$\tau_3 = \frac{v_X(t) - \sqrt{v_X(t)^2 - 2\bar{a}(c_1 + a_2 + c_2 + a_3 + c_3 + a_4)}}{\bar{a}} \quad (4.9)$$

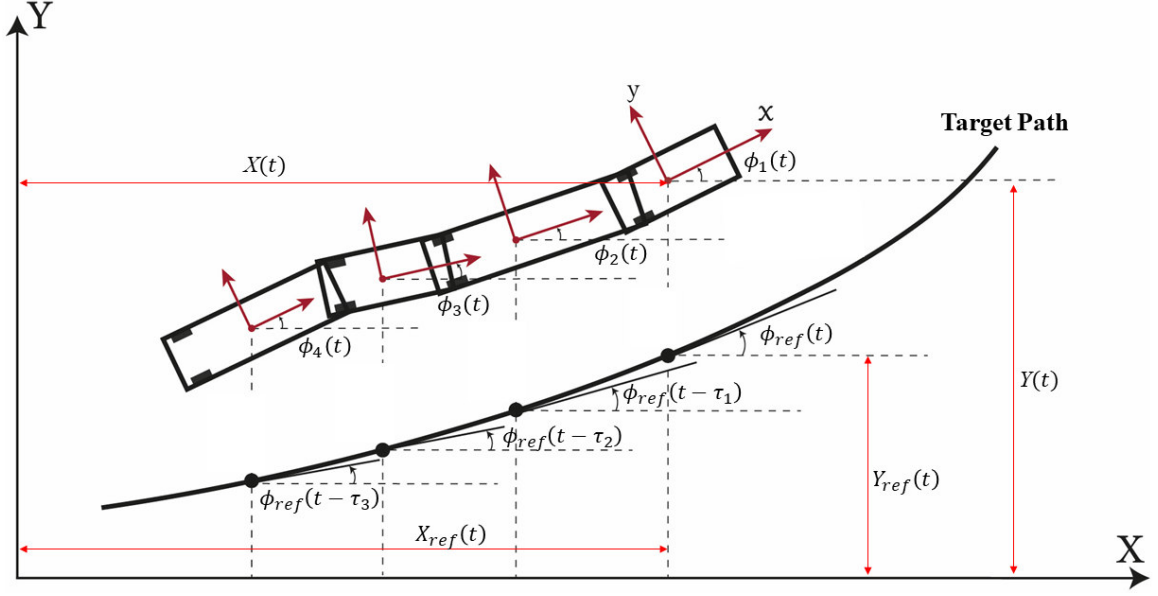


Figure 4-2: Geometry representation of the A-train double and target path.

4.4 Controller design

The main design objective of the trajectory-tracking controller is to determine steering angle for tractor front wheels and propulsion torque for tractor driving wheels. This ensures that the vehicle accurately follows the specified path and forward-speed profile[124, 125]. The design of the NLMPC controller is formulated as an optimization problem with the objective function expressed as:

$$J_{mpc} = \sum_{k=0}^{P_H-1} \|\mathbf{y}_{ref}(k) - \mathbf{y}(k)\|_{\mathbf{Q}}^2 + \sum_{k=0}^{C_H-1} \|\Delta \mathbf{u}(k)\|_{\mathbf{R}}^2 + \sum_{k=0}^{C_H-1} \|\mathbf{u}(k)\|_{\mathbf{S}}^2 \quad (4.10)$$

where P_H and C_H are the prediction horizon and control horizon, respectively, $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{S}$ are weighting matrices, which intend to minimize: 1) tracking error (the first term on the right-hand side of Equation (4.10), 2) control actions fluctuation (the second term on the right-hand side of Equation (4.10), and 3) control actions (the third term on the right-hand side of Equation (4.10), correspondingly.

4.5 Tuning weighting matrices

The weighting matrices are usually determined systematically using a normalization approach based on Bryson's rule. In this method, the weights are set inversely proportional to the square of the maximum acceptable deviation for each state and control input. Although normalization based on physical limits is a widely accepted baseline method, it is not an adaptive method; it relies on static weights that fail to dynamically prioritize parameters that may become critical in specific scenarios, such as prioritizing safety and comfort during aggressive maneuvers. Hence, to adjust the weighting matrices $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{S}$, a two-level multi-objective optimization problem is formulated[126, 127], as illustrated in Figure 4-3. At the top level, the weighting matrices are considered as design variables, $X_{D_{-}i}, i = 0, 1, 2, \dots, n - 1$. A meta-heuristic algorithm, specifically the NSDE, is chosen as the search algorithm for this multi-objective optimization problem. The objective function and constraints for this multi-objective optimization are defined as follows[128, 129]:

$$\begin{aligned}
 J_{mpc} \min_{\mathbf{Q}, \mathbf{R}, \mathbf{S}} J(\mathbf{Q}, \mathbf{R}, \mathbf{S}) &= \rho_1 J_1 + \rho_2 J_2 \\
 \text{subject to} \quad q_{li} &\leq Q_i \leq q_{ui} \quad i = 1, 2, \dots, 6 \\
 r_{li} &\leq R_i \leq r_{ui} \quad i = 1, 2 \\
 s_{li} &\leq S_i \leq s_{ui} \quad i = 1, 2
 \end{aligned} \tag{4.11}$$

where J_1 and J_2 correspond to the vehicle's dynamic responses associated with ride comfort and path-following accumulating error during the SLC maneuver. The weighting factors are denoted as ρ_1 and ρ_2 . The diagonal elements of the weighting matrices are represented by q_i , r_i , and s_i with $\{q_{li}, r_{li}, s_{li}\}$ and $\{q_{ui}, r_{ui}, s_{ui}\}$ being the lower and upper bounds for

these elements, respectively. Thus, the design variables are specified as $X_{D,i} = \{q_{mi}, r_{ni}, s_{oi}\}, m = 1, 2, \dots, 6, n = 1, 2, o = 1, 2$.

Given a reference motion trajectory (determined by Equations (4.1) and (4.2)) and a set of design variables, $X_{D,i}, i = 0, 1, 2, \dots, n - 1$, from the top level, a Sequential Quadratic Programming (SQP) algorithm is used to solve the NLMPC optimization problem and determine the optimal control inputs for the MTAHV at the bottom level. The resulting fitness values, $F(X_{D,i}), i = 0, 1, 2, \dots, n - 1$, are then returned to the top level. If the criteria are met, the optimization process terminates, and the optimal values for $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{S}$ are identified. Otherwise, another optimal algorithm continues to search for the design variables within the design space, repeating the process until the optimal solutions are found.

It should be mentioned that in this chapter, the default SQP search algorithm in Matlab is utilized to solve the NLMPC optimization problem expressed in Equation (4.10). It is observed that an effective and faster than real-time prediction vehicle model is critical to improve the computational efficiency of the SQP search algorithm. With this consideration, the simple, yet effective single-track MTAHV model is derived for the prediction model, and the Dugoff tire model is incorporated. Compared with the brush tire model[130] and the Magic Formula tire model[131], the Dugoff tire model is simpler in structure and more efficient in computation. Essentially, the Dugoff tire model is an analytical model, which is differentiable with respect to the state and control variables. Therefore, for the nonlinear programming problem, the Jacobian and Hessian matrices can be manipulated and predefined by using Maple software for the gradient evaluation of the

SQP search algorithm. This might further increase the computational efficiency of the nonlinear programming problem.

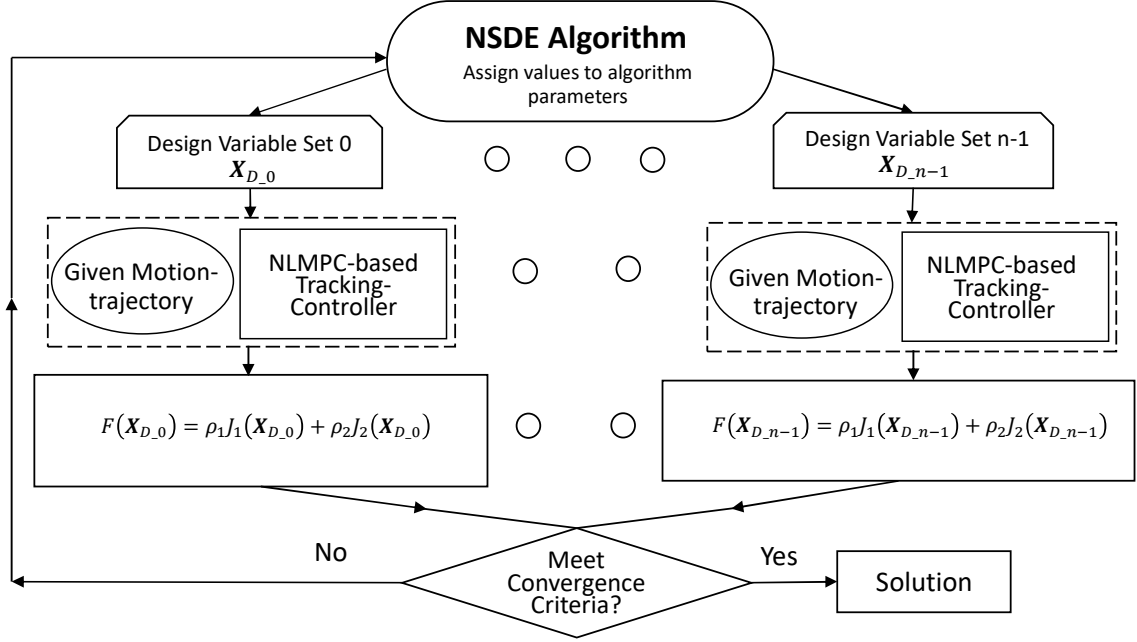


Figure 4-3: Two-level optimization method for tuning the weighting matrices for the NLMPC controller.

4.5.1 Ride comfort

To evaluate the passenger's ride comfort of the MTAHV during the specified SLC maneuver, the Root Mean Square (RMS) value of the lateral acceleration at tractor CG (a_y) can be used as an effective indicator. Assuming the duration of the SLC maneuver is T , the ride comfort performance measure is determined by[132]:

$$J_1 = a_{eq} = (a_y)_{RMS} = \sqrt{\frac{1}{T} \int_{t=0}^{t=T} a_y(t)^2 dt} \quad (4.12)$$

4.5.2 Accumulated path-following error

To assess the path-following performance of the designed NLMPC controller, the Integral of the Square Error (ISE) between the vehicle's output and its reference signal is used as the indicator[133]. The ISE index is defined as:

$$J_2 = ISE = \int_{t=0}^{t=T} \mathbf{e}(t)^T \cdot \mathbf{e}(t) dt \quad (4.13)$$

and

$$\mathbf{e}(t) = \mathbf{y}(t) - \mathbf{y}_{ref}(t) \quad (4.14)$$

4.5.3 NSED algorithm

The NSDE algorithm is used for the top-level optimization illustrated in Figure 4-3. Non-dominated sorting differential evolution is an adaptation of the Differential Evolution (DE) technique, aimed at identifying Pareto-optimal solutions for multi-objective optimization problems. It includes two primary components: 1) a DE optimization algorithm for performing mutation and crossover, and 2) an elitist non-dominated sorting algorithm for selection[134, 135].

4.6 Simulation results

To evaluate the designed NLMPC controller and the proposed method for tuning control parameters, a SLC testing maneuver with variable longitudinal speed is implemented using numerical simulation. Given the co-simulation platform shown in Figure 4-1Figure 3-4, the design optimization method seen in Figure 4-3, and the closed-

loop SLC testing maneuver by ISO-14791, the parameter values for the two MTAHV models and the NLMPC controller need to be determined to run numerical simulations. Table 4-1 lists the required parameter values for the nonlinear 11-DOF single-track yaw-plane model and the nonlinear 3-D TruckSim model.

Table 4-1: The parameter values for the A-train double models.

Parameter	Symbol	Value	Unit
Mass of i^{th} vehicle unit, $i=1,2,3,4$	m_i	{6310,24690,2354,24690}	kg
Yaw mass-moment of inertia of i^{th} vehicle unit	J_i	{19665,274658,1990,274658}	kg m ²
Distance from i^{th} vehicle unit CG to its front axle	a_i	{3.385,6.486,1.489,6.485}	m
Distance from i^{th} vehicle unit CG to its rear axle	b_i	{4.25,6.015,0,6.015}	m
Distance from i^{th} vehicle unit CG to its rear articulation joint	c_i	{4.25,8.8,0,0}	m
Frontal area of the tractor	A_a	3.2	m ²
Aerodynamic drag coefficient	C_D	0.66	
Longitudinal stiffness of j^{th} tire (Dugoff tire model), $j=1,2,3,4,5$	$C_{\sigma j}$	{200,756,680,739,680}	kN/rad
Lateral stiffness of j^{th} tire (Dugoff tire model)	$C_{\alpha j}$	{286,814,789,862,789}	kN/unitslip
Coefficient of j^{th} tire rolling resistance	f_{rj}	0.0041	
Effective rolling radius of j^{th} tire	R_{ej}	0.51	m
Friction coefficient	μ	0.5	

In MPC controller designs, the parameters involved may be divided into two groups, i.e., control-base and optimization-based. The first group consists of prediction horizon, P_H , control horizon, C_H , and sampling time, T_s ; the second group comprises output weighting matrix, $\mathbf{Q}$, control input variation weighting matrix, $\mathbf{R}$, and control input weighting matrix, $\mathbf{S}$, various constraint parameters, e.g., maximum tractor front-wheel steering angle, δ_{max} , and minimum tractor front-wheel steering angle, δ_{min} . Selecting the values for these parameters is important because an effective choice is likely to greatly improve control quality[136]. In the real-time execution of MPC controls, practical constraints often restrict the availability of the optimization-based constraint parameters. Table 4-2 lists the values for the steering input related constraint parameters. The values selection of the weighting matrices is the focus of this study, which are determined using the optimization method shown in Figure 4-3: Two-level optimization method for tuning

the weighting matrices for the NLMPC controller. The determination of the weighting matrices is case dependent, which will be discussed in the following context.

Since among the control-based parameters, sampling time is heavily dependent on vehicle forward speed, this chapter takes the following procedure for determining the values for these parameters: 1) specifying sampling time, T_s ; 2) choosing prediction and control horizons, P_H and C_H . It is justified that the sampling time of 0.05 s is feasible and effective for MPC controllers devised for passenger cars executing a SLC at a speed of 72 km/h[137]. Considering the facts: 1) a smaller sampling time enables the MPC controller to react more quickly to disturbance, thereby enhancing tracking performance; and 2) in this study, the NLMPC controller is devised for the MTAHV to execute the high-speed evasive maneuver. Thus, a shorter sampling time is chosen, i.e., $\Delta T_s = 0.01$ s. In selecting prediction and control horizons, reliable guidelines are well established[51]. The relationship between the two parameters needs to meet the requirements $C_H \leq P_H$. In general, the larger the prediction horizon is chosen, the more information about the future vehicle dynamics is achieved. However, a larger P_H usually results in a heavier computational burden on the controller. Moreover, a large C_H permits aggressive control actions but leads to low computational efficiency of the controller. In MPC controller designs for autonomous vehicles, P_H is frequently set to 10, which attains a compromised relation between tracking performance and computation time[137]. Thus, in this study, P_H is set to 10, and C_H takes the value of 2. Table 4-2 also lists the values of the control-based parameters.

Table 4-2: The constraint parameter values for the NLMPC controller design.

Parameter	Symbol	Value	Unit
Maximum Steering Angle	δ_{max}	$\pi/12$	rad
Minimum Steering Angle	δ_{min}	$-\pi/12$	rad
Maximum input's rate of change	$\Delta\delta_{max}$	$\pi/12$	rad/s
Minimum input's rate of change	$\Delta\delta_{min}$	$-\pi/12$	rad/s
Sampling Time	ΔT_s	0.01	sec
Prediction Horizon	P_H	10	
Control Horizon	C_H	2	

Figure 4-4 illustrates the optimal Pareto front for the two design criteria: J_1 , indicating ride comfort (a_{eq}), and J_2 , which indicates accumulating path-following error (ISE). This figure shows that the RMS acceleration of the MTAHV remains below 0.23 m/s^2 during the maneuver, which is lower than the 0.35 m/s^2 specified in Table 4-3. This table presents the evaluation levels of human ride comfort based on the equivalent RMS acceleration as specified by ISO 2631-1[132]. Thus, during the SLC maneuver, the passenger stays within the comfort zone at all points on the Pareto front. Hence, the design criterion J_2 is prioritized in the multi-objective optimization problem. The design variables, including the weighting matrices, $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{S}$, are optimized to minimize the performance index J_2 . Figure 4-5 depicts the fitness value of J_2 across different generations of the NSDE algorithm.

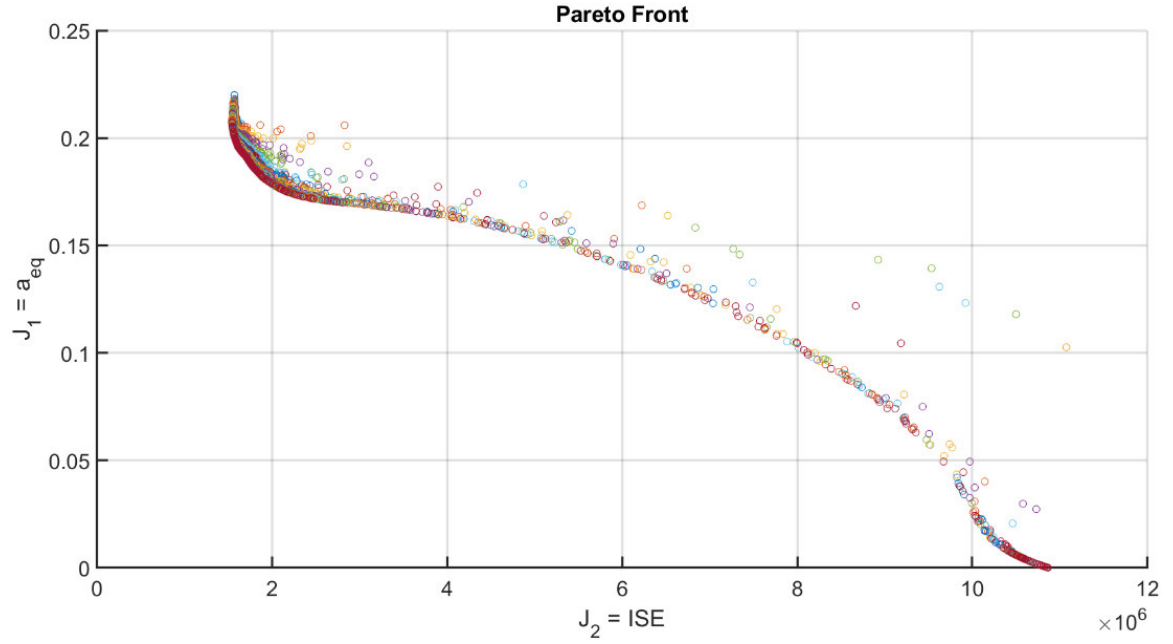


Figure 4-4: The optimal Pareto front for the NLMPC controller design for the MTAHV under SLC maneuver.

Table 4-3: Evaluation levels of human ride comfort by ISO-2631-1[132].

Equivalent Acceleration Range	Comfort Zone
$a_{eq} \leq 0.315 \text{ m/s}^2$	Not uncomfortable
$0.315 \text{ m/s}^2 \leq a_{eq} \leq 0.63 \text{ m/s}^2$	A little uncomfortable
$0.5 \text{ m/s}^2 \leq a_{eq} \leq 1.0 \text{ m/s}^2$	Fairly uncomfortable
$0.8 \text{ m/s}^2 \leq a_{eq} \leq 1.6 \text{ m/s}^2$	Uncomfortable
$1.25 \text{ m/s}^2 \leq a_{eq} \leq 2.5 \text{ m/s}^2$	Very uncomfortable
$a_{eq} \geq 2.0 \text{ m/s}^2$	Extremely uncomfortable

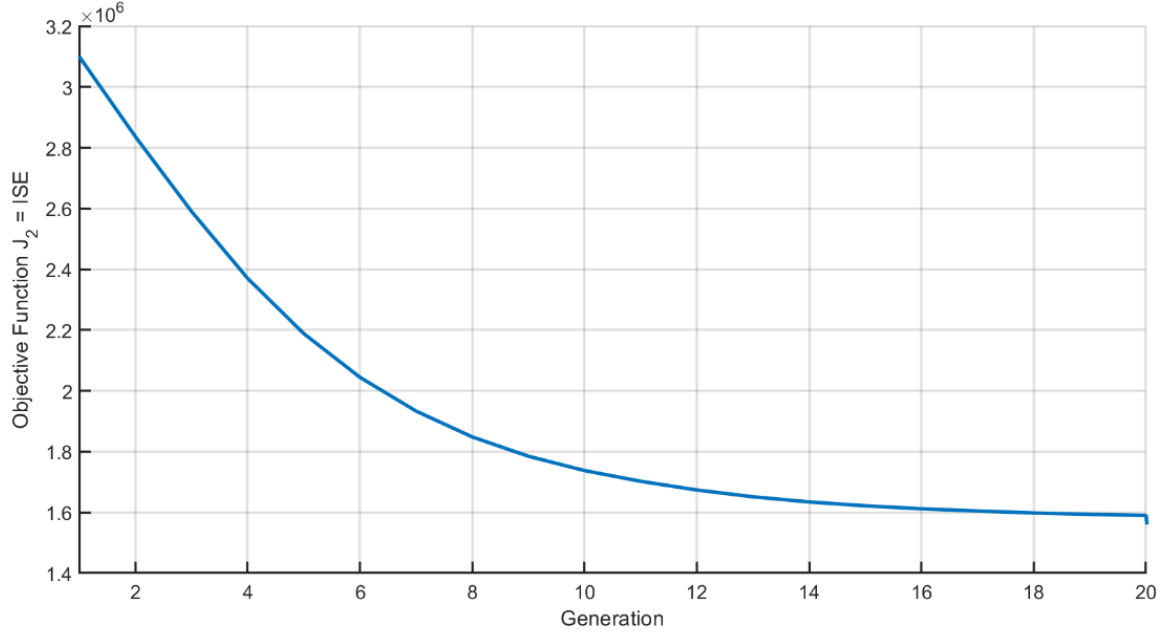


Figure 4-5: The fitness value of J_2 versus the generation number of the NSDE algorithm.

The simulation is conducted to find the weighting matrices, for which each element is allowed to vary within the range of 0 to 100, with an accuracy of one-tenth. The optimal weighting matrices with the values for all their elements for minimizing J_{MPC} are listed as follows:

$$Q = \begin{bmatrix} 74.8 & 0 & 0 & 0 & 0 & 0 \\ 0 & 70.1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 34.5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 30.3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 28.8 & 0 \\ 0 & 0 & 0 & 0 & 0 & 27.1 \end{bmatrix}, \quad R = \begin{bmatrix} 24.2 & 0 \\ 0 & 30.7 \end{bmatrix}, \quad S = \begin{bmatrix} 66.1 & 0 \\ 0 & 44.4 \end{bmatrix} \quad (4.15)$$

Now, to evaluate the performance of the finely tuned NLMPC controller, simulations are conducted considering two cases: 1) the baseline design of the NLMPC controller without tuning the weighting matrices (all weighting matrices are considered unity); and 2) the optimal design of the NLMPC controller with the weighting matrices set to the values specified in Equation (4.15).

Figure 4-6 displays the time history for reference speed ($v_{x,ref}(t)$) and the actual vehicle speeds for the cases with/without tuning weighting matrices (v_x). It demonstrates that the optimal NLMPC design more accurately tracks the reference forward speed.

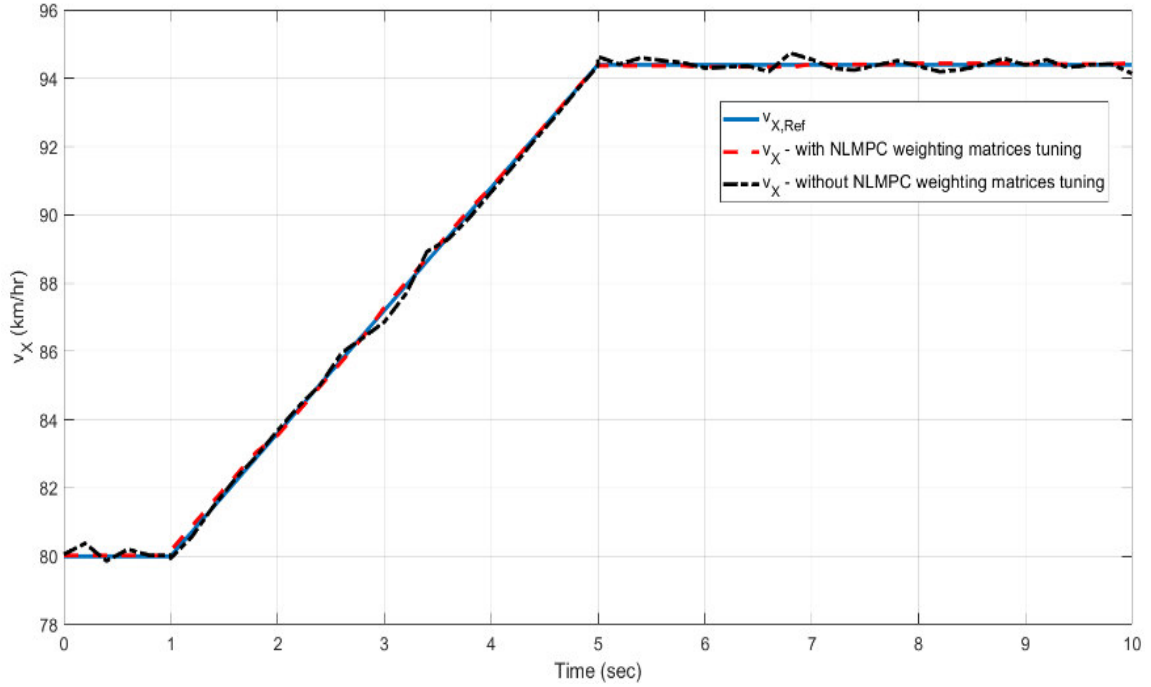
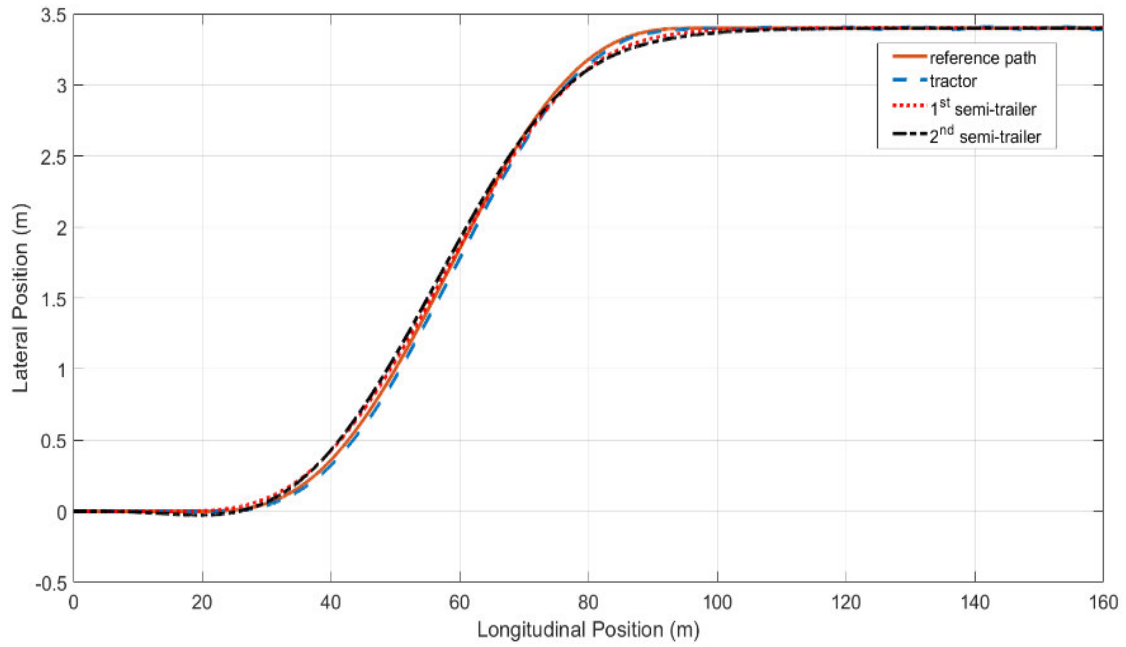


Figure 4-6: Time history of $v_{x,ref}(t)$ and $v_x(t)$ for the cases with and without NLMPC weighing matrices tuning.

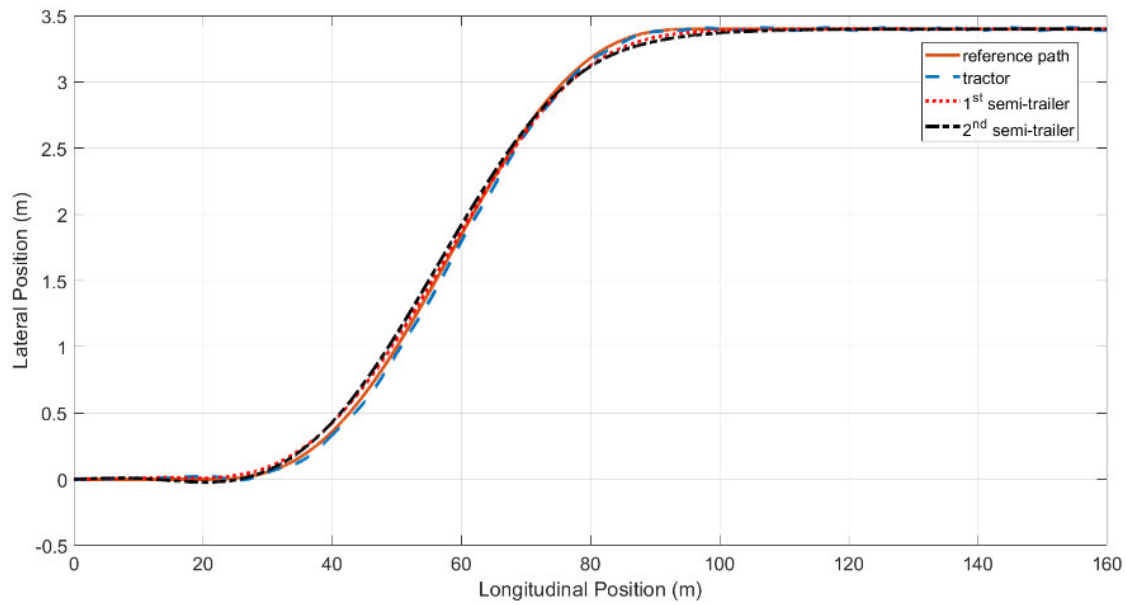
Figure 4-7 shows the reference SLC path and the actual paths for vehicle units for both the NLMPC controller designs with and without tuning weighting matrices. Figure 4-7 (a) and (b) illustrate that for both designs, the tractor can well track the target SLC path. In the meantime, the respective trailing units can well follow the leading unit track.

In the process of NLMPC control (with optimal weighting matrices tuning), the sideslip angle (β_i) and yaw rate ($\dot{\phi}_i$) for each vehicle unit are key state variables to describe the unit's lateral/yaw motions[138, 139]. Therefore, in the respective 2-D phase-plane representation, the x-axis represents the sideslip angle (β_i), and the y-axis the yaw rate ($\dot{\phi}_i$).

The phase plane can be used to evaluate the lateral/yaw stability of the vehicle unit over the SLC maneuver. Figure 4-8 depicts the phase-plane trajectories of the sideslip angle and yaw rate for both trailers. Each trajectory is linked to a particular vehicle forward speed, which clearly highlights the effect of the speed on the vehicle unit's lateral/yaw stability in the presence of the tuned NLMPC controller. When the forward speed is 110 km/h or less, all trajectories converge back to the origin of the coordinate system. However, if the forward speed exceeds 120 km/h, the trajectories begin to diverge (indicating the unstable yaw motion). In addition, Figure 4-9 shows the unstable motion mode of the A-train double (i.e., the nonlinear 3-D TruckSim model) during the SLC maneuver at forward speed of 120 km/h. Note that the SLC at 120 km/h is a very severe obstacle avoidance maneuver for MTAHVs in highway operations. It should be emphasized that the failure mode observed at high speeds is not a result of the control law becoming unbounded or unstable. Rather, the controller functions correctly by attempting to enforce the desired trajectory. The issue arises because the standard path-tracking controller does not inherently account for the vehicle's roll dynamics or physical constraints (such as the rollover threshold). Consequently, in its effort to eliminate lateral error during high-speed cornering, the controller commands steering angles that generate lateral accelerations beyond the physical capabilities of the vehicle platform, leading to dynamic instability and subsequent rollover.

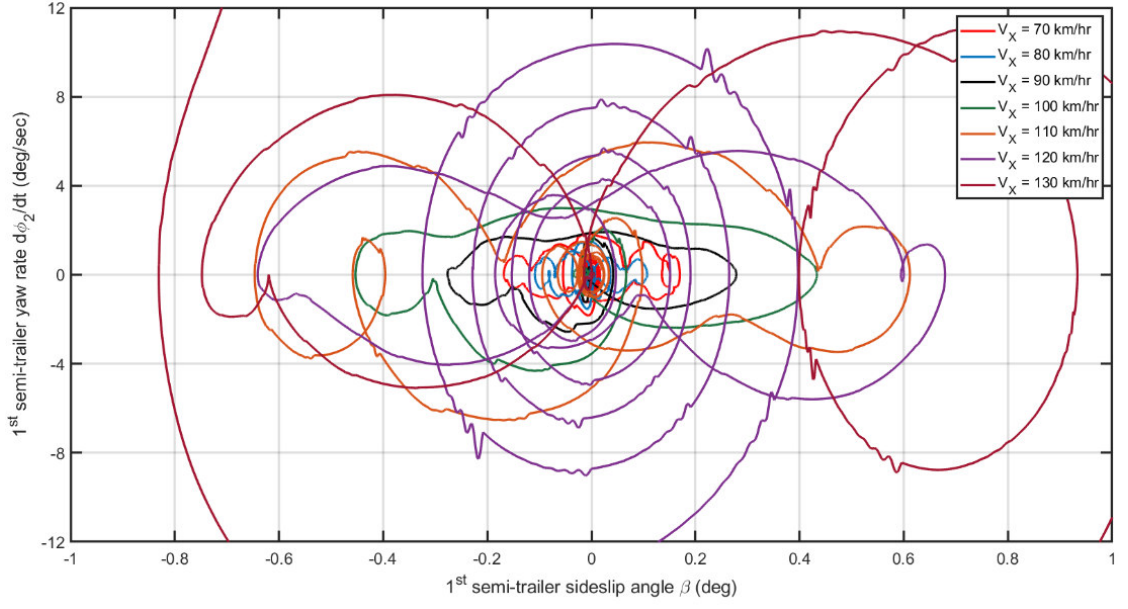


(a)

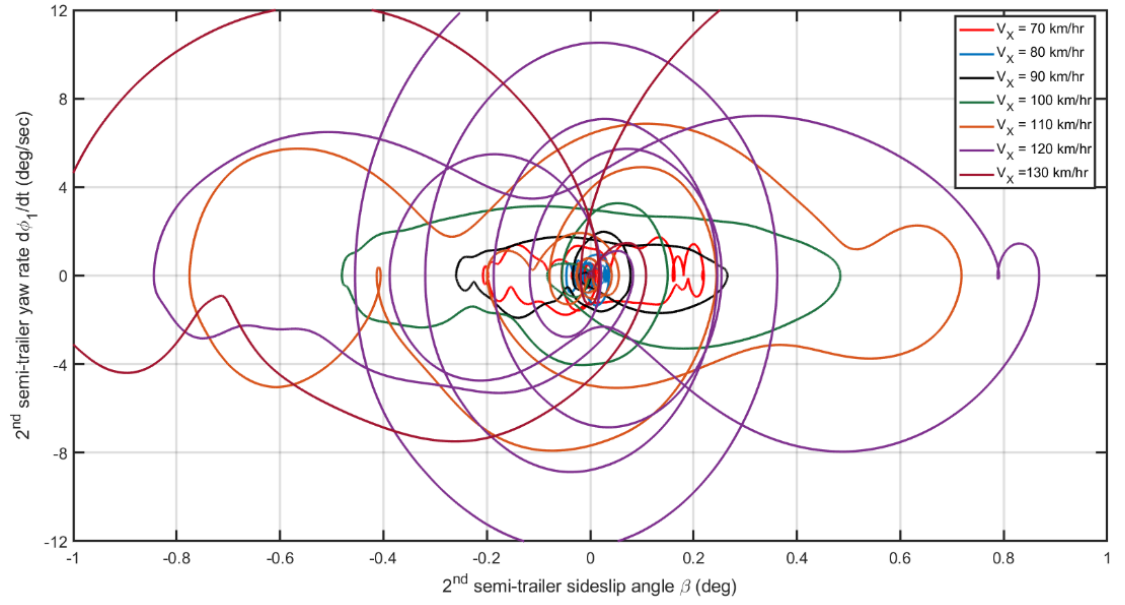


(b)

Figure 4-7: Reference SLC path and actual paths for vehicle units: (a) baseline NLMPC controller design without tuning weighting matrices; (b) optimal NLMPC controller design with tuning weighting matrices.



(a)



(b)

Figure 4-8: Phase-plane trajectories of $\beta_i - \dot{\phi}_i$ at different vehicle forward speed for trailers (with optimal NLMPC weighting matrices tuning): (a) trailer 1; (b) trailer 2.



Figure 4-9: Unstable motion mode of the A-train double (with optimal NLMPC weighting matrices tuning) during the SLC maneuver at the forward speed of 120 km/h.

Furthermore, Figure 4-10 shows the phase-plane trajectories of $\beta_4 - \dot{\phi}_4$ for trailer 2 (with optimal NLMPC weighting matrices tuning) under the SLC maneuver at the forward speed of 88 km/h with different ground adhesion coefficients. For the adhesion coefficient in the range of 0.4 to 0.8, all trajectories converge back to the origin. But, on slippery roads with $\mu = 0.2, 0.3$, although the vehicle does not lose its lateral/yaw stability since the trajectories don't diverge, the vehicle loses its steering capability because the trajectories don't converge to the origin.

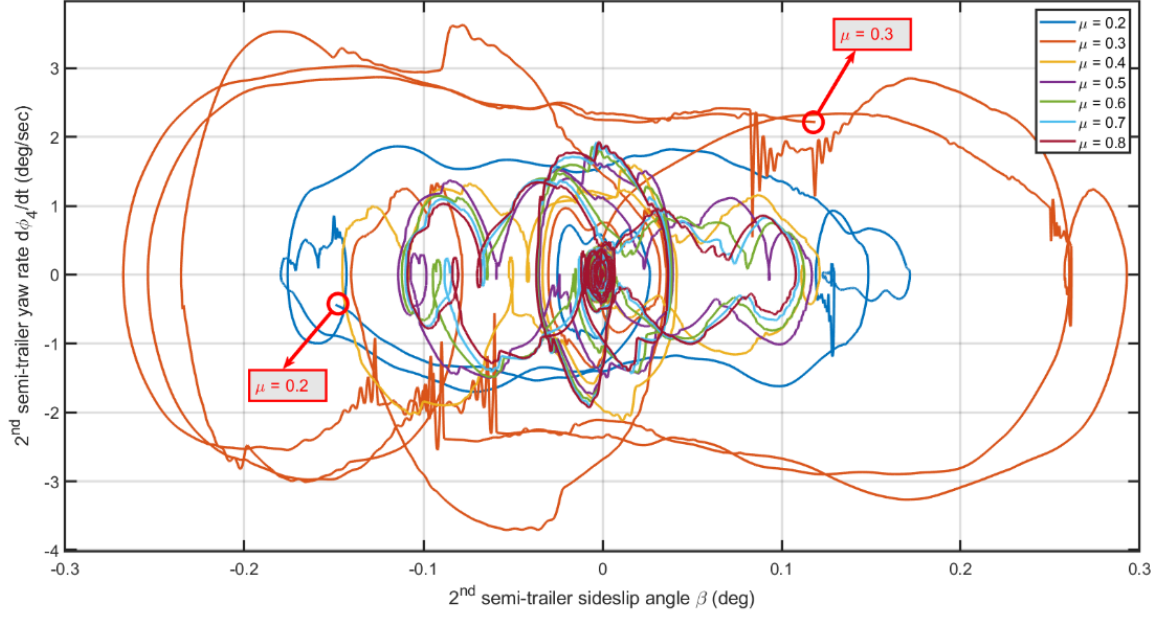


Figure 4-10: Phase-plane trajectories of $\beta_4 - \dot{\phi}_4$ for different ground adhesion coefficient for trailer 2 (with optimal NLMPC matrices tuning) under the SLC maneuver at the forward speed of 88 km/h.

To distinguish the baseline and optimal designs of the NLMPC controller, Figure 4-11 shows the lateral position deviation of the tractor CG from the target SLC path for both the designs. For the optimal design, the maximum tractor lateral position error is 7.5 cm, decreasing by 53.7% from its baseline value of 16.2 cm. For the optimal NLMPC controller design with tuned weighting matrices, the performance indices J_1 and J_2 achieve fitness values of 0.23 and 1.66×10^6 , respectively.

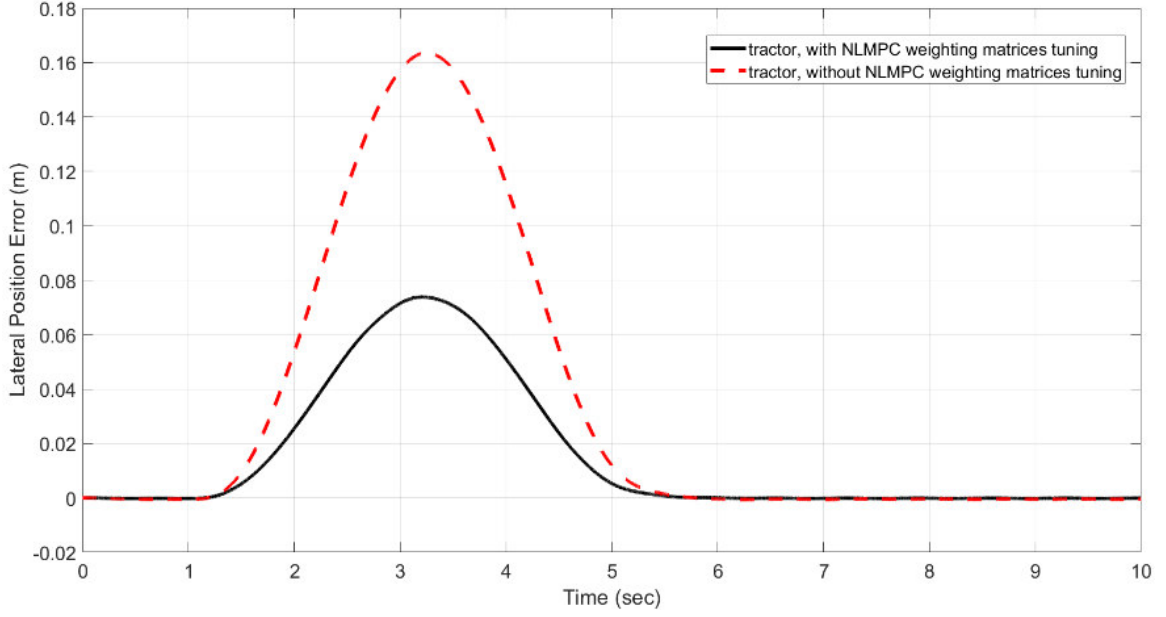


Figure 4-11: Time history of the tractor lateral position error for both NLMPC controller designs (with/without weighting matrices tuning) over the SCL maneuver at the speed of 88 km/h.

To evaluate the robustness of the NLMPC controller with the tuned weighting matrices to parameters uncertainty, two different scenarios are considered. In the first scenario, it is assumed that the actual payload of trailer 2 deviates from its nominal value listed in Table 4-1 within the ranges of 5 ~20% and -5 ~ -20%. In the second scenario, it is supposed that due to sensor or estimator errors, the measured state variables of trailer 2 deviates from their actual variables within the ranges of 5 ~20% and -5 ~ -20%. For the first scenario, Figure 4-12 shows the lateral position error for trailer 2, which is more susceptible to the payload variation with respect to other units. Results show that the controller is robust to the payload variation of trailer 2. Figure 4-13 illustrates that effect of the sensor errors on the path-tracking performance of trailer 2. It is observed that the larger the measured states in magnitude, the bigger the trailer lateral position becomes. Compared with the results shown in Figure 4-12 for the first scenario, the absolute lateral position error of trailer 2 in the second scenario seen in Figure 4-13 is increased. In both

scenarios, the maximum lateral position error of trailer 2 due to the uncertainties is less than 2.0 cm. The superior robustness of the NLMPC controller is partially attributed to the optimally tuned weighting matrices.

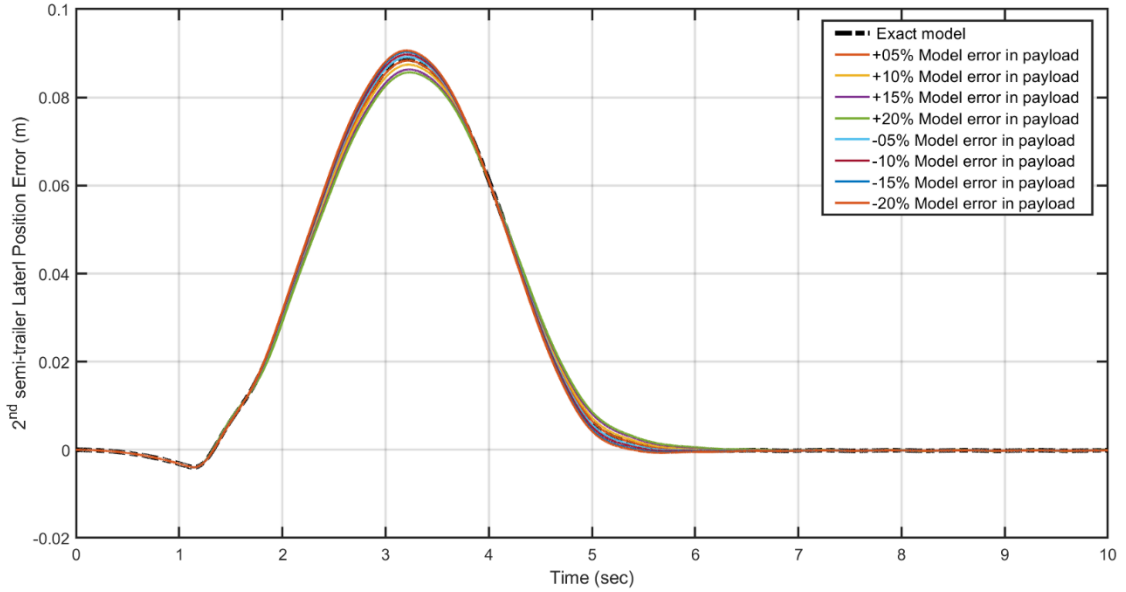


Figure 4-12: Time history of lateral position error for trailer 2 under the SLC maneuver at 88 km/h with different payloads for the trailer.

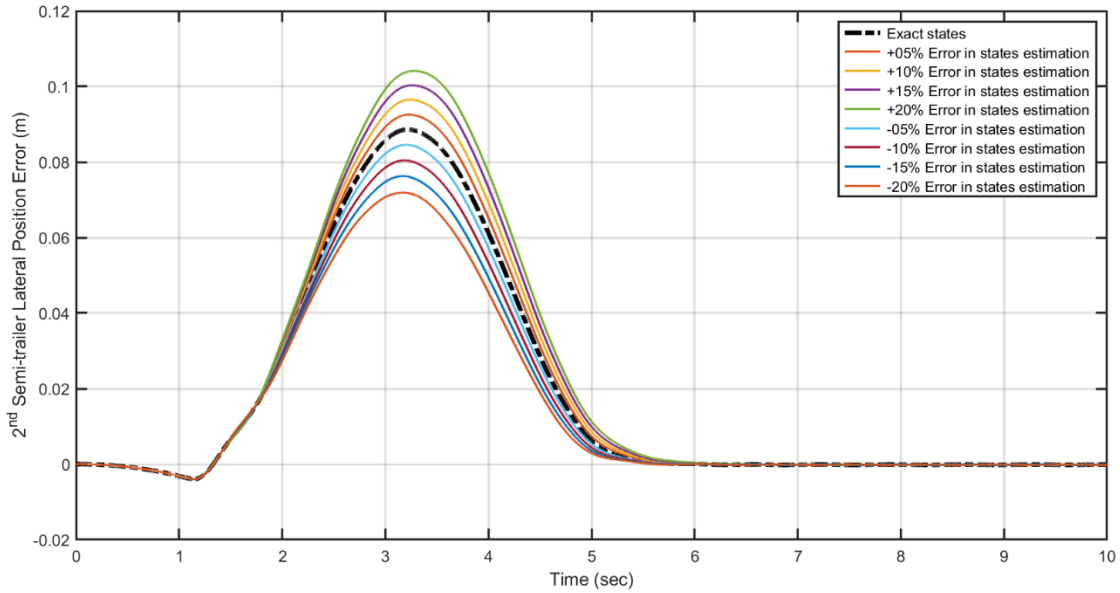


Figure 4-13: Time history of lateral position errors of trailer 2 under the SLC maneuver at 88 km/h with different sensor errors.

Note that this chapter presents the preliminary research results to examine the proposed method for devising autonomous driving controls for MTAHVs. In this preliminary research phase, considering the cost and safety concerns, it may not be practical to conduct field or road test to evaluate the proposed design method. To date, vehicle modeling and simulation is of importance for vehicle design and development. Simulation is increasingly used in the design and test of vehicles in an environment prior to physical full-size vehicle prototype be fabricated[140]. In the Australia performance-based standards for articulated vehicles, both in-vehicle tests and simulations are permitted to evaluate the lateral dynamics of these vehicles[141]. This section thus presents only numerical simulation results to evaluate the proposed design method.

4.7 Summary

This chapter proposes a novel design method for tuning a nonlinear model predictive control for autonomous (and semi-autonomous) driving of a multi-trailer articulated heavy vehicle with an A-train double configuration. Utilizing a multi-objective metaheuristic optimization approach, the weighting matrices in the NLMPC controller are automatically tuned to minimize both path-following error and the RMS value of vehicle lateral acceleration. The validation of the proposed controller framework was conducted through rigorous numerical simulations utilizing a standard Single Lane-Change (SLC) maneuver to stress-test lateral stability. To ensure reliability in real-world scenarios, the controller's robustness was extensively evaluated against operational uncertainties, including trailer payloads, road friction, and sensor noise. The comparative analysis focused on the efficacy of the novel multi-objective metaheuristic optimization tuning method versus conventional tuning strategies; the results demonstrated that the proposed

method superiorly balanced conflicting objectives by simultaneously minimizing path-following error and vehicle lateral acceleration (ride comfort), proving its effectiveness even in the presence of significant system uncertainties.

Future research should advance the proposed framework by moving from numerical simulation to Hardware-in-the-Loop (HIL) testing to verify real-time computational feasibility and address system latency. Additionally, the control strategy should evolve from static offline tuning to Adaptive Model Predictive Control (AMPC), allowing for dynamic weight adjustment based on driving conditions. finally, the validation scope must be expanded to include rigorous quantitative benchmarking against established methods (e.g., Bryson's Rule) and stress-testing in complex scenarios such as Double Lane Change maneuvers.

Chapter 5

An integrated modified visual driver model for highway driving

5.1 Introduction

In this chapter, the deficiencies mentioned in Section 2.8 will be addressed in the following sections. Built upon the previous research, a modified two-point visual driver model is generated based on the information of: 1) far point on the path, 2) near point on the path, and 3) yaw angle difference between the rearmost vehicle unit and the tractor. To find the optimal coefficient values for the driver model, two common scenarios are considered in the high-speed highway driving: 1) lane-change maneuver, and 2) lane-keeping maneuver. By means of the Particle Swarm Optimization (PSO) algorithm, the optimal coefficient values are extracted for the highway lane-change and lane-keeping maneuvers. Finally, to make the Modified Visual Driver Model (MVDM) model more realistic, the effects of driver's time-delay and arm NMS dynamics are incorporated into the modified driver model and the performance of the Integrated MVDM are evaluated based on numerical simulations.

In simulations of highway operations, the driver model is responsible for executing the actions generated by the upper level, i.e., decision-making, to guide the MTAHV to successfully and precisely perform lane-keeping or lane-change maneuver. In this chapter,

the driver model is built upon the essential concepts reported in [111, 142], which are associated with the visual and cognitive perceptions for passenger vehicles. In another word, the driver visually evaluates the road path ahead and continuously chooses the steering angle based on the visual information. The Integrated MVDM structure in the discrete-time format is depicted in Figure 5-1, where $\mathbf{x}(k)$ and $\mathbf{x}(k + 1)$ are the vehicle state vectors at time step k and $k + 1$, respectively, $\delta_{des}(k)$, $\delta_{des}^{-N_d}(k)$, and $\delta_{driver}^{NMS}(k)$ are the desired steering angle determined by driver visual perception, by delayed signal with N_d time step, and by arm NMS dynamics, respectively. It should be noted that the TruckSim model is identical to the one described in Section 3.3, with its parameters taken from Table 4-1.

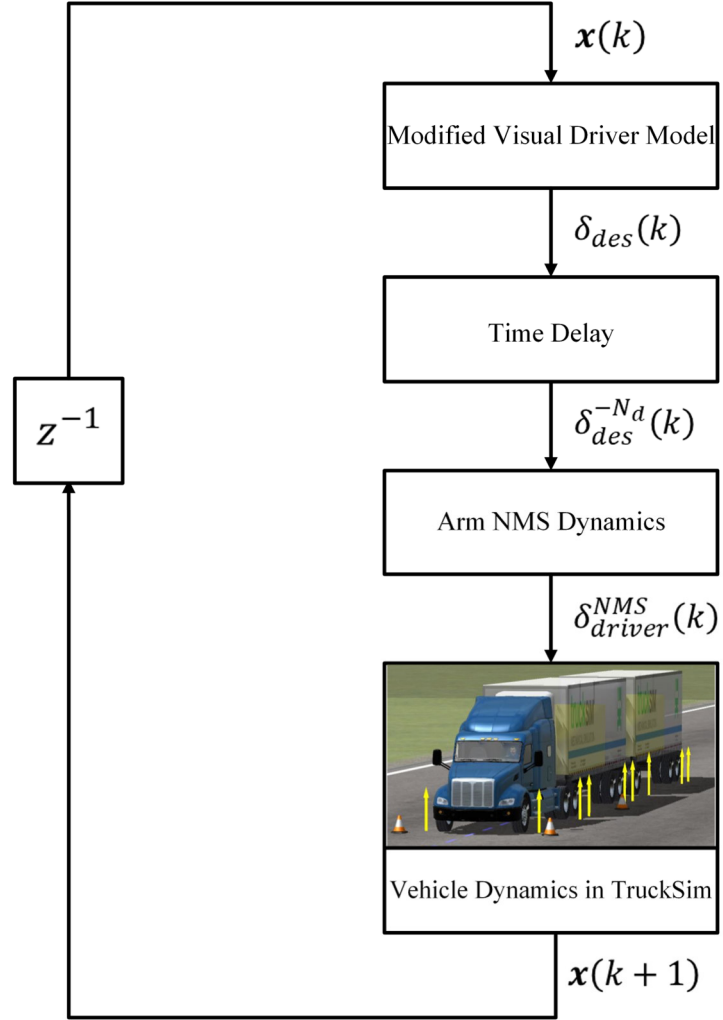


Figure 5-1: The integrated modified visual driver model structure for MTAHVs.

5.2 Integrated modified visual driver model

As shown in Figure 5-1, the Integrated MVDM consists of three parts: 1- visual driver model, 2- time delay, and 3- arm NMS dynamics, each of which is explained separately in detail in the following subsections.

5.2.1 Modified visual driver model

Several studies have shown that during driving tasks drivers are guided along the road by focusing on two specific reference points known as the near point and the far point[86, 143]. Specifically, for performing lane-keeping and lane-change maneuvers, this model can be used effectively. The near point corresponds to the one on the target lane centerline at a short distance ahead of the ego vehicle, representing the driver's perception of the target lane midpoint during the lane-change maneuver. Additionally, the far point corresponds to the one on the rear part of the preceding vehicle in the target lane if there is any vehicle in the field of view of the ego vehicle driver. Thus, based on[85], the driver steering utilization is formulated as:

$$\dot{\delta}_{des} = k_{P,f}\dot{\theta}_f + k_{P,n}\dot{\theta}_n + k_{I,n}\theta_n \quad (5.1)$$

where θ_n is the near angle with respect to first vehicle unit yaw angle (ϕ_1), $\dot{\theta}_n$ is the time derivative of the near angle, and $\dot{\theta}_f$ is the time derivative of the far angle, $k_{P,f}$, $k_{P,n}$, and $k_{I,n}$ are the driver model gains, which will be determined using an optimization algorithm. These angles and their time derivatives are calculated by:

$$\theta_n = \arctan\left(\frac{\Delta Y_n}{\Delta X_n}\right) - \phi_1 \quad (5.2)$$

$$\theta_f = \arctan\left(\frac{\Delta Y_f}{\Delta X_f}\right) - \phi_1 \quad (5.3)$$

$$\dot{\theta}_n = \frac{\Delta v_{Y_n} \cdot \Delta X_n - \Delta v_{X_n} \cdot \Delta Y_n}{\Delta Y_n^2 + \Delta X_n^2} - \dot{\phi}_1 \quad (5.4)$$

$$\dot{\theta}_f = \frac{\Delta v_{Y_f} \cdot \Delta X_f - \Delta v_{X_f} \cdot \Delta Y_f}{\Delta Y_f^2 + \Delta X_f^2} - \dot{\phi}_1 \quad (5.5)$$

where $\Delta X_n, \Delta Y_n, \Delta X_f, \Delta Y_f, \Delta v_{X_n}, \Delta v_{Y_n}, \Delta v_{X_f},$ and Δv_{Y_f} denote the relative longitudinal and lateral distances and velocities between the ego vehicle driver and the near and far points, respectively. Note that all the above kinematic variables are defined/measured in the inertial coordinate frame. Variables ϕ_1 and $\dot{\phi}_1$ denote the yaw angle and yaw rate of the tractor (see Figure 5-2 and Figure 5-3).

Although the above Proportional Integral (PI) visual driver model is widely used in research, and its effectiveness, particularly in passenger vehicles, is proven[75, 85, 86, 87, 143], further study and efforts are needed for exploring the applicability to MTAHVs since these large vehicles may intensify lateral movements and instability during high-speed maneuvers. As mentioned in Section 2.4, one critical measure of high-speed dynamic performance for MTAHVs is RWA. When a tractor initiates a turn, the yaw damping of the MTAHV decreases with vehicle forward speed, leading to the RWA effect. The rear units (trailers) may experience more pronounced lateral displacement than the tractor itself, especially at higher speeds. RWA is defined as the ratio between the peak motion variable, such as yaw rate, experienced by the rearmost unit and that of the first unit during a high-speed maneuver. Thus, in this section a new driver model entitled MVDM is proposed for MTAHVs in high-speed highway driving in the following format:

$$\begin{aligned} \delta_{des}(t) = & k_{P,f}\dot{\theta}_f(t) + k_{P,n}\dot{\theta}_n(t) + k_{I,n}\theta_n(t) + k_{P,p}\left(\dot{\phi}_4(t) - \dot{\phi}_1(t - \tau)\right) \\ & + k_{I,p}\left(\phi_4(t) - \phi_1(t - \tau)\right) \end{aligned} \quad (5.6)$$

where the last two terms on the right-hand side are the proportional and integral correction terms based on the driver experience and knowledge of the vehicle. In sharp turns, these terms try to minimize the instantaneous and accumulative yaw error between first and last vehicle unit to bring the vehicle RWA close to 1 as much as possible. By using these two

terms, the last vehicle unit tries to imitate the dynamic behavior of the first unit. In another word, the driver attempts to modify the tire steering angle during sharp turns in such a way that the last vehicle unit follows a path as closely as possible to that of the first vehicle unit. The variables ϕ_4 and $\dot{\phi}_4$ denote the yaw angle and yaw rate of the last unit (see Figure 5-2 and Figure 5-3). In Equation (5.6), τ is the time delay between first and last unit. With the assumption of small tractor yaw angle (with respect XY coordinate) and articulation angles, τ in general format can be approximately calculated as the positive compatible root of the following equation:

$$\int_{t-\tau}^t \left(\int_{\eta-\tau}^{\eta} \bar{a}_X(\xi) d\xi + v_X(\eta - \tau) \right) d\eta - l = 0, \quad l = c_1 + a_2 + c_2 + a_3 + c_3 + a_4 \quad (5.7)$$

where $\bar{a}_X(t)$ and $v_X(t)$ are the vehicle acceleration and velocity in the X direction, respectively. For a very common case, constant acceleration $\bar{a}_X(t) = \bar{a} = cte$, time delay τ can be approximately calculated as the positive compatible root of the following equation:

$$\frac{1}{2} \bar{a} \tau^2 + \sqrt{v_X(t)^2 - 2\bar{a}l} \tau - l = 0, \quad l = c_1 + a_2 + c_2 + a_3 + c_3 + a_4 \quad (5.8)$$

Thus, τ can be summarized in this format:

$$\begin{cases} \tau = \frac{v_X(t) - \sqrt{v_X(t)^2 - 2\bar{a}l}}{\bar{a}}, & \bar{a}_X = \bar{a} = cte \neq 0 \\ \tau = \frac{l}{v_X}, & \bar{a}_X = 0 \end{cases} \quad (5.9)$$

The details of highway operation modes of the driver model, i.e., 1) highway lane-change maneuver, and 2) highway lane-keeping maneuver, are explored below, and the relevant variables associated with them are extracted.

5.2.1.1 Highway lane-change maneuver

A highway lane-change maneuver for MTAHVs is a critical scenario that evaluates safety, efficiency, and compliance with traffic regulations. This maneuver involves the controlled transition of the vehicle from one lane to another, typically to overtake slower vehicles, respond to road conditions, or follow navigation instructions. Figure 5-2 shows the schematic representation of a lane change maneuver. The MTAHV as an Ego Vehicle (EV) is changing its current lane to the target lane with a Leading Vehicle (LV) inside. As shown in the figure, D_n is the visual near distance, D_f is the visual far distance, W is half width of the lane, and Y is the ego vehicle lateral position in the global coordinate frame $X - Y$. Thus, in this scenario, the relative longitudinal and lateral distances and velocities can be calculated as:

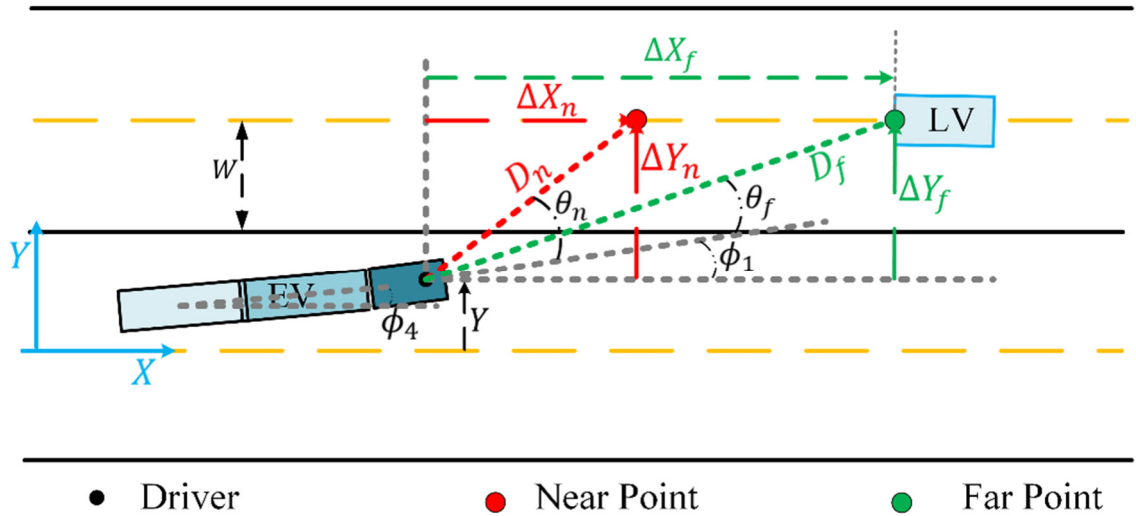


Figure 5-2: Visual variables utilized in the MVDM driver model for lane-change maneuvers.

$$\begin{aligned}
\Delta X_n &= \sqrt{D_n^2 - (2W - Y)^2}, & \Delta Y_n &= 2W - Y, & \Delta v_{X_n} &= \Delta \dot{X}_n, & \Delta v_{Y_n} &= \Delta \dot{Y}_n \\
\Delta X_f &= \sqrt{D_f^2 - (2W - Y)^2}, & \Delta Y_f &= 2W - Y, & \Delta v_{X_f} &= \Delta \dot{X}_f, & \Delta v_{Y_f} &= \Delta \dot{Y}_f
\end{aligned} \tag{5.10}$$

5.2.1.2 Highway lane-keeping maneuver

A Highway Lane-Keeping Maneuver for MTAHVs is a fundamental driver functionality in driving that results in maintain stability and accurate positioning within a designated lane while traveling on highways. Figure 5-3 shows the schematic representation of a lane-keeping maneuver. In the figure, R_r is the curve radius, and other variables are the same introduced in the Section 5.2.1.1. The relative longitudinal and lateral distances and velocities can be calculated as:

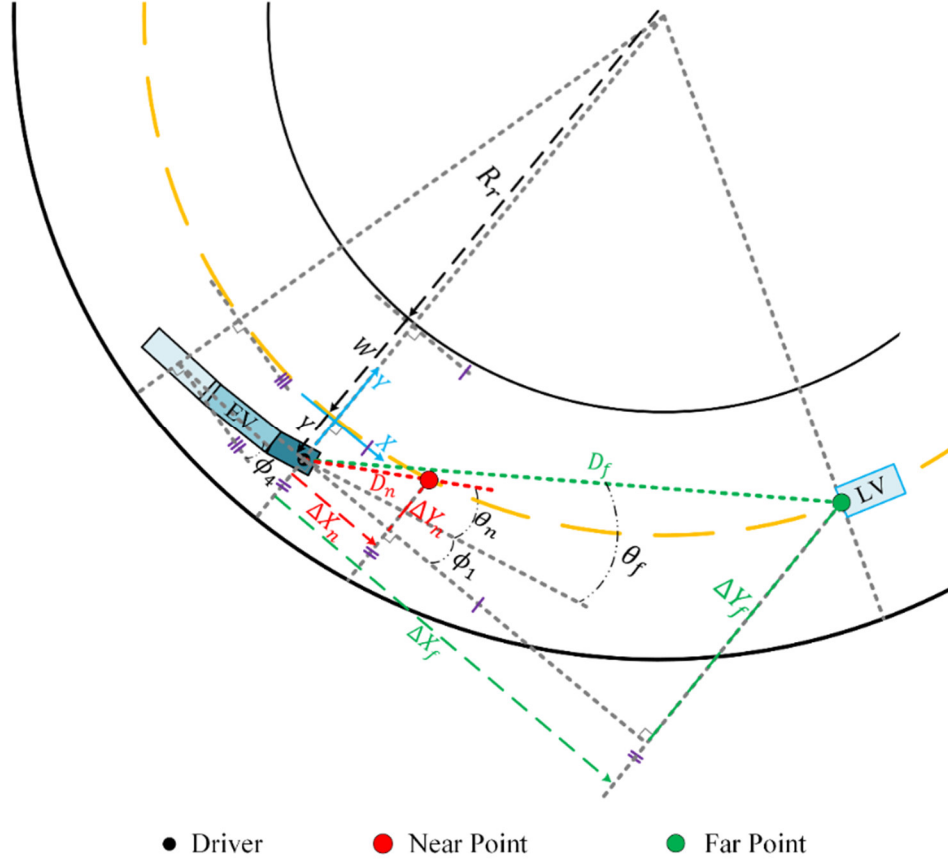


Figure 5-3: Visual variables utilized in the MVDM driver model for lane-keeping maneuvers.

$$\begin{aligned}
 \Delta X_n &= D_n \sqrt{1 - \left(\frac{Y^2 + D_n^2 - 2Y(W + R_r)}{2D_n(W + R_r - Y)} \right)^2}, \\
 \Delta Y_n &= (W + R_r) \left(1 - \sqrt{1 - \left(\frac{\Delta X_n}{W + R_r} \right)^2} \right) - Y, \quad \Delta v_{X_n} = \Delta \dot{X}_n, \quad \Delta v_{Y_n} = \Delta \dot{Y}_n \\
 \Delta X_f &= D_f \sqrt{1 - \left(\frac{Y^2 + D_f^2 - 2Y(W + R_r)}{2D_f(W + R_r - Y)} \right)^2}, \\
 \Delta Y_f &= (W + R_r) \left(1 - \sqrt{1 - \left(\frac{\Delta X_f}{W + R_r} \right)^2} \right) - Y, \quad \Delta v_{X_f} = \Delta \dot{X}_f, \quad \Delta v_{Y_f} = \Delta \dot{Y}_f
 \end{aligned} \tag{5.11}$$

5.2.1.3 Modified visual driver model gains optimization

Particle swarm optimization is a population-based optimization technique. In this study, this algorithm is used to find the optimal values of the control gains, $k_{P,f}$, $k_{P,n}$, $k_{I,n}$, $k_{P,p}$ and $k_{I,p}$, in Equation (5.6). In the PSO, each particle represents a potential solution and moves through the search space by updating its velocity and position based on its own experience and the experience of neighboring particles. These variables allow the evaluation of an individual's fitness function (cost function), which is defined as the Residual Sum of Squares (RSOS). This cooperative mechanism allows the PSO to efficiently explore and exploit the search space[144, 145]. In this chapter, two different reference paths are considered to find the optimal values for two different scenarios, highway lane-change and highway lane-keeping. The flow chart of the co-simulation algorithm (Matlab+TruckSim) for the PSO optimization is shown in Figure 5-4.

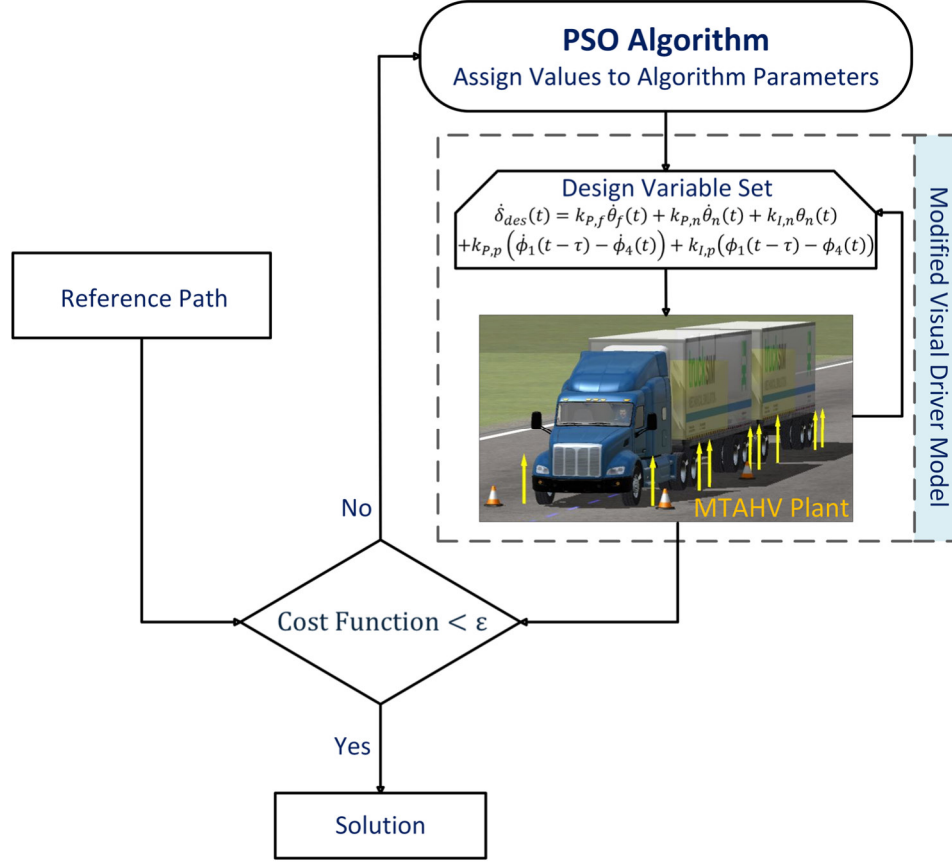


Figure 5-4: Co-simulation setup for the MVDM gains optimization

5.2.1.3.1 Highway single lane change maneuver with constant longitudinal acceleration

The SLC maneuver specified by ISO-14791 is selected as the testing maneuver, and the testing course is chosen as the reference path, which is defined by:

$$Y_{ref}(t) = \frac{L}{T} \left[t - \frac{T}{2\pi} \sin\left(\frac{2\pi t}{T}\right) \right] \quad (5.12)$$

$$X_{ref}(t) = v_{e,0}t + \frac{1}{2} \bar{a}t^2$$

where $Y_{ref}(t)$ and $X_{ref}(t)$ denote the lateral and longitudinal coordinates of the desired path at a given time t , L represents the maximum lateral displacement during the SLC

maneuver, T specifies the duration of the sine wave, $v_{e,0}$ shows the initial velocity, and $\bar{a}$ is the longitudinal constant acceleration. Parameters values are given in Table 5-1.

Table 5-1: The parameter value used for the highway single lane change maneuver[86, 129].

Parameter	Symbol	Value	Unit
Lane width	L	3.6	m
Sine wave duration	T	4	sec
Initial velocity	$v_{e,0}$	80 – 90	km/h
Horizontal acceleration	$\bar{a}$	1	m/s ²
LV Longitudinal velocity	v_l	80 – 100	km/h
Driver's Visual near point distance	D_n	10	m
Driver's visibility distance	D_v	100	m
Initial distance between EV and LV	$\Delta X_{f,0}$	40 – 100	m

The cost function (fitness function) used in this section and the next section is defined as the accumulative squared error between vehicle lateral position and lateral coordinate of the reference path based on the below equation:

$$J = \sum_i (Y_i - Y_{ref_i})^2 < \varepsilon \quad (5.13)$$

Thus, for the highway single lane-change maneuver with constant longitudinal acceleration, the optimal values for $k_{p,f}$, $k_{p,n}$, $k_{l,n}$, $k_{p,p}$ and $k_{l,p}$ are obtained as the output of the PSO algorithm (Table 5-2). The proposed method effectively mitigates the risk of overfitting by employing a stochastic optimization framework rather than relying on a single deterministic scenario. The reported control gains are not the result of a single run, but rather the ensemble average of 100 distinct simulations, each initialized with randomized values for $v_{e,0}$, v_l , and $\Delta X_{f,0}$. This approach forces the PSO to converge on a generalist solution that is robust across a wide operational envelope, rather than a specialist solution that exploits the quirks of one specific trajectory. The success of this strategy is quantitatively confirmed by the negligible standard deviation (0.013) observed across the

trials, demonstrating that the optimized gains remain stable and effective despite variations in the dynamic environment.

Table 5-2: Normalized Optimal values for MVDM in the highway single lane change maneuver.

$k_{p,f}$	$k_{p,n}$	$k_{l,n}$	$k_{p,p}$	$k_{l,p}$
0.19	0.54	0.91	0.58	1.00

To evaluate the performance of the MVDM in more detail, it is compared with the built-in driver model in TruckSim, i.e., the optimal preview driver model by McAdam[84] and the visual driver model reported in[86]. The McAdam driver model is an optimal preview control-based approach for linear systems. It uses preview information about the road ahead to make control decisions in an optimal way. The driver model minimizes a cost function that is typically a deviation from the desired path. The visual driver model is based on the Equation (5.1). Figure 5-5-a illustrates the paths of the tractor for the different driver models as well as the reference path. Although the performance of the three driver models is acceptable, the MVDM outperforms the other two models. Regarding HSTO, there is no overshoot for the MVDM and the deviation from the reference path is less than the others. Figure 5-5-b shows the path of the second semi-trailer (the rearmost vehicle unit) for the three different driver models. Although the MVDM has a small amount of undershoot at the beginning of the lane-change maneuver (8.6 cm), which is in the acceptable range, it can successfully and significantly reduce the overshoot at the end compared to the other two methods (30.02% with respect to the visual driver model and 59.03% with respect to McAdam model). This improvement is due to considering lateral motion correction terms in the driver model, thereby limiting the steering angle.

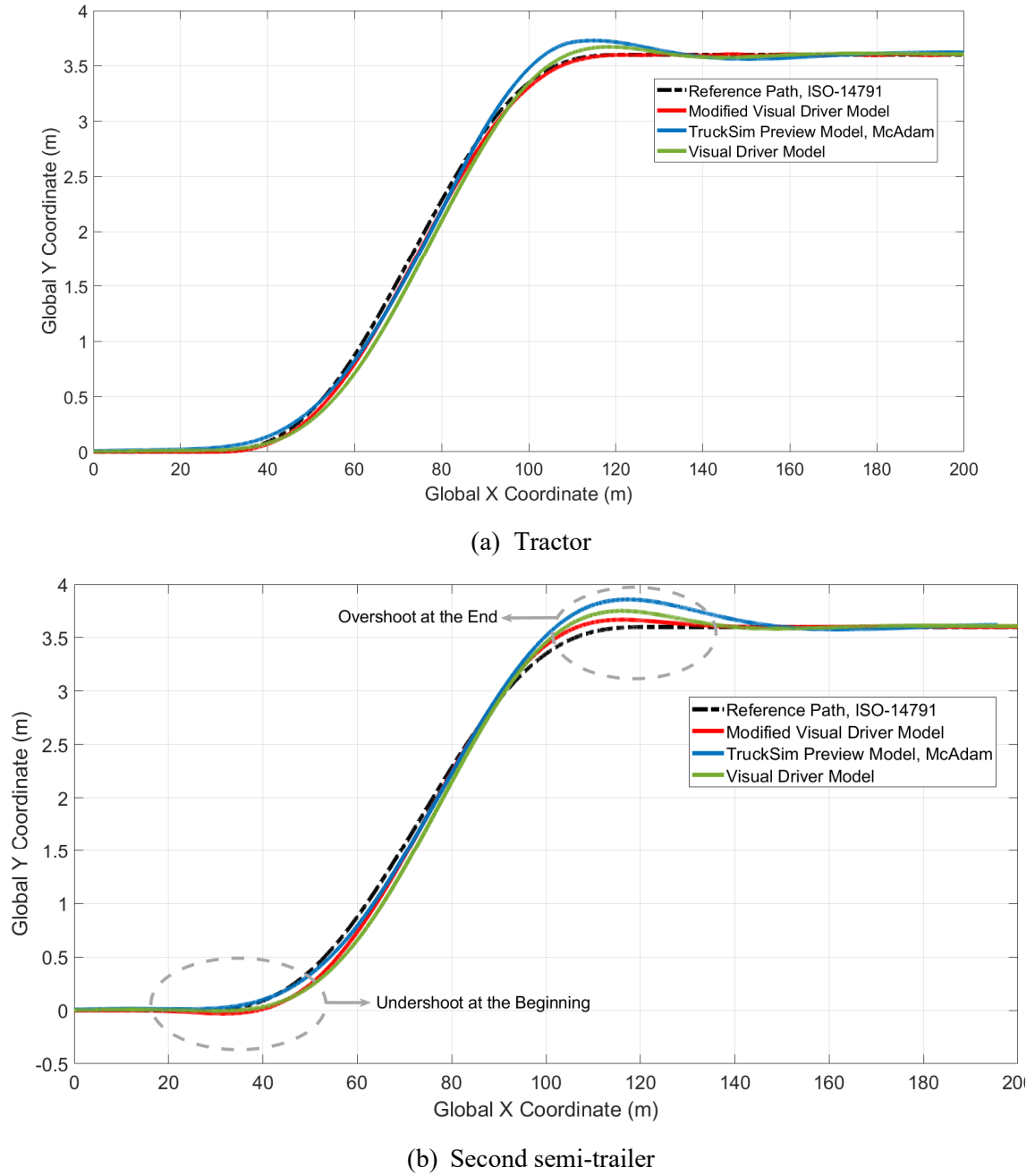


Figure 5-5: MTAHV single lane change path for different driver models (MVDM, McAdam, and visual driver model)

Figure 5-6 illustrates the lateral acceleration of the second semi-trailer, which experiences the worst condition in comparison to the other units. The MVDM surpasses the other two driver models, and the MVDM model exhibits a lower and more uniform

lateral acceleration, resulting to a reduced risk of rollover instability. It should be noted that by using the MVDM model, the maximum lateral acceleration is 0.15g, which reduces by 17.14% and 22.33% with respect to the McAdam and the visual driver model, respectively.

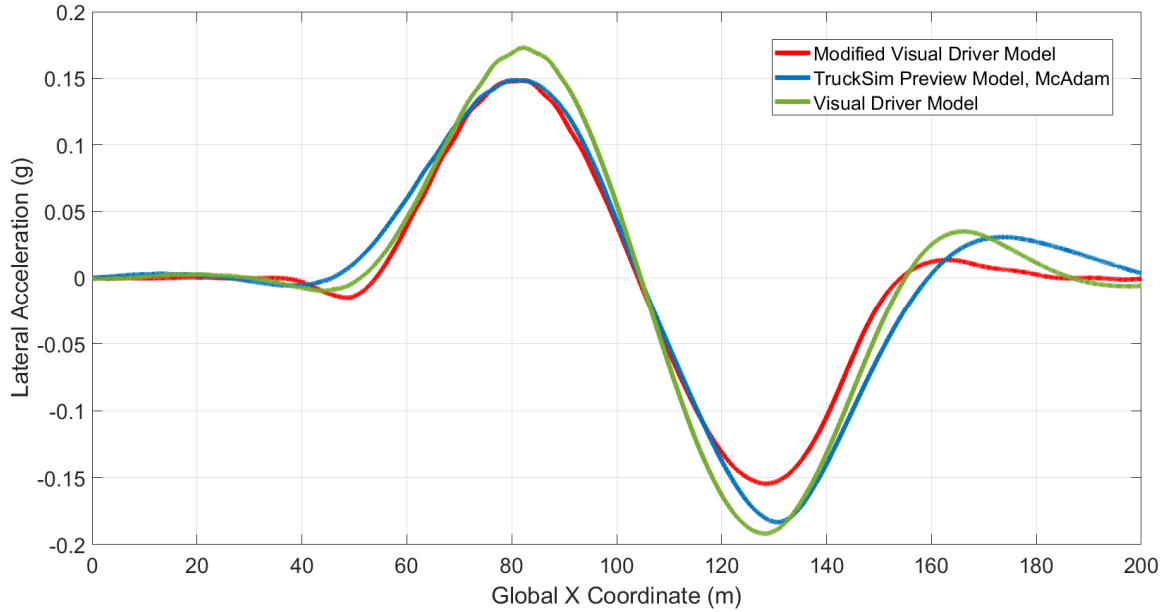


Figure 5-6: Lateral acceleration for the second semi-trailer

5.2.1.3.2 Highway lane-keeping maneuver

To evaluate the performance of the MVDM model in a highway-lane keeping maneuver, a curvy path has been considered in the TruckSim. According to[146], the permitted maximum lateral jerk for a long truck is 0.3 m/s^3 . Thus, based on[147], if the longitudinal velocity is less than or equal to 110 km/h, the permitted minimum curve radius is 615 m. Therefore, a 5-segment curvy path has been considered based on Table 5-3.

Table 5-3: 5-segment curvy path parameters[147].

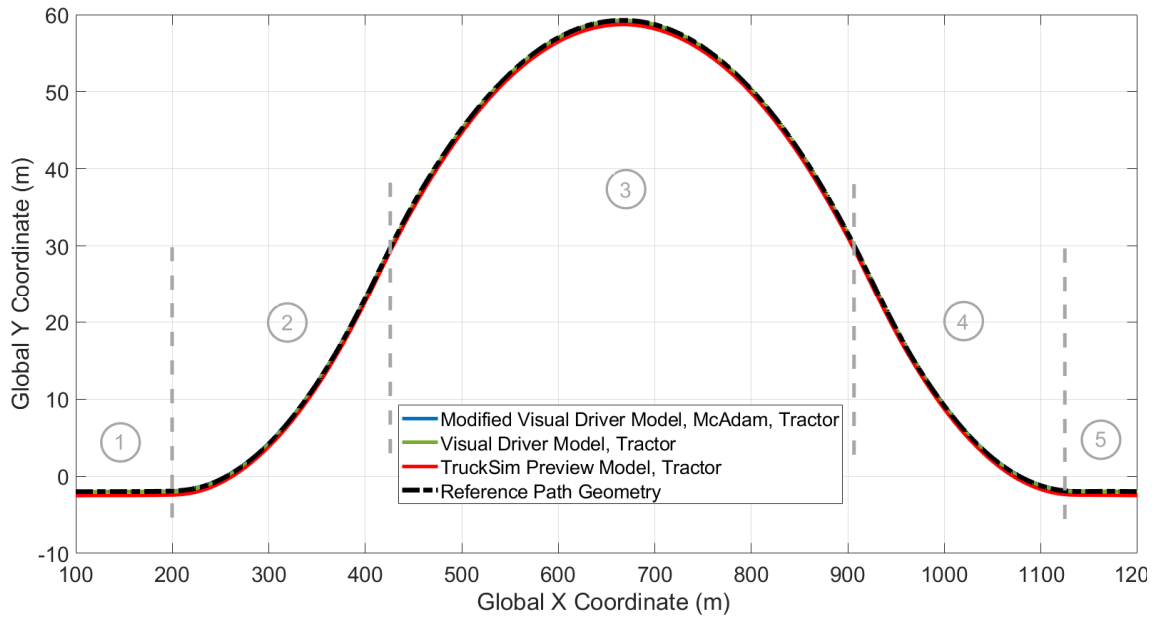
Section No.	Direction	Curve Radius	Unit
1	Ahead		
2	Left	615	m
3	Right	750	m
4	Left	615	m
5	Ahead		

For the highway lane-keeping maneuver with constant velocity (100 km/h), the optimal values for $k_{p,f}$, $k_{p,n}$, $k_{l,n}$, $k_{p,p}$ and $k_{l,p}$ are attained using the PSO algorithm, and these optimal values are listed in Table 5-4. The cost function is defined exactly same as Equation (5.13). These results were derived by averaging the outcomes of 100 simulations with random values for $v_{e,0}$, v_l , and $\Delta X_{f,0}$. The highest standard deviation among the five optimized parameters is 0.009, indicating that the relative speed between the ego and leading vehicles has very little effect on the results.

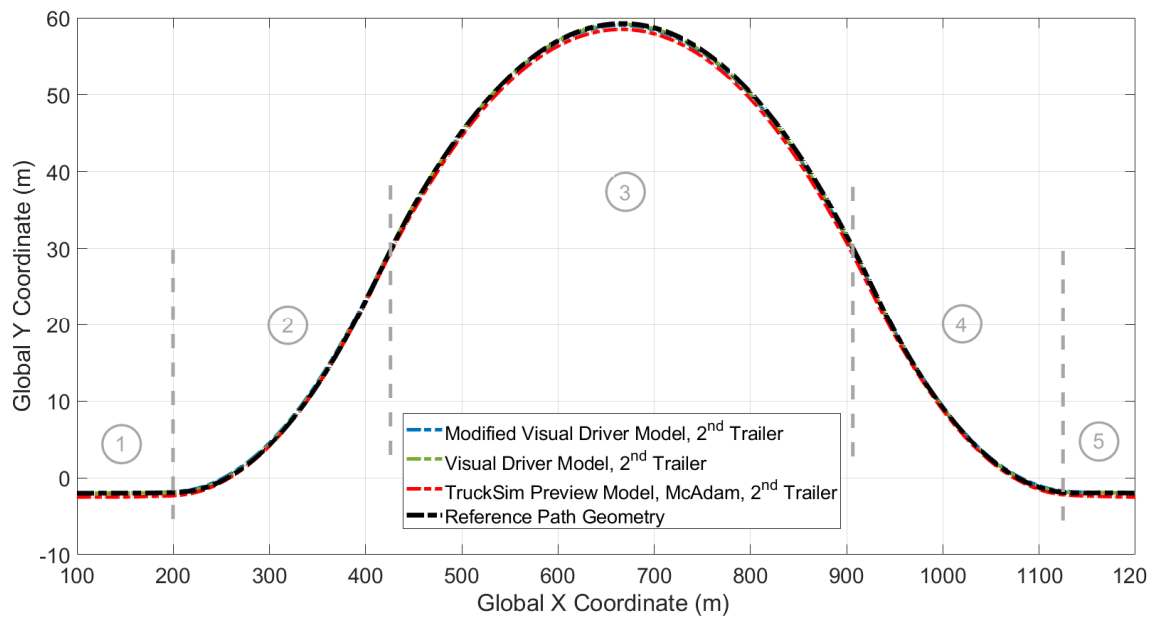
Table 5-4: Normalized optimal values for MVDM model for the highway lane keeping maneuver.

$k_{p,f}$	$k_{p,n}$	$k_{l,n}$	$k_{p,p}$	$k_{l,p}$
0.55	0.79	0.90	0.71	1.00

Figure 5-7 shows the performance of the three driver models in the MTAHV highway lane-keeping maneuver for both tractor and second semi-trailer with the constant longitudinal speed of 100 km/h. In this maneuver, all three models have a similar performance, and they can follow the lane centerline successfully. Interestingly, as shown in Figure 5-8, in the case of the MVDM model, the tractor front wheel steering angle changes more smoothly during turns, which in turn reduces the lateral acceleration exerted on the MTAHV.



(a) Tractor



(b) Second semi-trailer

Figure 5-7: MTAHV lane keeping path for the three driver models (MVDM, McAdam, and visual driver model)

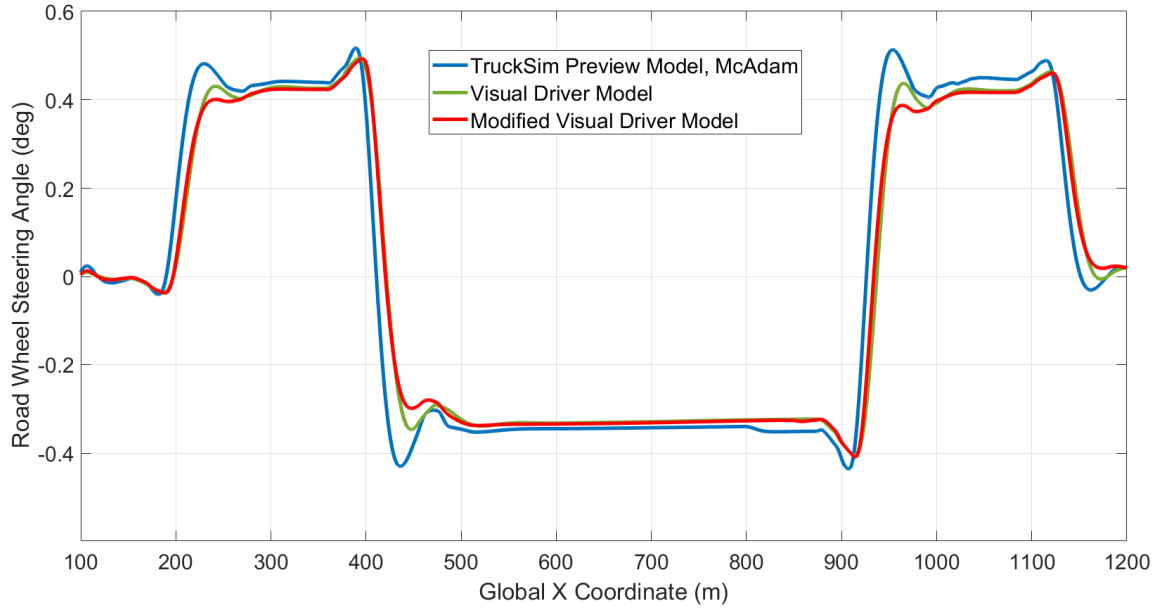


Figure 5-8: MTAHV road wheel steering angle for the three driver models (MVDM, McAdam, and visual driver model)

5.2.2 MVDM with time delay and arm NMS dynamics (Integrated MVDM)

As explained in Section 5.2.1, the MVDM driver model output angle is subsequently processed through the time delay block (simulating the delay in human steering response) and the arm NMS dynamics block (accounting for the effects of arm NMS dynamics). In this part these two blocks are added to the driver model to represent a more realistic driver model.

5.2.2.1 Time delay

There are two sources of delay, which have been considered in driver modeling, i.e., cognitive delay and actuation delay. These two sources are combined into a single unit of delay in the following format[111, 148]:

$$\begin{Bmatrix} \delta_{des}^{-1}(k+1) \\ \delta_{des}^{-2}(k+1) \\ \delta_{des}^{-3}(k+1) \\ \vdots \\ \delta_{des}^{-N_d}(k+1) \end{Bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & 0 \\ 0 & 0 & \dots & 1 & 0 \end{bmatrix} \begin{Bmatrix} \delta_{des}^{-1}(k) \\ \delta_{des}^{-2}(k) \\ \delta_{des}^{-3}(k) \\ \vdots \\ \delta_{des}^{-N_d}(k) \end{Bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \delta_{des}(k) \quad (5.14)$$

$$\delta_{des}^{-N_d}(k) = [0 \quad 0 \quad 0 \quad \dots \quad 1] \begin{Bmatrix} \delta_{des}^{-1}(k) \\ \delta_{des}^{-2}(k) \\ \delta_{des}^{-3}(k) \\ \vdots \\ \delta_{des}^{-N_d}(k) \end{Bmatrix}$$

where $\delta_{des}(k)$ is the neutral command, which is originated in the driver brain at the time step k which is modeled as the input of the above linear delay system. Additionally, $\delta_{des}^{-N_d}(k)$ as the output of the delay system denotes the N_d time-step delay to the neutral command, $\delta_{des}(k)$.

5.2.2.2 Arm NMS dynamics

The driver's arm NMS dynamics can be modeled as a second-order low-pass filter[149]. It can be written in the continuous time state-space as the following, but for the simulation it should be converted to the discrete time form.

$$\begin{Bmatrix} \dot{\delta}_{driver}^{NMS}(t) \\ \ddot{\delta}_{driver}^{NMS}(t) \end{Bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} \begin{Bmatrix} \delta_{driver}^{NMS}(t) \\ \dot{\delta}_{driver}^{NMS}(t) \end{Bmatrix} + \begin{bmatrix} 0 \\ \omega_n^2 \end{bmatrix} \delta_{des}^{-N_d}(t) \quad (5.15)$$

$$\delta_{driver}^{NMS}(t) = [1 \quad 0] \begin{Bmatrix} \delta_{driver}^{NMS} \\ \dot{\delta}_{driver}^{NMS} \end{Bmatrix}$$

Here, $\delta_{driver}^{NMS}(t)$ represents the driver's steering control action applied to the handwheel, responding to the delayed steering angle $\delta_{des}^{-N_d}(t)$. The neuromuscular dynamics of the arm are defined by the natural frequency ω_n and the damping ratio ζ .

5.2.3 Integrated MVDM evaluation

For evaluating the performance of the proposed driver model with the on-road measurement data, information of two single lane-change scenarios is adopted from[86]. The simulation variables values are given in the Table 5-5. It should be mentioned that for the simulation, the proposed driver model has been combined with a longitudinal driver model, adapted form that reported in[86], which is able to control the MTAHV velocity in the longitudinal direction. In the first testing scenario, the MTAHV changes its lane to the left with a constant forward-speed of 80 km/h and the initial near distance of 80 m. In the second testing scenario, the MTAHV changes its lane to the right with a constant forward-speed of 80 km/h and the initial near distance of 40 m.

Table 5-5: Simulation variable values[86, 111].

Parameter	Symbol	Value	Unit
MTAHV Longitudinal velocity	v_e	80	km/h
Driver's visibility distance	Dv	100	m
Driver's Visual near point distance	D_n	10	m
Sampling Time	Δt	0.01	sec
Number of time-step delay	N_d	10	
Natural frequency	ω_n	$1.5 \times 2\pi$	rad/s
Damping ratio	ζ	0.6	
Left vehicle velocity	$v_{l,l}$	80	km/s
Right vehicle velocity	$v_{l,r}$	80	km/s
Initial closing distance, left vehicle	$x_{l,l}$	80	m
Initial closing distance, right vehicle	$x_{l,r}$	40	m

The coefficient of determination (R^2) is defined to measure how well the output of the proposed driver model fits the on-road measurement data, and the coefficient is calculated by

$$R^2 = 1 - \frac{RSOS}{TSOS} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (5.16)$$

where y_i is measured data points (on-road measurements), $\hat{y}_i$ is predicted values from the proposed driver model (Integrated MVDM), and $\bar{y}$ is mean of the measured data. In other words, RSOS represents the residual sum of squares, the unexplained variance by the model, and TSOS is the total sum of squares, the total variance in the measured data. The result shows an acceptable improvement for the proposed driver model in fitting the on-road measurement data compared to the driver model presented in[86] for a MTAHV (Table 5-6).

Table 5-6: Comparison of R^2 for visual driver model[86] and Integrated MVDM.

On-road maneuver	Coefficient of Determination (R^2)	
	Visual driver model [86]	Integrated MVDM
Single lane change 1	0.87	0.93
Single lane change 2	0.56	0.71

5.3 Summary

This chapter proposes and develops an Integrated MVDM specifically designed for MTAHVs. The model mimics human driving behavior by incorporating a two-point visual perception strategy: a "far point" to anticipate upcoming road curvature and a "near point" to maintain lane position. Furthermore, to enhance physical realism, the framework integrates driver reaction time delays and arm Neuromuscular System dynamics. The control parameters were fine-tuned using PSO within a Matlab and TruckSim co-simulation environment, with the specific objective of manipulating the steering input so that the rearmost trailer accurately tracks the tractor's path.

Performance evaluations conducted under highway lane-changing and lane-keeping scenarios demonstrate the superiority of this approach. The trained MVDM

outperformed established benchmarks, such as the McAdam preview model and standard visual driver models, yielding lower lateral path deviation and acceleration. Crucially, the inclusion of NMS dynamics and time delays significantly improved the model's fidelity; validation against on-road measurement data confirmed that the proposed model aligns more closely with real-world driver behavior than previous iterations found in the literature. While the results are promising, it is noted that further validation through Software-in-the-Loop (SIL) and Hardware-in-the-Loop (HIL) testing is recommended to fully verify the model's robustness.

Despite these successes, the scalability of the Integrated MVDM is currently constrained by its specialized design for structured highway environments. The model operates within a defined operational envelope characterized by high longitudinal velocities and small steering angles, utilizing control gains that are optimized specifically for stability and comfort in these conditions. Consequently, the framework is not directly generalizable to diverse driving tasks such as urban navigation or emergency collision avoidance. Future expansion to these complex domains would necessitate a shift from the current fixed-gain approach to an adaptive control architecture capable of handling the nonlinearities and high-frequency inputs inherent in unstructured driving scenarios.

Chapter 6

Deep-reinforcement-learning based decision making for MTAHVs

6.1 Introduction

In this chapter, the deficiencies mentioned in Section 2.9 will be addressed in the following sections. Built upon the previous research, a Deep Reinforcement Learning (DRL) platform has been modified to employ Double Deep Q-Network (DDQN) as well as DDQN + Prioritized Experience Replay (DDQN+PER) for learning and decision-making for two different agents (autonomous controller and driver model). This platform is applicable to a dynamic environment where the ego vehicle (MTAHV) and the surrounding vehicles (regular passenger cars) are interacting with each other. Additionally, a fine-tuned MPC-based tracking-controller, which is built upon the one reported in[129], and the Integrated MVDM driver model generated in Chapter 5 are used to apply the relevant actions. Finally, to evaluate the DDQN+PER and controller/driver model performance together, the TruckSim model is used as the MTAHV plant. Figure 6-1 shows the proposed co-simulation framework for emulating a MTAHV decision making in Python, the autonomous controller/driver model in Python as well as the MTAHV model in TruckSim. The DRL technique is adapted to be the decision-maker to provide optimal actions for either the controller agent or the driver model agent. To explore the DRL

technique, co-simulations are conducted to examine the dynamic behavior of the MTAHV under the different highway operation, which is mimicked by SUMO, an open-source highway traffic simulation package. To create a simulated environment in SUMO, three lanes and N_{veh} randomly distributed surrounding vehicles[150] have been considered. Note that SUMO is an open source, portable, microscopic and continuous multi-modal traffic simulation package devised to handle large networks.

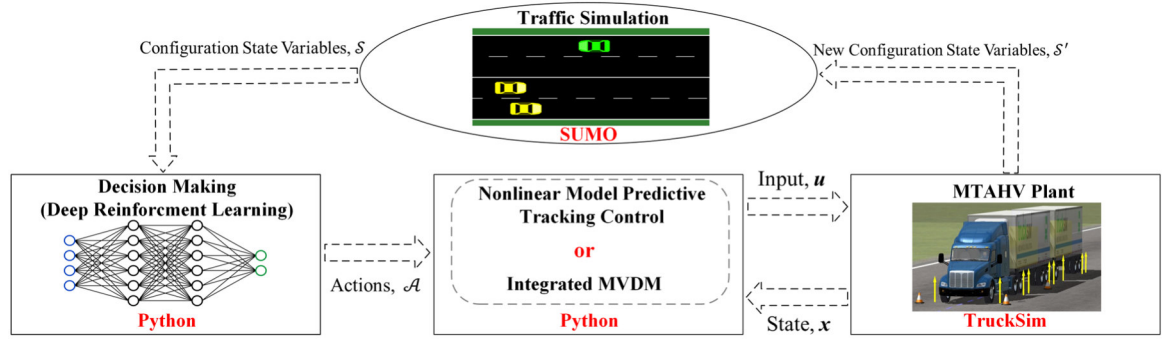


Figure 6-1: The proposed co-simulation framework for emulating autonomous/conventional MTAHVs in highway operations.

6.2 Reinforcement learning

Reinforcement Learning (RL) is a technique, which may combine machine learning and optimal control to determine how an intelligent agent can make decisions in a dynamic environment to maximize a given reward signal. RL essentially considers the relationship between an agent, environment, and goal. The RL agent learns from a problem by interacting with its environment. The environment provides information on its current state. The agent then uses the acquired information to determine which actions(s) to take. If the action obtains a reward signal from the surrounding environment, the agent is encouraged to take the action again while being in a similar future state. This process repeats for every new state thereafter. Over time, the agent learns from rewards and punishments to take

actions within the environment that meet a specified goal as shown in Figure 6-2. In the literature, it is common practice to formulate this relationship in terms of the Markov decision process (MDP).

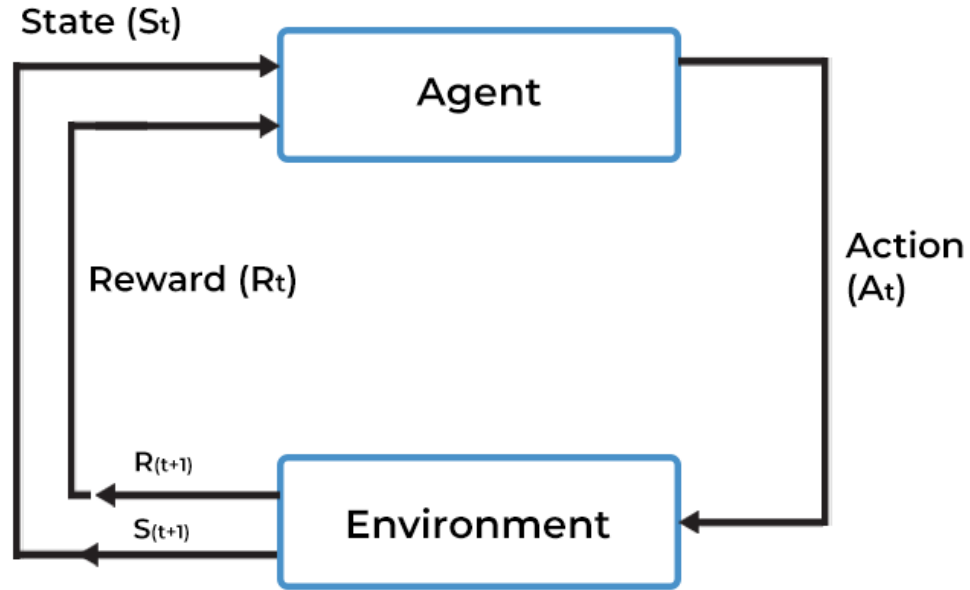


Figure 6-2: Reinforcement learning model[151].

6.3 Decision maker based on Markov decision process

In a general case, the RL problem can be formulated as a Markov Decision Process (MDP), which is defined by $\langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \mathcal{H}, \gamma \rangle$, where $\mathcal{S} \in \mathbb{R}^n$ is the configuration state space, $\mathcal{A}$ is the action space, $\mathcal{T}(s'|s, a): \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ is the configuration state transition model, $\mathcal{R}(s, a, s'): \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ is the user defined reward function, $\mathcal{H}$ is the horizon, and γ is the discount factor. In RL training phase, the agent (e.g., the MTAHV in this chapter) interacts with the environment (all surrounding vehicles on the highway) to learn an optimal policy $\pi(a|s)$. The goal of the agent is to choose an appropriate action (a) at any timestep to maximize the cumulative discounted reward[152]. Thus, the cumulative

reward should be represented in a formal way to use it in the calculations, which is called the *expected return*, and can be represented as:

$$G_t = \mathcal{R}_{t+1} + \gamma \mathcal{R}_{t+2} + \gamma^2 \mathcal{R}_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k \mathcal{R}_{t+k+1} \quad (6.1)$$

6.3.1 Policy

Policy in RL decides which action the agent will take in the current state. In other words, it tells the probability that an agent will select a specific action from a specific state. Policy is a function that maps a given state to probabilities of selecting each possible action from the given state. If at time t , an agent follows policy π , then $\pi(a|s)$ becomes the probability that the action at time step t is $\mathcal{A}_t = a$ if the state at time step t is $\mathcal{S}_t = s$. The meaning of this is the probability that an agent will take an action a in state s is $\pi(a|s)$ at time t with policy π .

6.3.2 Value function

The value functions are a simple measure of how good it is for an agent to be in a given state, or how good it is for the agent to perform a given action in a given state. There are two types of value functions:

- 1) **State-value function:** The state-value function for policy π denoted as $v_\pi(s)$ determines the goodness of any given state for an agent who is following policy π . This function gives us the value which is the expected return starting from state s at time step t and following policy π afterward.

$$v_\pi(s) = \mathbb{E}_s[G_t | s_t = s, \pi] = \mathbb{E}_s \left[\sum_{k=0}^{\infty} \gamma^k \mathcal{R}_{t+k+1} | s_t = s, \pi \right] \quad (6.2)$$

- 2) **Action-value function:** The action-value function determines the goodness of the action taken by the agent from a given state for policy π . This function gives the value, which is the expected return starting from state s at time step t , with action a , and following policy π afterward. The output of this function is also called as Q-value, where Q stands for quality.

$$Q_\pi(s, a) = \mathbb{E}_s[G_t | s_t = s, a_t = a, \pi] = \mathbb{E}_s \left[\sum_{k=0}^{\infty} \gamma^k \mathcal{R}_{t+k+1} | s_t = s, a_t = a, \pi \right] \quad (6.3)$$

6.3.3 Optimal policy

The optimal policy should have an associated optimal state-value function. The optimal state-value function $v^*(s)$ yields the largest expected return feasible for each of the state s with any policy π .

$$v^*(s) = \max_{\pi} v_\pi(s) \quad (6.4)$$

In addition, the optimal policy decides which action the agent will take in the current state. In a discrete action space, the optimal policy is derived by solving the optimal action-value function. In Q-learning[153], the agent attempts to learn the optimal action-

value function $Q^*_\pi(s, a)$, which is the largest expected return feasible for each state-action pair (s, a) with any policy π .

$$Q^*(s, a) = \max_{\pi} Q_{\pi}(s, a) = \max_{\pi} \mathbb{E}_s[G_t | s_t = s, a_t = a, \pi] \quad (6.5)$$

6.3.4 Bellman optimality equation

The optimal action-value function Q^* satisfies Equation (6.6). This is a fundamental property of the optimal action-value function. So, regarding the next state, an optimal Q^* function adheres to the Bellman equation:

$$Q^*_\pi(s, a) = \mathbb{E}_{s'} \left[r + \gamma \max_{a'} Q^*(s', a') | (s, a) \right] \quad (6.6)$$

6.3.5 RL formulation

As mentioned in Section 6.3, a RL problem can be modeled as an MDP. Hence, different parameters for the MDP have been defined for different agents (autonomous controller or driver model). It should be noted that where the formulations/configurations differ for two agents, both formulations/configurations are presented consecutively in the following subsections.

6.3.5.1 Configuration State Space, $\mathcal{S}$

- *Autonomous controller as an agent*

The agent is presumed to receive traffic data within x_{sensor} range, which encompasses the longitudinal position and velocity for the ego vehicle (index 0) as well as

all surrounding vehicles (index $1, \dots, N_{veh}$) in the sensor range. The longitudinal position and velocity for the ego vehicle $(x_0, v_{x,0})$ is the absolute value for the COG of the first unit (tractor), and for the others $(\{x_i, v_{x,i}\}_{i \in 1, \dots, N_{veh}})$ are the relative values with respect to the ego vehicle, as shown in Figure 6-3. The lane index of the vehicles $\{y_i\}_{i \in 0, 1, \dots, N_{veh}}$ is identified as 1, 2, or 3. The configuration state space can be expressed as:

$$\mathcal{S} = \left(\{x_i, y_i, v_{x,i}\}_{i \in 0, 1, \dots, N_{veh}} \right) \quad (6.7)$$



Figure 6-3: Typical configuration state variables, i.e., longitudinal and lateral positions.

- ***Driver model as an agent***

In real-world traffic conditions, a driver is unable to observe or process all the information regarding every vehicle on the road. Instead, they rely on perceiving and using information from vehicles within a certain range around them. The driver model (agent) is expected to receive traffic data within a defined range d_v , which is the maximum visibility distance. This data comprises longitudinal position and velocity for the ego vehicle and each of surrounding vehicles if they are within the driver's maximum visibility distance. The position and velocity of the ego vehicle are expressed as absolute values corresponding to the CG of its first unit (tractor), while for surrounding vehicles, these measurements are taken with respect to the ego vehicle. A driver is generally unable to precisely measure

his/her distance from other vehicles. Instead, they make approximate assessments and describe distances in relative terms as “*Close*”, “*Nominal*”, as well as “*Far*” and velocities in relative terms as “*Approaching*”, “*Stable*”, as well as “*Moving away*”[154]. As a result, the observation space for the driver model is defined as:

$$\mathcal{S} = \left(\begin{array}{c} x_0, y_0, v_{x,0}, \\ \{x_i\}_{i \in 1,2,3,4,5} \in \{Close, Nominal, Far\}, \\ \{y_i\}_{i \in 1,2,3,4,5} \\ \{v_{x,i}\}_{i \in 1,2,3,4,5} \in \{Approaching, Stable, Moving away\} \end{array} \right) \quad (6.8)$$

where x_0 and $v_{x,0}$ are the absolute values for longitudinal position and velocity of the ego vehicle, respectively, x_i and $v_{x,i}$ are the relative longitudinal position and velocity for the surrounding i^{th} vehicle respectively, and the lane index of the vehicles $\{y_i\}_{i \in 0,1,2,3,4,5}$ is identified as 1, 2, or 3. The configuration states can be assigned to the values in the following format:

- 1) Distance between the ego vehicle and the vehicle directly ahead is categorized as *Close* (when the distance $\leq d_c$), *Nominal* (when the $d_c \leq \text{distance} \leq d_f$), or *Far* (when the distance $\geq d_f$).
- 2) Distance to the vehicle in the front left lane is classified as *Close*, *Nominal*, or *Far*.
- 3) Distance to the vehicle in the front right lane is classified as *Close*, *Nominal*, or *Far*.
- 4) Distance to the vehicle in the rear left lane is classified as *Close*, *Nominal*, or *Far*.
- 5) Distance to the vehicle in the rear right lane is classified as *Close*, *Nominal*, or *Far*.
- 6) Distance rate of change to the vehicle directly ahead is categorized as *Approaching* (distance decreasing), *Stable* (distance remaining constant), or *Moving away* (distance increasing).

- 7) Distance rate of change to the vehicle in the front left lane is classified as *Approaching*, *Stable*, or *Moving away*.
- 8) Distance rate of change to the vehicle in the front right lane is classified as *Approaching*, *Stable*, or *Moving away*.
- 9) Distance rate of change to the vehicle in the rear left lane is classified as *Approaching*, *Stable*, or *Moving away*.
- 10) Distance rate of change to the vehicle in the rear right lane is classified as *Approaching*, *Stable*, or *Moving away*.

6.3.5.2 Action space, $\mathcal{A}$

- *Autonomous controller as an agent*

Four possible actions in the longitudinal direction are considered: 1) maintaining speed ($a_x = 0 \frac{m}{sec^2}$), 2) accelerating ($a_x = a_1 \frac{m}{sec^2}$), 3) decelerating ($a_x = -a_1 \frac{m}{sec^2}$), and 4) hard decelerating ($a_x = -a_2 \frac{m}{sec^2}$). In the lateral direction, three options are considered: 1) staying in the current lane (*CL*), 2) changing to the right lane (*RL*), and 3) changing to the left lane (*LL*). Thus, in total, the action space includes 10 elements, which are featured with the following patterns:

$$\begin{aligned}
 \mathcal{A} = & (\{a_x = 0, CL\}, \{a_x = 0, RL\}, \{a_x = 0, LL\}, \{a_x = a_1, CL\}, \{a_x = a_1, RL\}, \{a_x \\
 & = a_1, LL\}, \{a_x = -a_1, CL\}, \{a_x = -a_1, RL\}, \{a_x = -a_1, LL\}, \{a_x \\
 & = -a_2, CL\})
 \end{aligned} \tag{6.9}$$

- ***Driver model as an agent***

As introduced above, there exist four possible actions in the longitudinal direction and three possible actions in the lateral direction. The action space is expressed in the same as that described in Equation (6.9).

6.3.5.3 Configuration state transition model, $\mathcal{J}$

The generative simulation model implicitly determines the configuration state transition model. Additionally, because the MTAHVs feature complicated and nonlinear dynamics, the lateral and longitudinal dynamics for the ego vehicle during its maneuvers are considered[129]. In other words, instead of considering very simple models, such as a point mass model[155], a validated nonlinear vehicle model has been considered to reduce error sources. The model has been introduced in Chapter 3, and it is identical for the design of both agents (autonomous controller and driver model).

6.3.5.4 Reward Function, $\mathcal{R}$

For both agents, the main goal is to drive the ego vehicle safely and efficiently under various operating conditions. Thus, at each sampling time, the agent receives the rewards: 1) $\mathcal{R} = -15$, if the ego vehicle collides with other vehicles or drives off the road, and the episode will be terminated; 2) $\mathcal{R} = 1 - \frac{v_{max} - v_{EV}}{v_{max}}$, which is a positive reward to encourage the ego vehicle to drive at a possible speed approaching its maximum speed; 3) $\mathcal{R} = -1$, which is a negative reward to discourage the ego vehicle to change its lane or brake hard; 4) $\mathcal{R} = -10$, when time-to-collision (TTC) is less than 7 sec, but the episode will not be terminated.

6.4 Lower-level model

The lower-level model (i.e., tracking controller or driver model) takes the actions, which are the output of the upper-level model (i.e., decision maker) to navigate the MTAHV to the intended lateral position and at desired longitudinal velocity.

6.4.1 Lateral and longitudinal tracking controllers

The primary goal of the motion-trajectory tracking model is to calculate the steering angle for the tractor's front wheels and the propulsion torque for the driving wheels. This allows the vehicle to precisely take actions to follow the designated motion trajectory in terms of the desired geometric path and the requested forward speed profile[63, 124].

6.4.1.1 *NLMPC autonomous controller*

The MLMPC autonomous controller has been designed and introduced in Chapter 4. To perform the actions, the predefined path based on ISO 14791 is considered for the lateral direction control, along with a designated speed profile with a constant longitudinal acceleration based on the Equations (4.1) and (4.2). The NLMPC controller design is formulated as an optimization problem, with the objective function defined and expressed in Equation (4.10).

6.4.1.2 *Integrated MVDM driver model*

The human driver is represented by the Integrated MVDM driver model, which is designed and introduced in Section 5.2. This model considers driver time delay and arm neuromuscular system dynamics to enhance the control of MTAHVs. Utilizing visual

perception, the model integrates traffic information for near and far roadway sections; the information for far roadway section is used for motion trajectory planning, while the information for near roadway section is utilized for decision making to take an action of either lane-keeping or lane-changing. Additionally, the driver model strategically adjusts the steering wheel to make the trailers and the second trailer, in particular, closely follow the tractor's path, optimizing vehicle stability and path-following performance. In terms of forward speed control, this driver model leverages the built-in power-train model in TruckSim software to calculate the required driving torque.

6.4.2 Ego vehicle

A MTAHV with an A-train double configuration is selected as the ego vehicle. The subjective vehicle and corresponding model have been introduced in Section 3.2.

6.4.3 Surrounding vehicles

Each surrounding vehicle is represented by a computationally efficient point-mass model, where the longitudinal motion equations are governed by a discrete-time double integrator.

$$\begin{aligned} X(t+1) &= X(t) + v_X(t) \Delta T_s \\ v_X(t+1) &= v_X(t) + a_X(t) \Delta T_s \end{aligned} \tag{6.10}$$

where ΔT_s refers to the sampling time. Considering lateral motion, we assume a simple kinematic model,

$$Y(t + 1) = Y(t) + v_Y(t) \Delta T_s \quad (6.11)$$

It should be mentioned that the surrounding vehicles are operated using the standard SUMO driver model, which incorporates an adaptive cruise control system for speed management[156] and a lane change model for overtaking slower vehicles[157].

6.5 Implementation

In this section an implementation setup is designed for both controller and driver model decision makers. Thus, a one-way highway driving has been used to train and test the presented DDQN+PER algorithm for a MTHAV as the ego vehicle. A one-way highway with three lanes is selected for the SUMO, which is an open source, microscopic and continuous multi-modal traffic simulation package designed to handle large networks. The inertial and geometric parameters of the EV are listed in the Table 4-1. The maximum speed for the EV is $v_{max} = 25.00$ m/sec. At the beginning of each simulation episode, 20 passenger cars are incorporated per 100 meters of a 3-lane highway, reflecting a moderately dense traffic condition[153, 158], each assigned a random speed within a specified speed range, $[v_{min}, v_{max}] = [15, 35]$ m/sec. Each episode is concluded either after 1000-time steps or sooner if a collision occurs or the ego vehicle goes off the road. The training simulation time step is set to be $\Delta t = 1$ sec. For the EV, overtaking is permitted on both the left and right sides of other vehicles, with each lane-change taking $T = 4$ sec to complete. An example of an initial configuration space is shown in Figure 6-4.

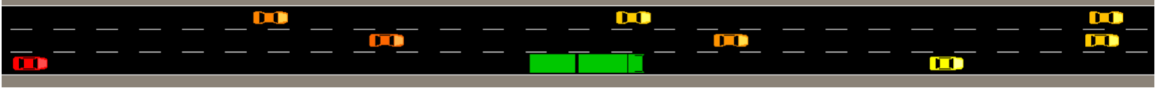


Figure 6-4: Example of an initial configuration space in the highway driving system: the green block denotes the ego vehicle, and the surrounding vehicles are color-coded according to their speeds, for which yellow represents 15 m/s, red 35 m/s, and different shades of orange indicate speeds between these two values.

6.5.1 Neural network architecture

Given the continuous and high-dimensional nature of the driving environment, using a tabular Q-learning approach is computationally infeasible. Instead, a Convolutional Neural Network (CNN) is utilized as a function approximator to estimate the Q-values for each state-action pair. While CNNs are traditionally associated with image processing, they are adapted here to process the structured state space of the traffic environment. The input to the network is not a visual image, but rather a state matrix representing the surrounding vehicles. This allows the network to exploit the spatial correlations between the ego vehicle and its neighbors. The architecture, adapted from [90, 101] is shown in Figure 6-5.

A CNN typically includes three primary operations: convolution, pooling, and fully connected layers, each contributing uniquely to the network's ability to analyze and learn from input data [159]. The convolutional layer serves as the cornerstone of the CNN architecture, focusing on feature extraction with multiple filters. These filters, characterized by their specific height and width dimensions, scan across the input image in a systematic manner [160, 161]. The convolution operation involves dividing the input image into smaller regions of a fixed size. Each region is represented as a small array of numbers, known as a kernel. By applying these kernels to the input data, the system extracts representative information from different regions, resulting in the creation of a feature map.

This map encapsulates the unique patterns and features present in the input data[160]. Subsequent pooling layers reduce the spatial dimensions of the feature map, enhancing computational efficiency and emphasizing dominant features. Fully connected layers, at the final stages of the network, integrate these extracted features to make accurate predictions. In this structure, the network's output becomes unaffected by the order of surrounding vehicles in the input vector, and the architecture supports an input vector of varying size. The neural network architecture is shown in Figure 6-5.

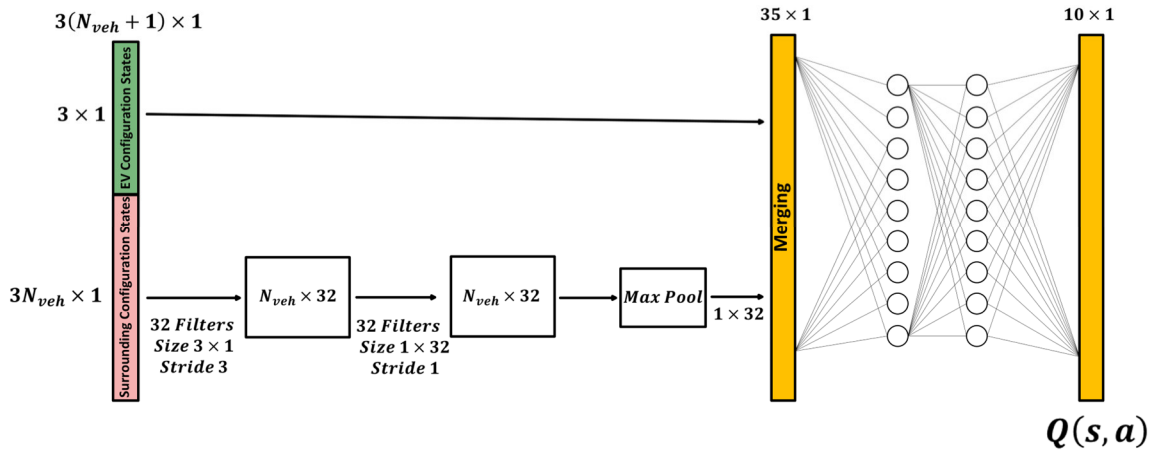


Figure 6-5: The neural network architecture.

Both convolutional layers contain 32 filters. The first layer has a filter size and stride of three, corresponding to the number of state inputs for each surrounding vehicle, while the second layer's filter size and stride are set to one. The two FC layers contain 64 units each. Rectified Linear Units (ReLUs) serve as activation functions for all layers, except the final one, which uses a linear activation function.

6.5.1.1 Training by deep reinforcement learning

In the past few years, deep learning techniques have become very popular. Deep reinforcement learning combines reinforcement learning with deep learning techniques to

solve challenging sequential decision-making problems. The use of deep learning is most useful in problems with high-dimensional state space. With deep learning, reinforcement learning solves complicated tasks with lower prior knowledge. This is because of its ability to learn different levels of abstraction from data.

In DRL, instead of forming a lookup table for $Q(s, a)$ which is highly time-consuming, a neural network with weights θ is used to approximate $Q(s, a)$ with a parametrized Q function, $Q_\theta(s, a)$. The DRL agent is built upon a modified version of the DDQN algorithm, originally presented in [162, 163]. Double deep Q-network corrects the overestimation problem found in DQN while also boosting performance across numerous benchmark challenges [164]. The loss function for parameter updates which should be minimized in the DDQN is calculated by:

$$L(\theta) = \mathbb{E}_{(s, a, r, s') \sim D} \left[\left(r + \gamma Q_{\theta^-} \left(s', \arg \max_{a' \in \mathcal{A}} Q_\theta(s', a') \right) - Q_\theta(s, a) \right)^2 \right] \quad (6.12)$$

where (s, a, r, s') are batches of samples taken from the replay buffer D , which is structured to retain the last N samples. Additionally, θ represents the current or online network, and θ^- refers to the target network. Both networks utilize the same architecture ($Q_\theta(s, a)$ and $Q_{\theta^-}(s, a)$). Furthermore, instead of random selection from the replay buffer D , the DDQN is used with Prioritized Experience Replay (PER) to guarantee efficient learning by focusing on the value of experiences that challenge and refine the agent's strategy. Thus, the elements will be selected from the buffer based on the magnitude of their Temporal-difference (TD) error determined by:

$$\delta_i = \left| r + \gamma Q_{\theta} \left(s', \arg \max_{a' \in \mathcal{A}} Q_{\theta}(s', a') \right) - Q_{\theta}(s, a) \right| \quad (6.13)$$

6.6 Simulation results

Table 6-1 lists the values for the controller and driver model parameters, as well as the values for the hyperparameters utilized in the controller, driver model, and training process. The prediction horizon, control horizon, MPC sampling time, sensor range, lower band, upper band, and visibility distance are chosen from the similar cases in the field of autonomous driving[129, 137, 165]. Additionally, due to the computational demands, the hyperparameter values are determined through an informal search instead of a comprehensive grid search.

Table 6-1: Values for controller and driver model parameters and hyperparameters.

Autonomous Controller Parameter	Symbol	Value	Unit
Acceleration	a_1	1	m/s ²
Deceleration	$-a_1$	-1	m/s ²
Hard deceleration	$-a_2$	-4	m/s ²
Lane width	L	3.6	m
SLC time	T	4	sec
Prediction horizon	P_H	10	
Control horizon	C_H	2	
Sampling time	ΔT_s	0.01	sec
MTAHV sensor range	x_{sensor}	300	m
Driver Model Parameter	Symbol	Value	Unit
Acceleration	a_1	1	m/s ²
Deceleration	$-a_1$	-1	m/s ²
Hard deceleration	$-a_2$	-4	m/s ²
Lane width	L	3.6	m
Lower bound	d_c	21	m
Upper bound	d_f	42	m
Visibility distance	D_v	100	m
Hyperparameter	Symbol	Value	Unit
Discount factor	γ	0.9	
Replay memory size	M_{replay}	50,000	
Learning rate	η	0.001	
Mini-batch size	M_{mini}	32	
Target network update frequency	N_{update}	1,000	Time step
Initial exploration constant	ϵ_{start}	1	
Final exploration constant	ϵ_{end}	0.05	
Final exploration iteration	$N_{\epsilon-end}$	50,000	Time step
Training time step	Δt	0.1	sec

The results indicate that the DDQN+PER method outperforms the DDQN and shows improved performance compared to the DDQN method when evaluated on scenarios like the ones used for agent training. This section provides further details on the results, along with a brief analysis and discussion of some key characteristics of the findings.

6.6.1 NLMPC autonomous controller

Figure 6-6 presents the collision-free step number during training episodes progresses for both the ensemble DDQN and DDQN+PER agents. It is calculated from 20 random seeds. Although the DDQN+PER agent learns the policy faster than the DDQN agent (1128 episodes vs. 1471 episodes), it suffers from more oscillation during its initial episodes. This is likely because of the DDQN+PER preference, which focuses more on the challenging (higher TD error) experiences. As training continues, both methods learn to make the decisions with 94.6 % accuracy for the DDQN+PER agent and 92.2% accuracy for the DDQN agent. Accuracy is defined as the ratio of collision-free steps to the total number of steps, normalized to a scale of 0 to 100.

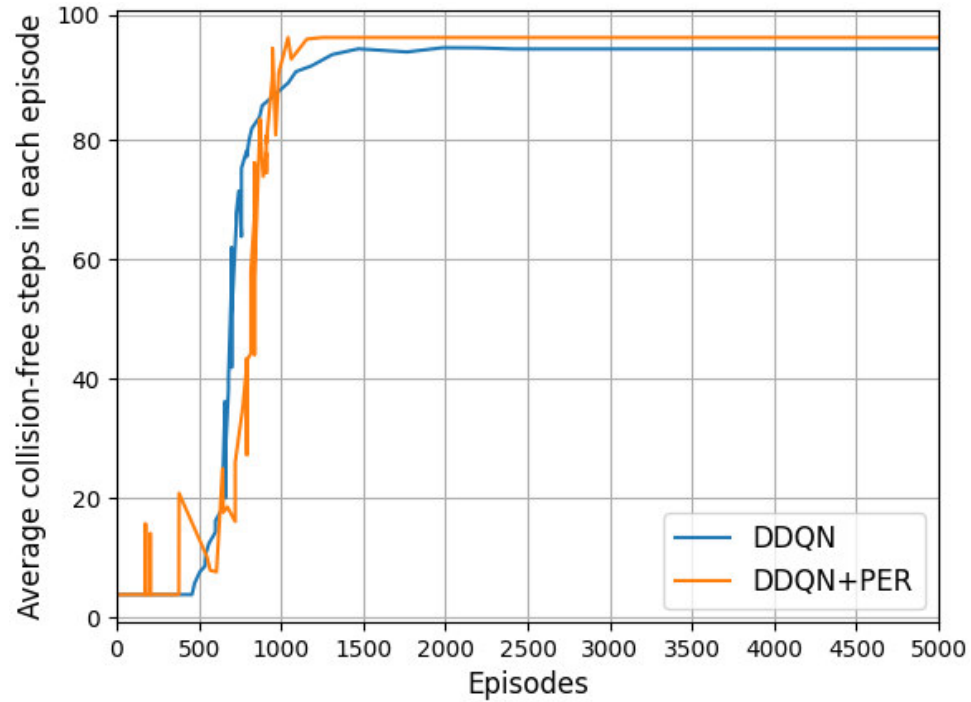


Figure 6-6: Average collision-free steps in each episode.

The unique challenges posed by the MTAHV's size are also considered in the decision-making process. Unlike passenger cars, the MTAHV's large dimensions increase the likelihood of spatial conflicts during maneuvers, particularly lane changes. For example, when the MTAHV transitions between lanes, the front section may enter the target lane while the rear remains in the initial lane. This spatial discrepancy may lead to unsafe conditions for surrounding vehicles, especially those traveling at high speeds, as shown in Figure 6-7 (left).



Figure 6-7: Collision between the MTAHV and a high-speed vehicle (left); Initial and target lanes reservation during the lane change maneuver (right).

Thus, to address this issue during the lane change maneuver, without expanding the Configuration State Space $\mathcal{S}$, both lanes (initial and target) have been reserved for the MTAHV during the maneuver to decrease the chance of collision, as seen in Figure 6-7 (right). By doing this, the final accuracy of DDQN+PER and DDQN agents have been successfully increased to 96.1% and 93.8%, respectively.

Table 6-2 presents the training and testing times for the DDQN and DDQN+PER agent. Although the training process can only be performed offline, the learned parameters and policies can be executed in real time. This enables the decision-making module to be integrated as a unified control algorithm within an experimental setup, such as a hardware-in-the-loop (HIL) real-time simulation for future implementation.

Table 6-2: Time for training and testing.

Approach	Training Time (h)	Testing Time (sec)
DDQN	24.3	54.21
DDQN+PER	24.1	50.73

Figure 6-8 illustrates the average normalized cumulative reward during the training phase from the 20 random seeds. It can be concluded that the DDQN+PER agent outperforms the DDQN, implying that the DDQN+PER agent makes more safe and efficient decisions. This conclusion can be validated by the 1000 test runs for two different agents. The results clearly show that not only the average number of lanes change for DDQN+PER is less than DDQN, but also the average speed is higher for DDQN+PER, as shown in Figure 6-9 and Table 6-3.

Table 6-3: Average number of lanes change and average speed for 1000 test runs.

Approach	Number of Lane Change	Average Speed (m/s)	Maximum Speed (m/s)
DDQN	4.27	21.01	25
DDQN+PER	4.03	22.33	25

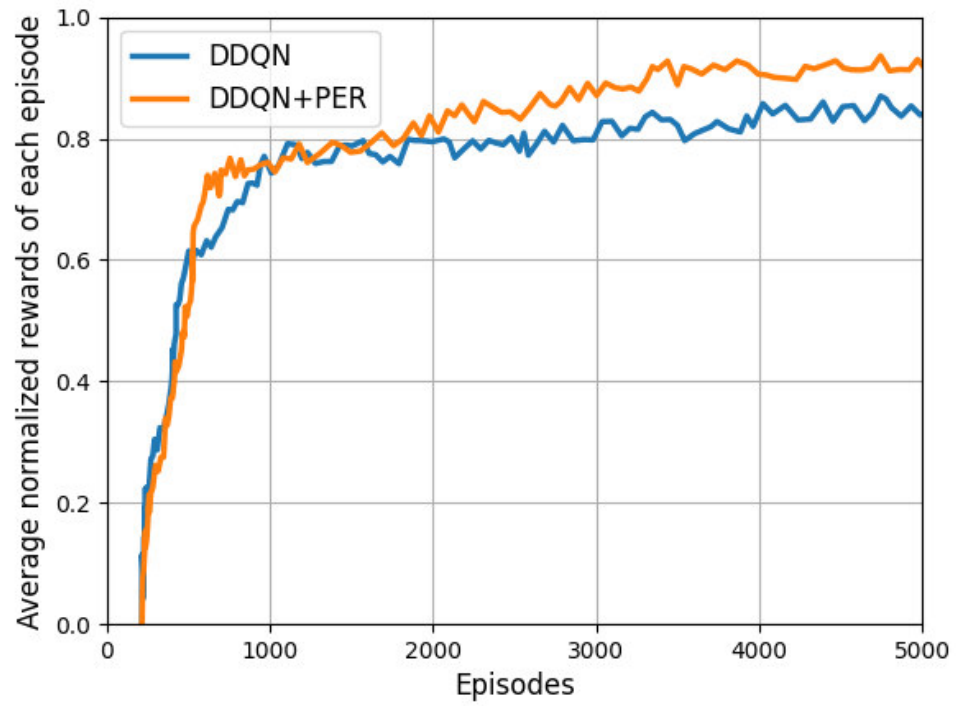


Figure 6-8: Average normalized reward in each episode.

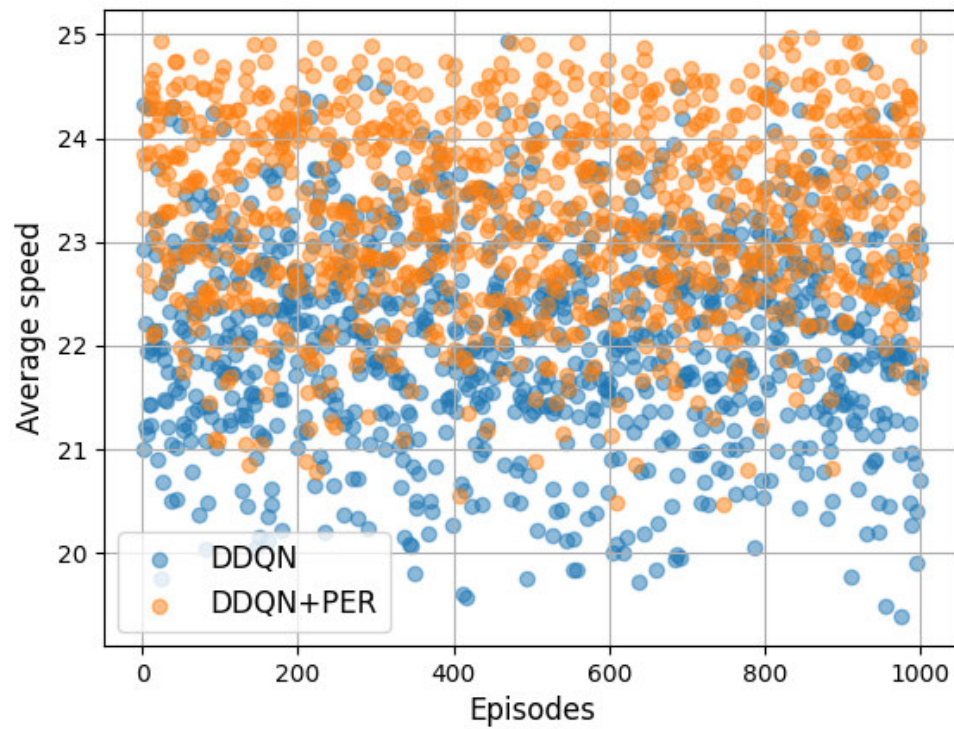


Figure 6-9: Average speed distribution for 1000 test runs.

In the next step, the effect of number of surrounding vehicles on the average number of collisions in the training phase for the 20 random seeds is investigated. Table 6-4 shows the different traffic condition classifications for the highway driving. The number of surrounding vehicles in 100 meters for a 3-lane highway is within the range of 10-25 vehicles, which varies from the moderate to the heavy density traffic condition specified in Table 6-4. It clearly shows that the average number of collisions for the DDQN agent is larger than that for the DDQN+PER agent for almost any number of surrounding vehicles, meaning that the DDQN+PER learns the policy faster than the DDQN agent. In addition, an increase in the number of surrounding vehicles (heavier traffic) poses a sharp effect on the average number of accident before the complete learn, which results in much more time-consuming training phase, as seen in Figure 6-10.

Table 6-4: Traffic density classification for a 3-lane highway[158].

Traffic Condition	Total Density for 3 Lanes (Vehicles/100 m)	Characteristics
Heavy Traffic	~ 30-36	Near-capacity conditions with frequent stops
Moderate Traffic	~ 9-21	Smooth flow with moderate spacing
Low Traffic	< 9	Sparse traffic with free-flow conditions

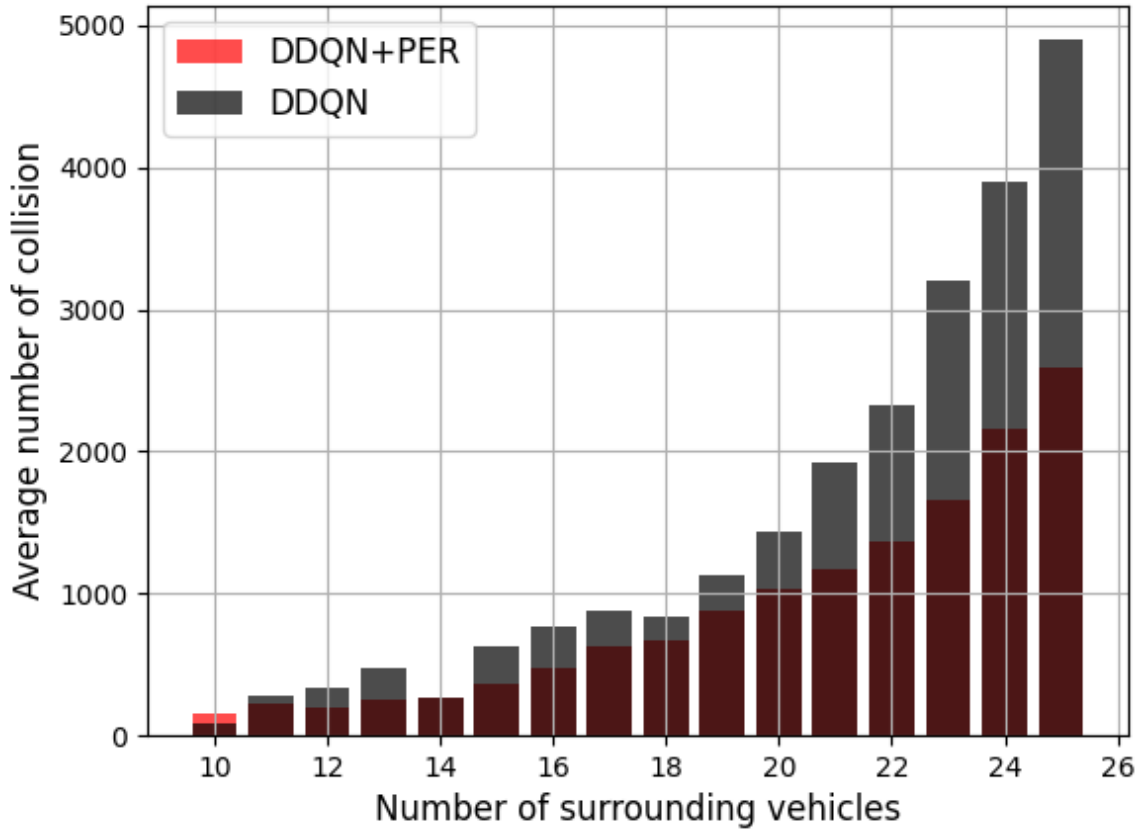


Figure 6-10: Average cumulative number of collisions for different number of surrounding vehicles in 100 meters of a 3-lane highway.

To evaluate the performance of decision making and controller subsystems, the path-following performance of the EV has been examined, as seen in Figure 6-11. The DDQN+PER agent decides to change the lane to the left and accelerate the EV over the maneuver. Thus, the NLMPC controller must navigate the vehicle in the proper manner. Figure 6-11 shows that all units well follow the reference path over the 4-seconds lane-change maneuver, over which the longitudinal speed of the vehicle has been increased based on the command, as shown in Figure 6-12.

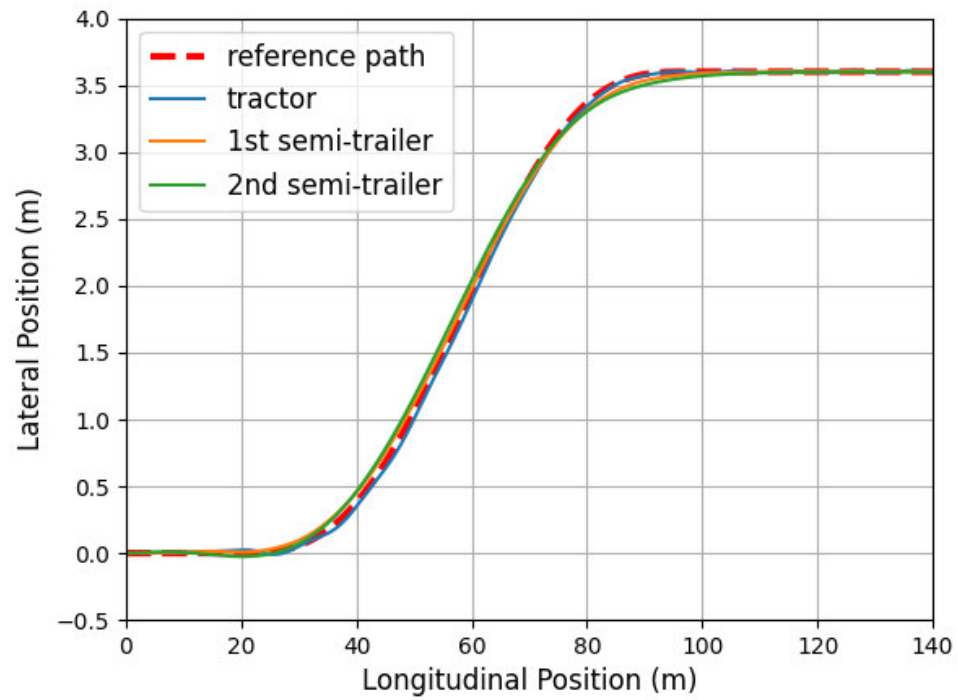


Figure 6-11: TruckSim output for the left lane change.

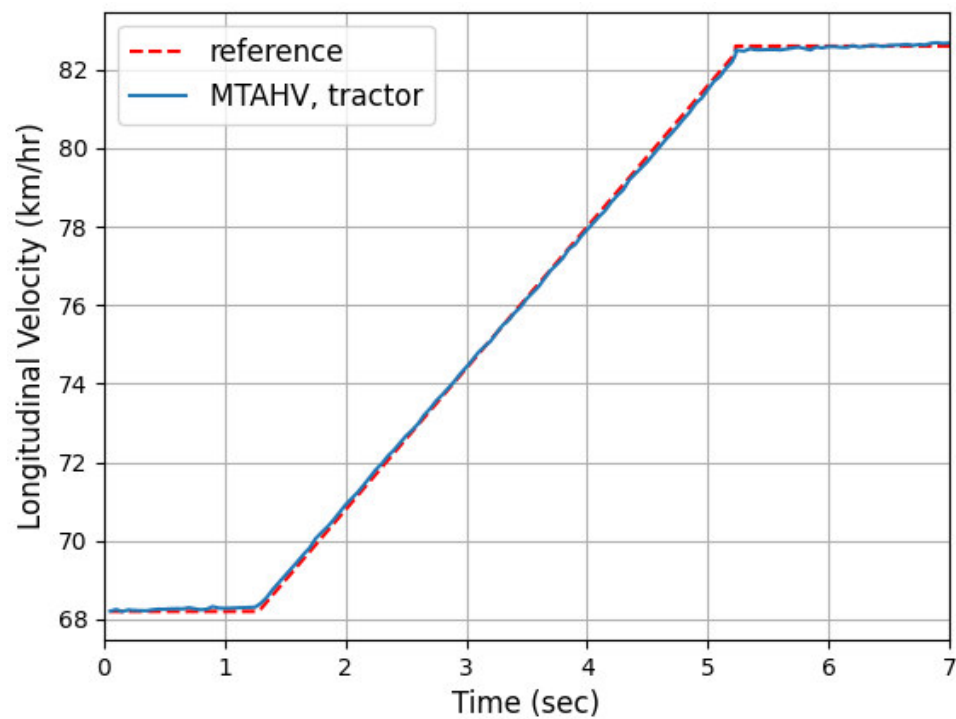


Figure 6-12: TruckSim output for the longitudinal speed.

6.6.2 Integrated MVDM Driver model

Figure 6-13 illustrates the progression of collision-free decision steps across training episodes for both the DDQN and DDQN+PER agents, using the data from the 20 random seeds. The results reveal that the DDQN+PER achieves policy optimization more quickly than the DDQN, requiring 121 episodes versus 262 episodes to reach a stable performance level. However, during early training, the DDQN+PER agent exhibits greater variability due to its focus on high-priority experiences with large TD errors. Over time, this prioritization allows the agent to achieve a decision accuracy of 95.7%, surpassing the 93.2% accuracy achieved by DDQN.

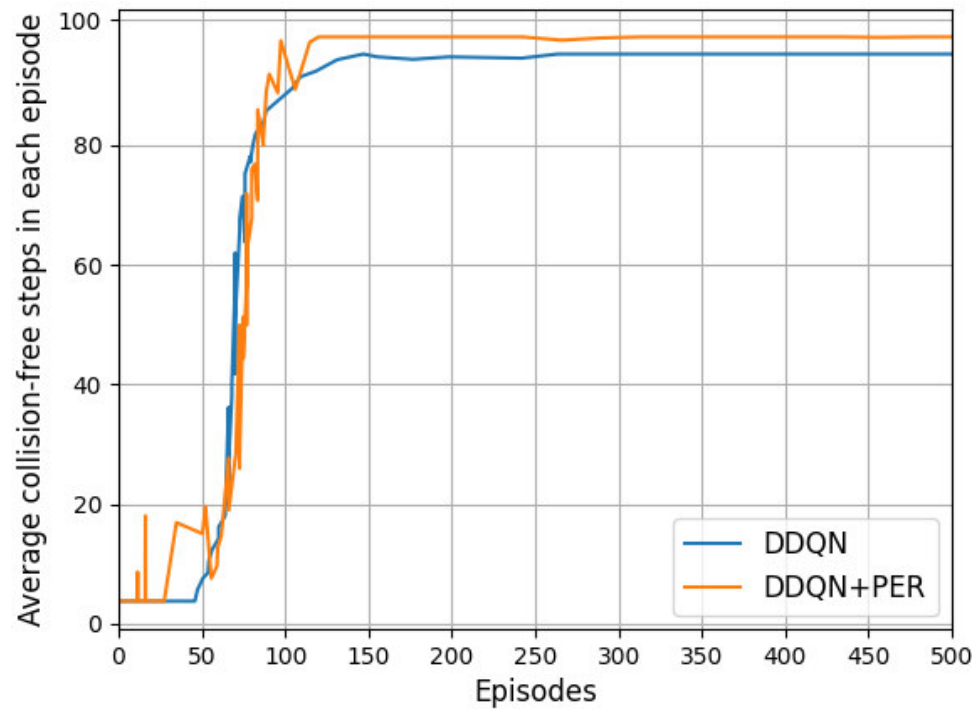


Figure 6-13: Average collision-free steps in each episode.

The large size of the MTAHV can cause potentially hazardous situations for nearby vehicles, especially when traveling at high speeds, as seen in Figure 6-7. To mitigate this issue, a novel strategy is proposed and implemented: reserving both the initial and target

lanes during lane-change maneuvers. By pre-emptively reserving these lanes, the likelihood of collisions was significantly reduced without requiring an expansion of the configuration state space $\mathcal{S}$. This approach improved the final decision accuracy to 98.2% for the DDQN+PER agent and 95.1% for the DDQN agent.

Training and testing times for the decision-making subsystem are summarized in Table 6-5. While the training is conducted offline due to its computational intensity, the learned policies are suitable for real-time implementation, making them viable for deployment in experimental setups, such as hardware-in-the-loop (HIL) simulations.

Table 6-5: Time for training and testing.

Approach	Training Time (h)	Testing Time (sec)
DDQN	2.4	48.01
DDQN+PER	2.2	44.38

Figure 6-14 shows the average normalized cumulative reward during the training for the driver model subsystem. The results clearly indicate that the DDQN+PER agent outperforms the DDQN in terms of safety and efficiency. This conclusion is reinforced by a 1000-test-run analysis, where the DDQN+PER requires fewer lane-changes and maintains higher average speeds than the DDQN, as summarized in Table 6-6 and shown in Figure 6-14.

Table 6-6: Average number of lanes change and average speed for 1000 test runs.

Approach	Number of Lane Change	Average Speed (m/s)	Maximum Speed (m/s)
DDQN	4.71	19.87	25
DDQN+PER	4.59	21.18	25

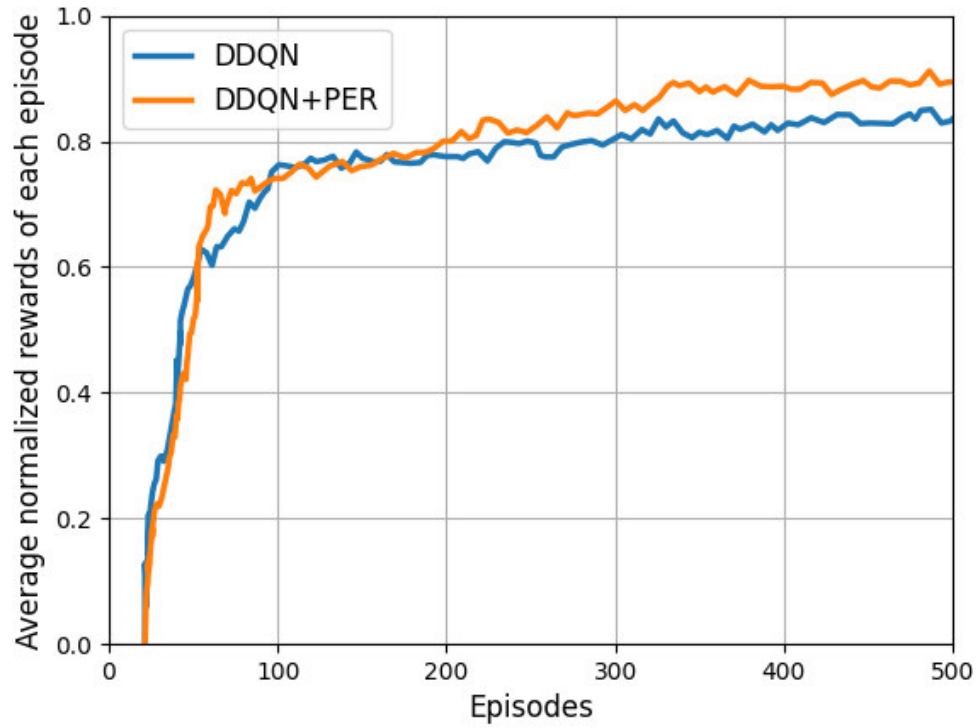


Figure 6-14: Average normalized reward in each episode.

6.6.3 Comparison between autonomous controller and driver model

In this section, the performance of the NLMPC autonomous controller is compared with that of the Integrated MVDM driver model, which combines decision-making at the upper level and controller/driver model functionality at the lower level. The comparison aims to evaluate their efficiency from various perspectives. It is important to note that the evaluation is based on the DDQN+PER decision-making method.

6.6.3.1 Training time

By comparing Figure 6-6 and Figure 6-13, it can be observed that the number of episodes required for agent training for the NLMPC autonomous controller is 1128, while for the Integrated MVDM, it is 121. Therefore, the time required for the entire training

process for the latter is approximately one-tenth of the former, as indicated in Table 6-2 and Table 6-5. The primary reason for this phenomenon is the larger configuration state space, $\mathcal{S}$, in the case of the NLMPC autonomous controller compared to the Integrated MVDM driver model. Since the autonomous controller receives information about all vehicles, N_{veh} , within the sensor range, x_{sensor} , whereas the driver only perceives information about up to five vehicles within their visibility distance, d_v . The network training process for the driver model is thus significantly faster.

6.6.3.2 Final accuracy

As mentioned earlier, the final accuracy is 96.1% for the NLMPC autonomous controller and 98.2% for the Integrated MVDM driver model. This difference can be attributed to the much smaller configuration state space, $\mathcal{S}$, for the driver model, resulting in simpler, less risky, and less error-prone decision-making process.

6.6.3.3 Average number of lane changes and average speed

As previously mentioned, the average number of lane changes for the NLMPC autonomous controller is 4.03, while for the Integrated MVDM driver model, the counterpart is 4.59. It is worth noting that when a vehicle is forced to change lanes, its average speed decreases. So, this results in a higher average speed for the autonomous controller of 22.33 m/s with respect to the respective average speed for the driver model of 21.18 m/s. This phenomenon can be attributed to the fact that the autonomous controller has access to information about a larger number of vehicles compared to the driver model. Consequently, it can consider the data from vehicles several rows ahead when making

decisions. This enables the autonomous controller to reduce the number of lane changes, optimize their execution, and ultimately maintain a higher average speed.

6.6.3.4 Traffic condition

In this section, the effect of number of surrounding vehicles on the ego vehicle average speed for the 1000 test runs is investigated. The number of surrounding vehicles in 100 meters for a 3-lane highway varies in the range of 10-25 vehicles, which is from the moderate to the heavy dense traffic as defined in In the next step, the effect of number of surrounding vehicles on the average number of collisions in the training phase for the 20 random seeds is investigated. Table 6-4 shows the different traffic condition classifications for the highway driving. The number of surrounding vehicles in 100 meters for a 3-lane highway is within the range of 10-25 vehicles, which varies from the moderate to the heavy density traffic condition specified in Table 6-4. It clearly shows that the average number of collisions for the DDQN agent is larger than that for the DDQN+PER agent for almost any number of surrounding vehicles, meaning that the DDQN+PER learns the policy faster than the DDQN agent. In addition, an increase in the number of surrounding vehicles (heavier traffic) poses a sharp effect on the average number of accident before the complete learn, which results in much more time-consuming training phase, as seen in Figure 6-10.

Table 6-4. In scenarios with low traffic and heavy traffic, there is little difference between the ego vehicle average speed of the NLMPC autonomous controller and the Integrated MVDM driver model. This can be explained as follows. In low-traffic conditions, both entities have sufficient space to make decisions, and they are aware of all surrounding vehicles or only a subset of them has minimal impact on decision-making and average speed. Conversely, this holds true for heavy-traffic conditions as well. However,

in the moderate to heavy traffic scenarios, having information about a greater number of surrounding vehicles ahead enables more optimal decision-making of the NLMPC autonomous controller and results in its higher average speed.

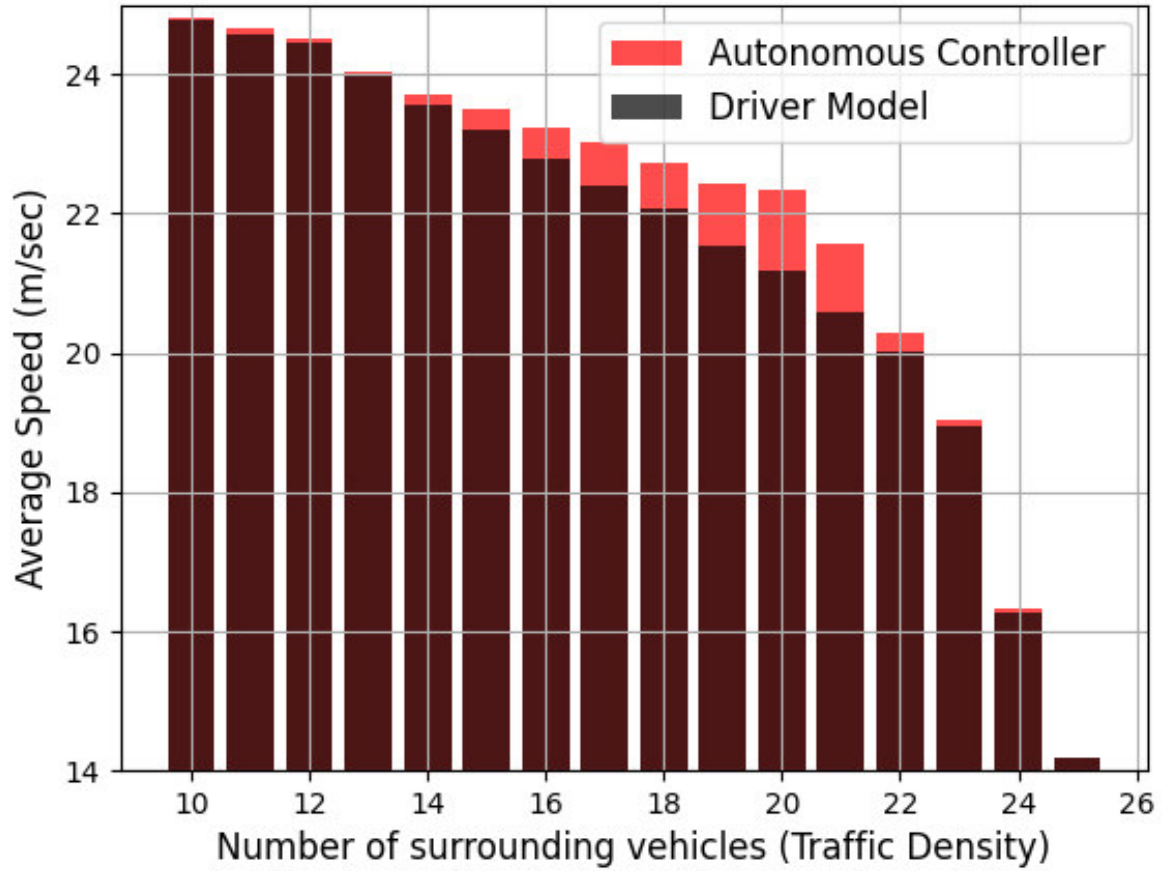


Figure 6-15: Ego vehicle's average speed for different number of surrounding vehicles in 100 meters of a 3-lane highway.

6.7 Summary

This chapter establishes a novel closed-loop Software-in-the-Loop (SIL) platform to evaluate a hierarchical control architecture for an A-train double MTAHV. The design integrates an upper-level decision-making module, utilizing a DDQN+PER algorithm to

enhance learning speed and safety, with a lower-level execution layer. For maneuver execution, the study compares a tuned NLMPC against an Integrated MVDM driver model, both simulated within the high-fidelity TruckSim environment to assess performance in mixed-traffic highway operations.

The comparative analysis reveals a distinct trade-off between safety accuracy and operational efficiency. While both systems achieve high decision-making fidelity, the Integrated MVDM attains a slightly higher collision-avoidance accuracy (98.2%) compared to the NLMPC (96.1%), primarily due to its reduced state space complexity which minimizes decision noise in the training phase. Conversely, in the test phase, the NLMPC demonstrates superior traffic flow performance, particularly in moderate traffic densities; by leveraging broader situational awareness, it maintains higher average speeds (22.33 m/s) and executes fewer, more optimized lane changes compared to the MVDM, proving more effective at navigating congestion without sacrificing stability.

However, a significant limitation identified is the inability of the DDQN+PER framework to achieve 100% collision-free reliability, attributed to the inherent stochastic nature of function approximation in DRL. To address this non-zero failure rate and the observed performance trade-offs, future research should investigate a hybrid control architecture that dynamically switches between the robust Integrated MVDM for critical safety maneuvers and the efficient NLMPC for traffic optimization. Furthermore, implementing a deterministic safety shield layer is recommended to override probabilistic AI errors with guaranteed safe fallback maneuvers, ensuring strict adherence to safety constraints in real-world deployment.

Chapter 7

Semi-autonomous driving control

7.1 Introduction

Motivated by the research gaps mentioned in Section 2.10, this chapter introduces a novel nonlinear non-cooperative shared control framework. This innovative approach addresses critical deficiencies in existing methodologies by integrating advanced modeling, dynamic control, and validation techniques tailored to MTAHVs. Thus, the primary contributions are as follows:

1) Modeling complex nonlinear dynamics of MTAHVs

Unlike traditional methods that assume linear vehicle dynamics, this approach incorporates the inherently nonlinear behavior of MTAHVs. The dynamics is characterized by the interplay of multiple trailer units, articulated joints, and the vehicle's response under various driving conditions. By capturing these complexities, the framework enables more realistic and robust control strategies that are directly applicable to heavy vehicle operation.

2) Variable forward speed with constant acceleration

The framework accommodates MTAHVs operating at variable forward speed with constant acceleration, reflecting real-world driving scenarios, which are more accurate and realistic than the prevalent assumption of constant forward speed. This enhancement enables the system to respond dynamically to changing traffic and environmental

conditions, ensuring smoother transitions and safer interactions between human drivers and automated control systems.

3) Integrated co-simulation setup for validation

To validate the proposed approach, a comprehensive co-simulation environment is developed. This setup integrates:

- MPC for autonomous commands, offering optimal control solutions considering the nonlinear dynamics.
- Integrated MVDM driver model to simulate realistic driver behavior, bridging the gap between theoretical control strategies and practical application.
- TruckSim for high-fidelity simulation of MTAHVs, ensuring that the results are applicable to real-world scenarios.

In the semi-autonomous control system, where both a human driver and an autonomous controller are present, each controller's steering actions contribute to changes in the vehicle's state, $(\mathbf{x}(k))$, and thereby affecting the vehicle's motion-trajectory tracking performance. If the two controllers aim for different target paths, each may adjust its strategy by accounting for the other's steering behavior. For instance, the autonomous controller can be designed to mitigate the driver's input, especially in situations where human errors or inattentiveness could lead to a collision[57]. Conversely, the human driver may attempt to partially remove the autonomous actions and counteract them to achieve their own path-following objectives. This dynamic interaction reflects the characteristics of a theoretical game and can be considered as non-cooperative shared control. A conceptual representation of this driver-controller non-cooperative game framework is

illustrated in Figure 7-1. This configuration includes the Integrated MVDM driver model (presented in Chapter 5) and the NLMPC tracking controller (presented in Chapter 4), each of which individually attempts to control and navigate the MTAHV (modeled in TruckSim in Chapter 3) toward its own reference trajectory. Naturally, the command from one controller will influence the another, and this interaction, represented by the dashed line, is referred to as the non-cooperative controller. Therefore, the co-simulation framework not only demonstrates the feasibility of the shared control strategy but also provides insights into the interaction between automated and human inputs. In summary, this chapter pushes the boundaries of shared autonomous control for heavy vehicles by overcoming the limitations of traditional game-theoretical approaches. By addressing the unique challenges of MTAHVs, this research sets the stage for safer and more efficient semi-autonomous driving solutions in complex, real-world environments.

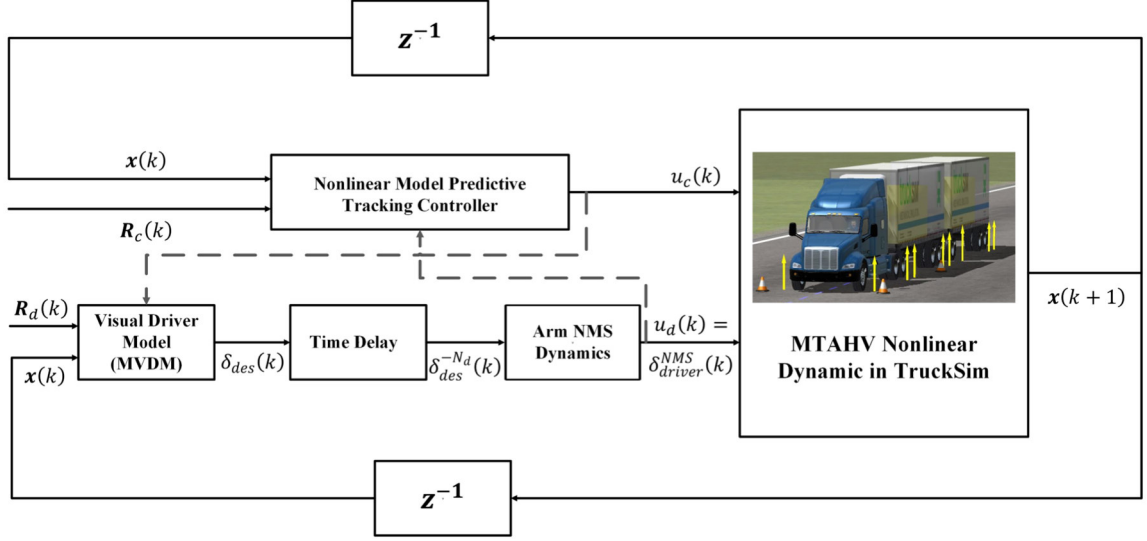


Figure 7-1: The co-simulation setup for semi-autonomous driving control.

7.2 Semi-autonomous shared control design

This section presents the design of a semi-autonomous shared control framework in which both the human driver and the automated controller contribute to vehicle guidance.

7.2.1 11-DOF single track nonlinear MTAHV dynamic model

The nonlinear 11-DOF single-track yaw-plane model for representing the lateral dynamics of the A-train double is generated and introduced in Section 3.2, and the state-space description of the model is given by:

$$\dot{\mathbf{x}}(t) = \mathbf{f}[\mathbf{x}(t), \mathbf{u}(t)] \quad (7.1)$$

where $\mathbf{x}(t)$ and $\mathbf{u}(t)$ are the state and control variable vectors expressed as:

$$\mathbf{x}(t) = [v_y, \phi_1, \dot{\phi}_1, \theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2, \theta_3, \dot{\theta}_3, v_x, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5]^T \quad (7.2)$$

$$\mathbf{u}(t) = [\delta_{1f}, T_{x1f,r}]^T \quad (7.3)$$

In the context of an acceleration scenario, the traction torque, $T_{x1f,r}$, is calculated using a simplified tractor powertrain model as described in[63]. Thus, as introduced in Section 3.2, the nonlinear equations of motion of the MTAHV in the semi-matrix and discrete-time format is cast as:

$$M\mathbf{x}(k+1) = A(\mathbf{x}(k)) + B(\mathbf{x}(k))\mathbf{u}(k) \quad (7.4)$$

Considering two different inputs from two different sources (autonomous controller and driver model), Equation (7.4) can be revised as:

$$\begin{aligned} M\mathbf{x}(k+1) &= A(\mathbf{x}(k)) + B_c(\mathbf{x}(k))\mathbf{u}_c(k) + B_d(\mathbf{x}(k))u_d(k) \\ M\mathbf{x}(k+1) &= A(\mathbf{x}(k)) + [B_c(\mathbf{x}(k)) \quad B_d(\mathbf{x}(k))] \begin{Bmatrix} \mathbf{u}_c(k) \\ u_d(k) \end{Bmatrix} \end{aligned} \quad (7.5)$$

where $B_c(\mathbf{x}(k))$ and $B_d(\mathbf{x}(k))$ are the controller and driver input matrices, respectively, and $B(\mathbf{x}(k)) = [B_c(\mathbf{x}(k)) \quad B_d(\mathbf{x}(k))]$ is the combination input matrix. It should be mentioned that $\mathbf{u}_c(k) = [\delta_{1f}(k), T_{x1f,r}(k)]^T$ is the input signal generated by the autonomous controller.

7.2.2 Integrated MVDM driver model

The Integrated MVDM driver model is generated and introduced in Section 5.2. Figure 5-1 describes the structure of the driver model.

7.2.3 Integration of MTAHV vehicle model and Integrated MVDM

driver model

As shown in Figure 7-2, the equations of motion for the MTAHV and the driver model are integrated to form a unified generalized system, which will be used to design a semi-autonomous driving control.

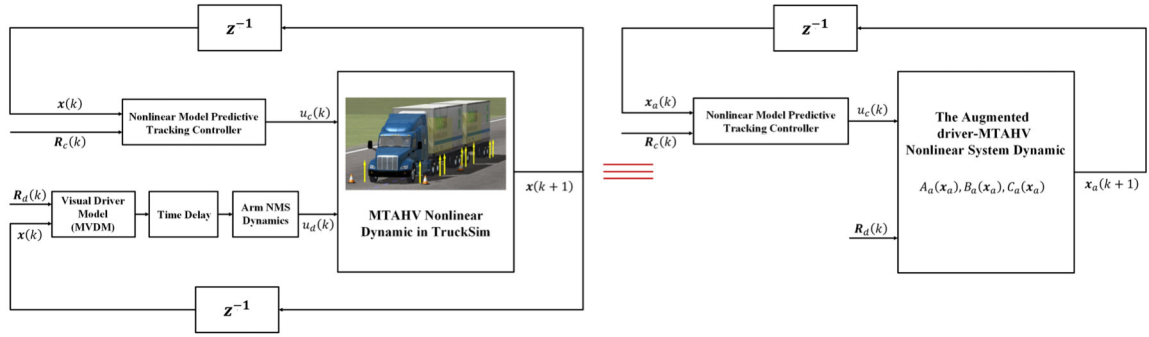


Figure 7-2: The augmented model integrating the MTAHV vehicle model and the Integrated MVDM driver model.

7.2.3.1 Integration of arm NMS and MTAHV vehicle model

The arm NMS dynamics, time delay block, and MVDM driver model are integrated with the MTAHV model. Based on Equation (5.15), $u_d(t)$ is equal to the $\delta_{driver}^{NMS}(t)$; moreover, the continuous-time dynamic equation of motion for the arm NMS block is (Equation (5.15)):

$$\begin{aligned} \begin{Bmatrix} \dot{\delta}_{driver}^{NMS}(t) \\ \ddot{\delta}_{driver}^{NMS}(t) \end{Bmatrix} &= \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} \begin{Bmatrix} \delta_{driver}^{NMS}(t) \\ \dot{\delta}_{driver}^{NMS}(t) \end{Bmatrix} + \begin{bmatrix} 0 \\ \omega_n^2 \end{bmatrix} \delta_{des}^{-N_d}(t) \\ \delta_{driver}^{NMS}(t) &= [1 \quad 0] \begin{Bmatrix} \delta_{driver}^{NMS} \\ \dot{\delta}_{driver}^{NMS} \end{Bmatrix} \end{aligned} \quad (7.6)$$

To enhance clarity and improve readability, the state variables in continuous time for the arm NMS block is defined as:

$$\mathbf{x}_{1,c}(t) = \begin{cases} \delta_{driver}^{NMS}(t) \\ \dot{\delta}_{driver}^{NMS}(t) \end{cases} \quad (7.7)$$

Thus, the dynamics of the arm NMS, represented in Equation (7.6), can be reformulated as:

$$\begin{aligned} \dot{\mathbf{x}}_{1,c}(t) &= A_{1,c}\mathbf{x}_{1,c}(t) + B_{1,c}u_1(t), \quad A_{1,c} = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix}, \quad B_{1,c} = \begin{bmatrix} 0 \\ \omega_n^2 \end{bmatrix} \\ y_{1,c} &= C_{1,c}\mathbf{x}_{1,c}(t), \quad C_{1,c} = [1 \quad 0] \end{aligned} \quad (7.8)$$

The discretization process is essential for enabling numerical implementation and facilitating the integration of arm NMS dynamics with the MTAHV system. To integrate the governing equations of motion of the arm NMS, i.e., Equation (7.8), with the MTAHV model, that is, Equation (7.5), we have Equation (7.8), which is discretized as:

$$\begin{aligned} \mathbf{x}_1(k+1) &= A_1\mathbf{x}_1(k) + B_1u_1(k), \quad A_1 = I + T_s A_{1,c}, \quad B_1 = T_s B_{1,c} \\ y_1 &= C_1\mathbf{x}_1(k), \quad C_1 = C_{1,c} \end{aligned} \quad (7.9)$$

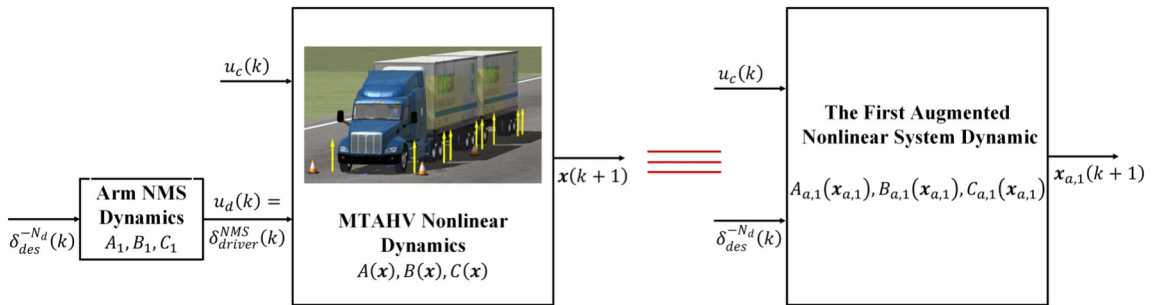


Figure 7-3: The first augmented model combining the MTAHV model and the arm NMS dynamics.

Figure 7-3 visualizes the integration of the MTAHV model and the arm NMS dynamics.

The resulting governing equations of motion are given by:

$$\begin{cases} \mathbf{x}_1(k+1) = A_1 \mathbf{x}_1(k) + B_1 u_1(k) \\ y_1(k) = u_d(k) = C_1 \mathbf{x}_1(k) \\ \mathbf{x}(k+1) = M^{-1}A(\mathbf{x}(k)) + M^{-1}B_c(\mathbf{x}(k))\mathbf{u}_c(k) + M^{-1}B_d(\mathbf{x}(k))u_d(k) \end{cases} \quad (7.10)$$

The matrix representation provides a compact and structured way to describe the equations of motion, facilitating analytical and computational processes. Equation (7.10) can be expressed in matrix form as:

$$\begin{Bmatrix} \mathbf{x}_1(k+1) \\ \mathbf{x}(k+1) \end{Bmatrix} = A_{a,1}(\mathbf{x}_{a,1}(k)) + B_{a,1}(\mathbf{x}_{a,1}(k)) \begin{Bmatrix} \mathbf{u}_c(k) \\ \delta_{des}^{-N_d}(k) \end{Bmatrix} \quad (7.11)$$

where:

$$\begin{aligned} \mathbf{x}_{a,1}(k) &= \begin{Bmatrix} \mathbf{x}_1(k) \\ \mathbf{x}(k) \end{Bmatrix}, \quad A_{a,1}(\mathbf{x}_{a,1}(k)) = \begin{bmatrix} A_1 \mathbf{x}_1(k) \\ M^{-1}B_d(\mathbf{x}(k))C_1 \mathbf{x}_1(k) + M^{-1}A(\mathbf{x}(k)) \end{bmatrix}, \\ B_{a,1}(\mathbf{x}_{a,1}(k)) &= \begin{bmatrix} \mathbf{0} & B_1 \\ M^{-1}B_c(\mathbf{x}(k)) & \mathbf{0} \end{bmatrix} \end{aligned} \quad (7.12)$$

7.2.3.2 Integration of first augmented nonlinear system and time delay

This step enables the incorporation of time delay effects into the overall system dynamics, enhancing the model's realism and applicability in practical scenarios. Thus, the first augmented nonlinear system dynamics and the time delay block are integrated to create the second nonlinear system dynamics, as illustrated in Figure 7-4.

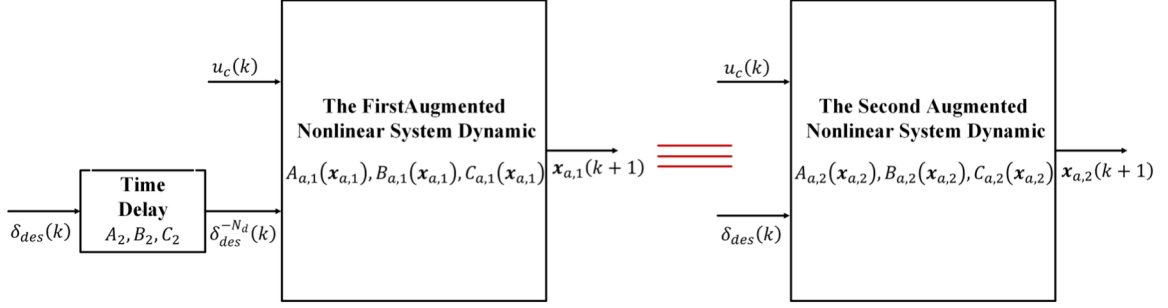


Figure 7-4: The second augmented model including time delay effect and the first augmented model.

Based on Equation (5.14), the dynamic equation for the time delay effect is expressed as:

$$\begin{Bmatrix} \delta_{des}^{-1}(k+1) \\ \delta_{des}^{-2}(k+1) \\ \delta_{des}^{-3}(k+1) \\ \vdots \\ \delta_{des}^{-N_d}(k+1) \end{Bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & 0 \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix} \begin{Bmatrix} \delta_{des}^{-1}(k) \\ \delta_{des}^{-2}(k) \\ \delta_{des}^{-3}(k) \\ \vdots \\ \delta_{des}^{-N_d}(k) \end{Bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \delta_{des}(k) \quad (7.13)$$

$$\delta_{des}^{-N_d}(k) = [0 \quad 0 \quad 0 \quad \cdots \quad 1] \begin{Bmatrix} \delta_{des}^{-1}(k) \\ \delta_{des}^{-2}(k) \\ \delta_{des}^{-3}(k) \\ \vdots \\ \delta_{des}^{-N_d}(k) \end{Bmatrix}$$

For convenience, the third set of state variables is defined as:

$$\mathbf{x}_2(k) = \begin{Bmatrix} \delta_{des}^{-1}(k) \\ \delta_{des}^{-2}(k) \\ \delta_{des}^{-3}(k) \\ \vdots \\ \delta_{des}^{-N_d}(k) \end{Bmatrix} \quad (7.14)$$

Thus, the time delay dynamic equation, i.e., Equation (7.13), can be reformulated in the following format.

$$\mathbf{x}_2(k+1) = A_2 \mathbf{x}_2(k) + B_2 u_2(k), \quad A_2 = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & 0 \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (7.15)$$

$$y_2(k) = C_2 \mathbf{x}_2(k), \quad C_2 = [0 \quad 0 \quad 0 \quad \cdots \quad 1]$$

Hence, the second augmented nonlinear system dynamics is formulated as:

$$\begin{cases} \mathbf{x}_2(k+1) = A_2 \mathbf{x}_2(k) + B_2 u_2(k) \\ y_2(k) = u_1(k) = C_2 \mathbf{x}_2(k) \\ \mathbf{x}_1(k+1) = A_1 \mathbf{x}_1(k) + B_1 u_1(k) \\ y_1(k) = u_d(k) = C_1 \mathbf{x}_1(k) \\ \mathbf{x}(k+1) = M^{-1}A(\mathbf{x}(k)) + M^{-1}B_c(\mathbf{x}(k))\mathbf{u}_c(k) + M^{-1}B_d(\mathbf{x}(k))u_d(k) \end{cases} \quad (7.16)$$

which can be expressed the matrix format as:

$$\begin{Bmatrix} \mathbf{x}_2(k+1) \\ \mathbf{x}_1(k+1) \\ \mathbf{x}(k+1) \end{Bmatrix} = A_{a,2}(\mathbf{x}_{a,2}) + B_{a,2}(\mathbf{x}_{a,2}) \begin{Bmatrix} \mathbf{u}_c(k) \\ \delta_{des}(k) \end{Bmatrix} \quad (7.17)$$

where:

$$\mathbf{x}_{a,2}(t) = \begin{Bmatrix} \mathbf{x}_2(t) \\ \mathbf{x}_1(t) \\ \mathbf{x}(t) \end{Bmatrix}, \quad A_{a,2}(\mathbf{x}_{a,2}) = \begin{bmatrix} A_2 \mathbf{x}_2(k) \\ B_1 C_2 \mathbf{x}_2(k) + A_1 \mathbf{x}_1(k) \\ M^{-1}B_d(\mathbf{x}(k))C_1 \mathbf{x}_1(k) + M^{-1}A(\mathbf{x}(k)) \end{bmatrix}, \quad (7.18)$$

$$B_{a,2}(\mathbf{x}_{a,2}) = \begin{bmatrix} \mathbf{0} & B_2 \\ \mathbf{0} & \mathbf{0} \\ M(\mathbf{x})^{-1}B_c(\mathbf{x}) & \mathbf{0} \end{bmatrix}$$

7.2.3.3 Integration of second augmented nonlinear system and MVDM driver model

So far, the MTAHV dynamics equations, the arm NMS dynamics, and the time delay model have been combined to form Equations (7.17) and (7.18). At this stage, the MVDM driver model must be integrated with Equations (7.17) and (7.18) to generate the

final augmented model. Based on the Equation (5.6), the time derivative of $\delta_{des}(t)$ is calculated by:

$$\begin{aligned} \dot{\delta}_{des}(t) = & k_{P,f}\dot{\theta}_f(t) + k_{P,n}\dot{\theta}_n(t) + k_{I,n}\theta_n(t) + k_{P,p}\left(\dot{\phi}_1(t-\tau) - \dot{\phi}_4(t)\right) + \\ & k_{I,p}(\phi_1(t-\tau) - \phi_4(t)) \end{aligned} \quad (7.19)$$

Thus, integrating Equation (7.19) leads to:

$$\begin{aligned} \int_0^t \dot{\delta}_{des}(\xi) d\xi = & k_{P,f} \int_0^t \dot{\theta}_f(\xi) d\xi + k_{P,n} \int_0^t \dot{\theta}_n(\xi) d\xi + k_{I,n} \int_0^t \theta_n(\xi) d\xi \\ & + k_{P,p} \int_0^t \left(\dot{\phi}_1(\xi - \tau(\xi)) - \dot{\phi}_4(\xi) \right) d\xi \\ & + k_{I,p} \int_0^t \left(\phi_1(\xi - \tau(\xi)) - \phi_4(\xi) \right) d\xi \end{aligned} \quad (7.20)$$

In highway operations, the tractor's heading angle (ϕ_1) and its rate of change with respect to time ($\dot{\phi}_1$) are typically small [166]. This small-angle assumption simplifies the mathematical modeling of the system. It allows for the application of first-order Taylor expansions to approximate delayed terms like $\phi_1(\xi - \tau(\xi))$ and $\dot{\phi}_1(\xi - \tau(\xi))$. These approximations are particularly useful for dynamic modeling and control design, as they reduce complexity without significantly compromising accuracy under highway operations. Thus, the terms $\phi_1(\xi - \tau(\xi))$ and $\dot{\phi}_1(\xi - \tau(\xi))$ can be approximated and substituted as:

$$\begin{aligned} \phi_1(\xi - \tau) &= \phi_1(\xi) - \dot{\phi}_1(\xi)\tau \\ \dot{\phi}_1(\xi - \tau) &= \dot{\phi}_1(\xi) - \ddot{\phi}_1(\xi)\tau \end{aligned} \quad (7.21)$$

Substituting Equation (7.21) into Equation (7.20) leads to:

$$\begin{aligned}
\int_0^t \dot{\delta}_{des}(\xi) d\xi &= k_{P,f} \theta_f(t) + k_{P,n} \theta_n(t) + k_{I,n} \int_0^t \theta_n(\xi) d\xi \\
&+ k_{P,p} \int_0^t \left(\{\dot{\phi}_1(\xi) - \dot{\phi}_4(\xi)\} - \ddot{\phi}_1(\xi) \tau(\xi) \right) d\xi \\
&+ k_{I,p} \int_0^t \left(\{\phi_1(\xi) - \phi_4(\xi)\} - \dot{\phi}_1(\xi) \tau(\xi) \right) d\xi
\end{aligned} \tag{7.22}$$

$$\begin{aligned}
&= k_{P,f} \theta_f(t) + k_{P,n} \theta_n(t) + k_{P,p} (\phi_1(t) - \phi_4(t)) + k_{I,n} \int_0^t \theta_n(\xi) d\xi + k_{I,p} \int_0^t (\phi_1(\xi) - \\
&\phi_4(\xi)) d\xi - k_{I,p} \int_0^t \dot{\phi}_1(\xi) \tau(\xi) d\xi - k_{P,p} \int_0^t \ddot{\phi}_1(\xi) \tau(\xi) d\xi
\end{aligned}$$

Using the formula for integration by parts, the sixth and seventh terms of Equation (7.22) can be simplified. The integration by parts formula is given by:

$$\int u dv = uv - \int v du \tag{7.23}$$

Using the formula twice, the sixth and seventh term of Equation (7.22) can be calculated as:

$$\begin{aligned}
&k_{I,p} \int_0^t \dot{\phi}_1(\xi) \tau(\xi) d\xi \\
&= k_{I,p} \left(\phi_1(t) \tau(t) - \dot{\tau}(t) \int_0^t \phi_1(\xi) d\xi \right. \\
&\quad \left. + \int_0^t \left(\int_0^\xi (\phi_1(\eta) d\eta) \right) \ddot{\tau}(\xi) d\xi \right)
\end{aligned} \tag{7.24}$$

$$k_{P,p} \int_0^t \ddot{\phi}_1(\xi) \tau(\xi) d\xi = k_{P,p} \left(\dot{\phi}_1(t) \tau(t) - \phi_1(t) \dot{\tau}(t) + \int_0^t \phi_1(\xi) \ddot{\tau}(\xi) d\xi \right)$$

Thus, Equation (7.22) can be simplified into the following format (further details are provided in appendix B).

$$\begin{aligned}
\delta_{des}(t) = & k_{P,f}\theta_f(t) + k_{P,n}\theta_n(t) + k_{P,p}(\phi_1(t) - \phi_4(t)) + k_{I,n}\int_0^t \theta_n(\xi)d\xi + \\
& k_{I,p}\int_0^t (\phi_1(\xi) - \phi_4(\xi))d\xi - k_{I,p}(\phi_1(t)\tau(t) - \\
& \dot{\tau}(t)\int_0^t \phi_1(\xi)d\xi) - k_{P,p}(\dot{\phi}_1(t)\tau(t) - \phi_1(t)\dot{\tau}(t))
\end{aligned} \tag{7.25}$$

The above equation can be reformulated based on the state variables and their integrals as:

$$\begin{aligned}
\delta_{des}(t) = & k_{P,f}\theta_f(t) + k_{P,n}\theta_n(t) + k_{P,p}(\phi_1(t) - \phi_4(t)) + k_{I,n}\int_0^t \theta_n(\xi)d\xi + \\
& k_{I,p}\int_0^t (\phi_1(\xi) - \phi_4(\xi))d\xi + \phi_1(t)(-k_{I,p}\tau(t) + k_{P,p}\dot{\tau}(t)) + \dot{\phi}_1(t)(-k_{P,p}\tau(t)) + \\
& \int_0^t \phi_1(\xi)d\xi(k_{I,p}\dot{\tau}(t))
\end{aligned} \tag{7.26}$$

In Equation (7.30), based on Equations (5.2), (5.3), (5.10), and (5.11), $\theta_n(t)$ and $\theta_f(t)$ represent functions corresponding to the state variables ($\mathbf{x}_{a,2}$) in both highway lane-change and highway lane-keeping maneuvers as:

$$\begin{aligned}
\theta_n(t) &= f_1(Y) + f_2(\phi_1) \\
\theta_f(t) &= f_3(Y) + f_4(\phi_1)
\end{aligned} \tag{7.27}$$

where:

$$f_1(Y) = \begin{cases} \arctan\left(\frac{2W - Y}{\sqrt{D_n^2 - (2W - Y)^2}}\right) & \begin{array}{l} \text{highway lane-change} \\ \text{maneuver} \end{array} \\ \arctan\left(\frac{(W + R_r)\left(1 - \sqrt{1 - \left(\frac{\Delta X_n}{W + R_r}\right)^2}\right) - Y}{D_n \sqrt{1 - \left(\frac{Y^2 + D_n^2 - 2Y(W + R_r)}{2D_n(W + R_r - Y)}\right)^2}}\right) & \begin{array}{l} \text{highway lane-keeping} \\ \text{maneuver} \end{array} \end{cases} \quad (7.28)$$

$$f_2(\phi_1) = f_4(\phi_1) = -\phi_1 \quad \begin{array}{l} \text{both highway lane-} \\ \text{change and lane-keeping} \\ \text{maneuvers} \end{array} \quad (7.29)$$

$$f_3(Y) = \begin{cases} \arctan\left(\frac{2W - Y}{\sqrt{D_f^2 - (2W - Y)^2}}\right) & \begin{array}{l} \text{highway lane-change} \\ \text{maneuver} \end{array} \\ \arctan\left(\frac{(W + R_r)\left(1 - \sqrt{1 - \left(\frac{\Delta X_f}{W + R_r}\right)^2}\right) - Y}{D_f \sqrt{1 - \left(\frac{Y^2 + D_f^2 - 2Y(W + R_r)}{2D_f(W + R_r - Y)}\right)^2}}\right) & \begin{array}{l} \text{highway lane-keeping} \\ \text{maneuver} \end{array} \end{cases} \quad (7.30)$$

It should be noted that in Equation (7.30), in the general case of varying horizontal acceleration, $\tau(t)$ and $\dot{\tau}(t)$ can be calculated using Equation (5.7). However, in the case of constant horizontal acceleration, $\tau(t)$ and $\dot{\tau}(t)$ can be determined by:

$$\begin{cases} \tau = \frac{v_X(t) - \sqrt{v_X(t)^2 - 2\bar{a}l}}{\bar{a}} \\ \dot{\tau} = 1 - \frac{v_X(t)}{\sqrt{v_X(t)^2 - 2\bar{a}l}} \end{cases} \quad \bar{a}_X = \bar{a} = cte \neq 0 \quad (7.31)$$

$$\begin{cases} \tau = \frac{l}{v_X} \\ \dot{\tau} = 0 \end{cases}, \quad l = c_1 + a_2 + c_2 + a_3 + c_3 + a_4 \quad \bar{a}_X = 0 \quad (7.32)$$

where $\bar{a}_X(t)$ and $v_X(t)$ are the vehicle acceleration and velocity in the X direction, respectively. In Equation (7.26), all terms are expressed as summation functions of the state variables ($\mathbf{x}_{a,2}$) except $k_{I,n} \int_0^t \theta_n(\xi) d\xi$, $k_{I,p} \int_0^t (\phi_1(\xi) - \phi_4(\xi)) d\xi$, and $\int_0^t \phi_1(\xi) d\xi$. Now, Equation (7.26) can be rewritten in discrete format as:

$$\begin{aligned} \delta_{des}(k) = & k_{P,f} \theta_f(k) + k_{P,n} \theta_n(k) + k_{P,p} (\phi_1(k) - \phi_4(k)) + k_{I,n} \theta_{n,int}(k) \\ & + k_{I,p} (\phi_{1,int}(k) - \phi_{4,int}(k)) + \phi_1(k) (-k_{I,p} \tau(k) + k_{P,p} \dot{\tau}(k)) \\ & + \dot{\phi}_1(k) (-k_{P,p} \tau(k)) + k_{I,p} \phi_{1,int}(k) \dot{\tau}(k) \end{aligned} \quad (7.33)$$

The integral of a state variable in the discrete format can be expressed as[167]:

$$x_{int}(k+1) = x_{int}(k) + x(k) \Delta T_s \quad (7.34)$$

where $x_{int}(k+1)$ represents the integral of the state variable up to the discrete time step $k+1$, $x_{int}(k)$ is the integral at the current time step k , $x(k)$ is the value of the state variable at the current time step k , and ΔT_s is the time step size. The dynamics of the three new state variables are described as:

$$\theta_{n,int}(k+1) = \theta_{n,int}(k) + \theta_n(k) \Delta T_s = \theta_{n,int}(k) + (f_1(Y) + f_2(\phi_1)) \Delta T_s \quad (7.35)$$

$$\phi_{1,int}(k+1) = \phi_{1,int}(k) + \phi_1(k)\Delta T_s$$

$$\begin{aligned}\phi_{1-4,int}(k+1) &= \phi_{1,int}(k+1) - \phi_{4,int}(k+1) = \phi_{1-4,int}(k) + \phi_{1-4}(k)\Delta T_s \\ &= \phi_{1-4,int}(k) - (\theta_1(k) + \theta_2(k) + \theta_3(k))\Delta T_s\end{aligned}$$

where $\phi_{1-4}(k) = \phi_1(k) - \phi_4(k)$. Finally, the system defined by Equation (7.18), will be augmented into the following system, represented by Equations (7.36) and (7.37), by incorporating three additional state variables, $\theta_{n,int}$, $\phi_{1,int}$, and $\phi_{1-4,int}$. This results in a compound nonlinear system that integrates the nonlinear MTAHV model with the Integrated MVDM driver model, operating under a single NLMPC controller to form a semi-autonomous driving controller, which is represented by:

$$\begin{Bmatrix} \mathbf{x}_2(k+1) \\ \mathbf{x}_1(k+1) \\ \mathbf{x}(k+1) \\ \theta_{n,int}(k+1) \\ \phi_{1,int}(k+1) \\ \phi_{1-4,int}(k+1) \end{Bmatrix} = A_a(\mathbf{x}_a) + B_a(\mathbf{x}_a)\mathbf{u}_c(k) \quad (7.36)$$

where:

$$\begin{aligned}\mathbf{x}_a(k) &= \begin{Bmatrix} \mathbf{x}_2(k) \\ \mathbf{x}_1(k) \\ \mathbf{x}(k) \\ \theta_{n,int}(k) \\ \phi_{1,int}(k) \\ \phi_{1-4,int}(k) \end{Bmatrix}, \\ A_a(\mathbf{x}_a) &= \begin{bmatrix} A_2\mathbf{x}_2(k) + B_2\delta_{des}(k) \\ B_1C_2\mathbf{x}_2(k) + A_1\mathbf{x}_1(k) \\ M^{-1}B_a(\mathbf{x}(k))C_1\mathbf{x}_1(k) + M^{-1}A(\mathbf{x}(k)) \\ \theta_{n,int}(k) + (f_1(Y) + f_2(\phi_1))\Delta T_s \\ \phi_{1,int}(k) + \phi_1(k)\Delta T_s \\ \phi_{1-4,int}(k) - (\theta_1(k) + \theta_2(k) + \theta_3(k))\Delta T_s \end{bmatrix},\end{aligned} \quad (7.37)$$

$$B_a(\mathbf{x}_a) = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ M^{-1}B_c(\mathbf{x}(k)) \end{bmatrix}$$

7.2.4 Semi-autonomous nonlinear MPC controller design

The main design objective of the trajectory-tracking controller is to automatically determine the steering angle for the tractor front wheels and the propulsion torque for the tractor driving wheels. It should be emphasized that the driver's steering angle ($\delta_{des}(k)$) is a function of the augmented states $\mathbf{x}_a(k)$. Additionally, as a non-cooperative semi-autonomous controller, the driver and autonomous controller might bear different objectives in controlling the vehicle. The driver and autonomous controller systems described in Section 7.2 are incorporated into the MTAHV model, enabling strategic interaction through shared access to each other's steering actions. In this framework, the driver and autonomous controller steering decisions are made based on three critical pieces of information: (i) the intended path, (ii) the vehicle's state and dynamic behavior, and (iii) the steering input from the other agent. This ensures that the vehicle tries to follow both specified paths based on the players (i.e., the autonomous controller and the driver) engagement in the game. The design of the NLMPC motion-trajectory-tracking controller is formulated as an optimization problem with the objective function expressed as

$$\begin{aligned}
J_{mpc} = & \sum_{k=0}^{P_H-1} \|\mathbf{y}_{ref,c}(k) - \mathbf{y}_{a,c}(k)\|^2_{\mathbf{Q}_c} + \sum_{k=0}^{P_H-1} \|\mathbf{y}_{ref,d}(k) - \mathbf{y}_{a,d}(k)\|^2_{\mathbf{Q}_d} \\
& + \sum_{k=0}^{C_H-1} \|\Delta \mathbf{u}_c(k)\|^2_{\mathbf{R}_c} + \sum_{k=0}^{C_H-1} \|\mathbf{u}_c(k)\|^2_{\mathbf{S}_c} + \sum_{k=0}^{P_H-1} \|\Delta u_d(k)\|^2_{\mathbf{R}_d} \\
& + \sum_{k=0}^{P_H-1} \|u_d(k)\|^2_{\mathbf{S}_d}
\end{aligned} \tag{7.38}$$

where P_H and C_H are the prediction horizon and control horizon, respectively, $\mathbf{Q}_c$, $\mathbf{Q}_d$, $\mathbf{R}_c$, $\mathbf{R}_d$, $\mathbf{S}_c$ and $\mathbf{S}_d$ are weighting matrices, which intend to minimize: 1) trajectory-tracking errors (the first and second terms on the right-hand side of Equation (7.38), 2) control actions fluctuation (the third and fifth terms on the right-hand side of Equation (7.38), and 3) control actions (the fourth and sixth terms on the right-hand side of Equation (7.38), correspondingly. It should be noted that $\mathbf{Q}_c$ and $\mathbf{Q}_d$ are associated with the NLMPC controller and the driver, respectively. The matrices $\mathbf{Q}_c$, $\mathbf{Q}_d$, $\mathbf{R}_c$, $\mathbf{R}_d$, $\mathbf{S}_c$ and $\mathbf{S}_d$ are given by:

$$\mathbf{Q}_c = \begin{bmatrix} q_{c1} & 0 & 0 & 0 \\ 0 & q_{c2} & 0 & 0 \\ 0 & 0 & q_{c3} & 0 \\ 0 & 0 & 0 & q_{c4} \end{bmatrix}, \quad \mathbf{Q}_d = \begin{bmatrix} q_{d1} & 0 & 0 \\ 0 & q_{d2} & 0 \\ 0 & 0 & q_{d3} \end{bmatrix}, \tag{7.39}$$

$$\mathbf{R}_c = \begin{bmatrix} r_{c1} & 0 \\ 0 & r_{c2} \end{bmatrix}, \quad \mathbf{S}_c = \begin{bmatrix} s_{c1} & 0 \\ 0 & s_{c2} \end{bmatrix}, \quad \mathbf{R}_d = [r_d], \quad \mathbf{S}_d = [s_d],$$

7.2.4.1 Reference motion trajectories for both semi-autonomous controller and driver

As mentioned earlier, in a non-cooperative semi-autonomous shared driving control, the driver and the controller might pursue different objectives in controlling the vehicle. For example, the controller tends to do the lane-change, but the driver wants to maintain the current lane. Thus, in this situation, there are two different reference motion

trajectories. It should be noted that, in this chapter, lateral position control is solely the responsibility of the controller. Consequently, the lateral position control for the driver is not considered. In the general case, the vector of reference motion trajectories can be defined as

$$\begin{aligned}\mathbf{R}_c(t) &= \begin{bmatrix} X_{ref,c}(t) \\ Y_{ref,c}(t) \end{bmatrix} \\ \mathbf{R}_d(t) &= Y_{ref,d}(t)\end{aligned}\tag{7.40}$$

where $\mathbf{R}_c(t)$ denotes the reference trajectory for the controller, while $\mathbf{R}_d(t)$ represents the target path for the driver. For instance, in highway driving, an SLC maneuver is conducted using a predefined testing course specified by ISO 14791 and a designated speed profile[123]. It is assumed that the vehicle follows the predefined path with a constant longitudinal acceleration. The reference motion trajectory is specified in[63] as:

$$\begin{aligned}Y_{ref,c/d}(t) &= \frac{L}{T} \left[t - \frac{T}{2\pi} \sin\left(\frac{2\pi t}{T}\right) \right] \\ X_{ref,c}(t) &= v_{e,0}t + \frac{1}{2}\bar{a}t^2\end{aligned}\tag{7.41}$$

where $Y_{ref,c/d}(t)$ is the lateral position of the target path at time t , which could be used by the controller or driver, $X_{ref,c}(t)$ represents the longitudinal position of the target path at time t , which could be used by the controller, L denotes the maximum lateral displacement during the SLC maneuver, T the time period of the sinewave lateral acceleration input by the tractor, $v_{e,0}$ the initial vehicle forward speed at the start of the maneuver, and $\bar{a}$ the constant longitudinal acceleration throughout the maneuver.

7.2.5 Kinematics for the MTAHV with semi-autonomous driving control

Figure 7-5 shows the geometry representation of the A-train double and the target path. At an arbitrary time instant t , the longitudinal (X) and lateral (Y) components for the velocity of at tractor CG can be calculated by

$$\begin{aligned}\dot{X} &= v_x \cos \phi_1 - v_y \sin \phi_1 \\ \dot{Y} &= v_x \sin \phi_1 + v_y \cos \phi_1\end{aligned}\quad (7.42)$$

where the output and reference vectors are defined as

$$\begin{aligned}\mathbf{y}_{a,c} &= [X(t), Y(t), \phi_1(t), \phi_4(t)]^T \\ \mathbf{y}_{ref,c} &= [X_{ref,c}(t), Y_{ref,c}(t), \phi_{ref,c}(t), \phi_{ref,c}(t - \tau)]^T \\ \mathbf{y}_{a,d} &= [Y(t), \phi_1(t), \phi_4(t)]^T \\ \mathbf{y}_{ref,d} &= [Y_a(t), \phi_{a_1}(t), \phi_{a_4}(t)]^T \Big|_{\mathbf{u}_c=0}\end{aligned}\quad (7.43)$$

where the time delay, τ is calculated by

$$\tau = \frac{v_X(t) - \sqrt{v_X(t)^2 - 2\bar{a}(c_1 + a_2 + c_2 + a_3 + c_3 + a_4)}}{\bar{a}} \quad (7.44)$$

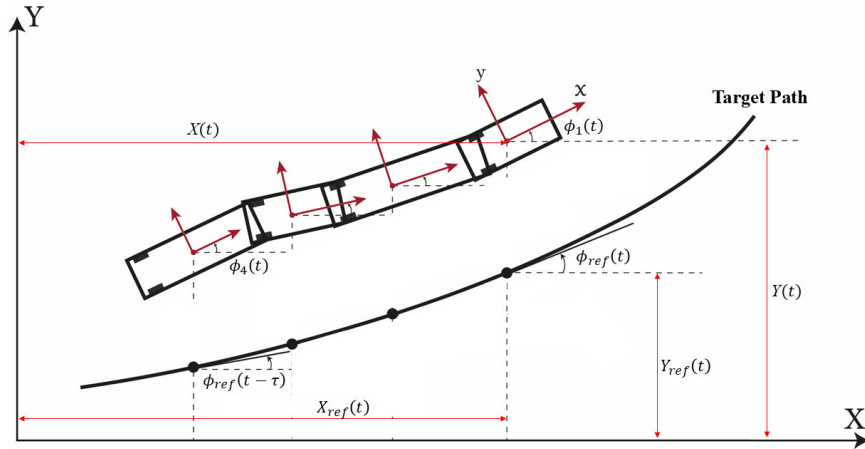


Figure 7-5: Geometry representation of the A-train double and target path.

7.3 Simulation results

In this section, a simulation study is conducted to evaluate the performance of the semi-autonomous driving control involving the two agents, i.e., the controller and the driver. A driving scenario is generated to illustrate a conflict between the driver and the controller regarding vehicle trajectory tracking on a highway. In this scenario, the controller detects a potential rear-end collision with a preceding vehicle and decides to execute a lane-change maneuver. The controller's target trajectory, which shifts the MTAHV from its current lane to the adjacent left lane, is shown in Figure 7-6. Conversely, the driver identifies a different risk — a potential collision with a motorcycle in the adjacent left lane — and opts to continue driving straight, even though this choice increases the risk of colliding with the vehicle in front. The driver's intended trajectory is also depicted in Figure 7-6. In this situation, the driver's decision to drive straight is ethically more justifiable, prioritizing the safety of the cyclist over other considerations.

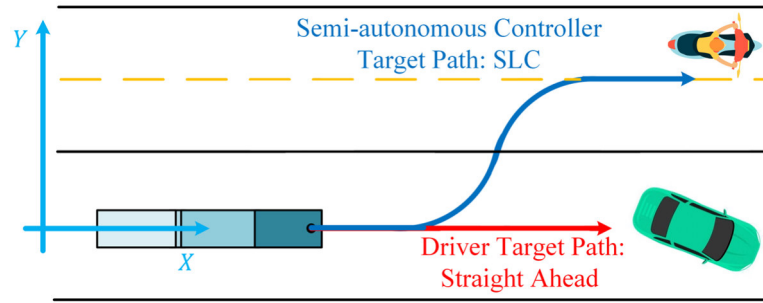


Figure 7-6: Semi-autonomous controller and driver target trajectories for a MTAHV.

Based on Figure 7-6 and Equations (7.40) and (7.41), $\mathbf{R}_c(t)$ and $\mathbf{R}_d(t)$ are given by

$$\mathbf{R}_c(t) = \begin{bmatrix} v_{e,0}t + \frac{1}{2}\bar{a}t^2 \\ \frac{L}{T} \left[t - \frac{T}{2\pi} \sin\left(\frac{2\pi t}{T}\right) \right] \end{bmatrix} \quad (7.45)$$

$$\mathbf{R}_d(t) = 0$$

The co-simulation platform depicted in Figure 7-1 and Figure 3-4 is designed to assess the performance of the non-cooperative semi-autonomous driving control in the described scenarios, which is characterized by $\mathbf{R}_c(t)$ and $\mathbf{R}_d(t)$. Table 4-1 lists the necessary parameter values for the nonlinear 11-DOF single-track yaw-plane model and the nonlinear 3-D TruckSim model. In the MPC controller design, parameters are categorized into control-based (e.g., prediction horizon P_H , control horizon C_H , and sampling time ΔT_s) and optimization-based (e.g., weighting matrices $\mathbf{Q}_c$, $\mathbf{Q}_d$, $\mathbf{R}_c$, $\mathbf{R}_d$, $\mathbf{S}_c$, $\mathbf{S}_d$ and constraints like steering angle limits δ_{max} and δ_{min}). Proper selection of these parameters significantly enhances control performance. However, in real-time applications, practical constraints often limit the flexibility in choosing optimization-based parameters[51]. In this chapter, determining the control-based parameters for the MPC controller follows the following steps:

- 1) Specify sampling time (ΔT_s): Sampling time is crucial as it depends on vehicle speed. A value of 0.05 s is effective for passenger cars performing SLC at 72 km/h[137]. However, since this chapter involves high-speed evasive maneuvers for MTAHVs, a shorter sampling time of 0.01 s is selected to improve responsiveness and performance.
- 2) Choose prediction horizon (P_H) and control horizon (C_H): The relationship $C_H \leq P_H$ must be satisfied. A larger P_H provides better anticipation of future dynamics but increases computational demands, while a larger C_H allows aggressive control actions but reduces efficiency. In autonomous vehicle MPC designs, $P_H = 10$ is commonly

used for a balance between performance and computation[137]. Following this guideline, P_H is set to 10 and C_H to 2 in this chapter as in Section 4.7.

The above guideline ensures effective parameter selection for the NLMPC controller while maintaining a balance between performance and computational feasibility. The vectors of reference motion trajectories and NLMPC controller parameters are defined in Table 7-1.

Table 7-1: Values for target path and NLMPC controller parameters and hyperparameters.

Reference Motion and Controller Parameter	Symbol	Value	Unit
Lane width	L	3.6	m
SLC time	T	4	sec
Initial velocity	$v_{e,0}$	24.44	m/s
Acceleration	$\bar{a}$	-1	m/s ²
Prediction horizon	P_H	10	
Control horizon	C_H	2	
Sampling time	ΔT_s	0.01	sec
Maximum Steering Angle	δ_{max}	$\pi/12$	rad
Minimum Steering Angle	δ_{min}	$-\pi/12$	rad
Maximum input's rate of change	$\Delta\delta_{max}$	$\pi/12$	rad/s
Minimum input's rate of change	$\Delta\delta_{min}$	$-\pi/12$	rad/s

Additionally, the values for the Integrated MVDM driver model are listed in Table 7-2.

Table 7-2: Integrated MVDM driver model parameters[86, 111].

Parameter	Symbol	Value	Unit
Driver's visibility distance	D_v	100	m
Driver's Visual near point distance	D_n	10	m
Number of time-step delay	N_d	10	
Natural frequency	ω_n	$1.5 \times 2\pi$	rad/s
Damping ratio	ζ	0.6	

To assess the performance of the NLMPC controller for the noncooperative semi-autonomous driving control, simulations are performed under two scenarios:

- 1) All six matrices $\mathbf{Q}_c$, $\mathbf{Q}_d$, $\mathbf{R}_c$, $\mathbf{R}_d$, $\mathbf{S}_c$ and $\mathbf{S}_d$ have normalized elements, and each equal to 0.5.
- 2) Since the focus of this chapter is to evaluate the final performance of the semi-autonomous driving control for steering control, all elements of the four matrices $\mathbf{R}_c$,

R_d , S_c and S_d are set to 0.5. However, matrices Q_c and Q_d are adjusted to analyze their impact on the controller's performance.

Figure 7-7 shows the reference SLC path for the autonomous controller, the reference path for the driver, and the actual paths for vehicle units. In this scenario, all the values of the six matrices Q_c , Q_d , R_c , R_d , S_c and S_d are assumed to be equal, with a value of 0.5. In addition, driver time delay and NMS dynamics are removed. It appears that the two players (i.e., autonomous controller and driver), having equal weights, have reached a balance in pulling the vehicle toward themselves. Therefore, the final path of the vehicle is positioned midway between the two target paths.

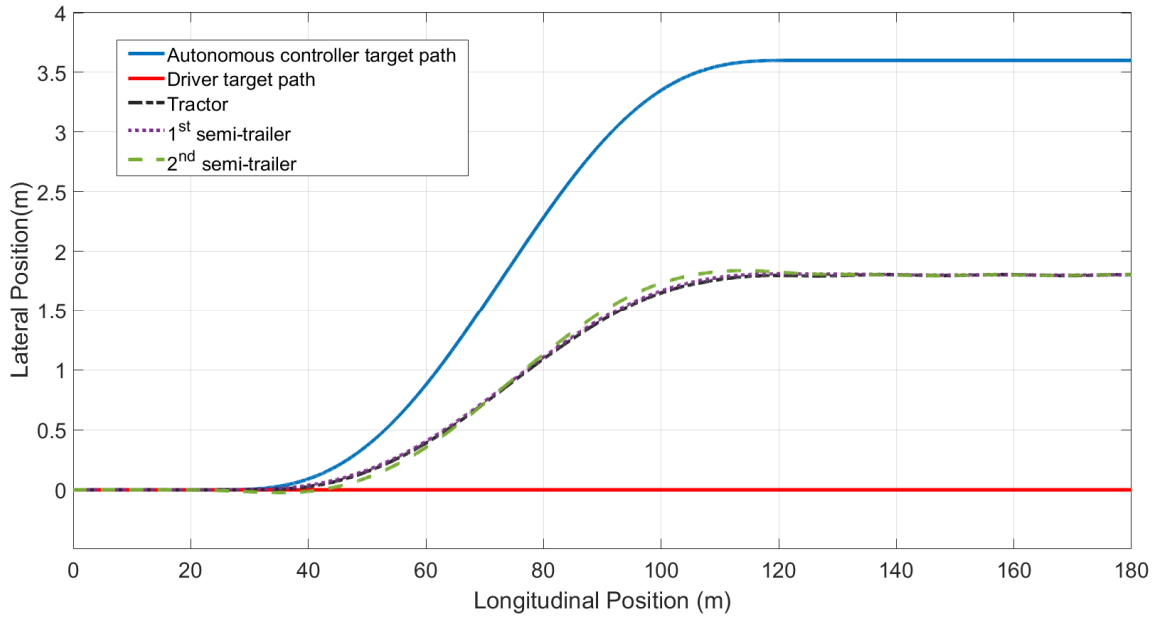


Figure 7-7: Autonomous controller and driver reference paths as well as different vehicle units' actual paths based on the non-cooperative semi-autonomous driving control.

Figure 7-8 illustrates the variations in yaw angle for different units of the MTAHV during the maneuver. At the end of the maneuver, very small oscillations are observed,

which may result from the highly nonlinear dynamics of the TruckSim model or other factors.

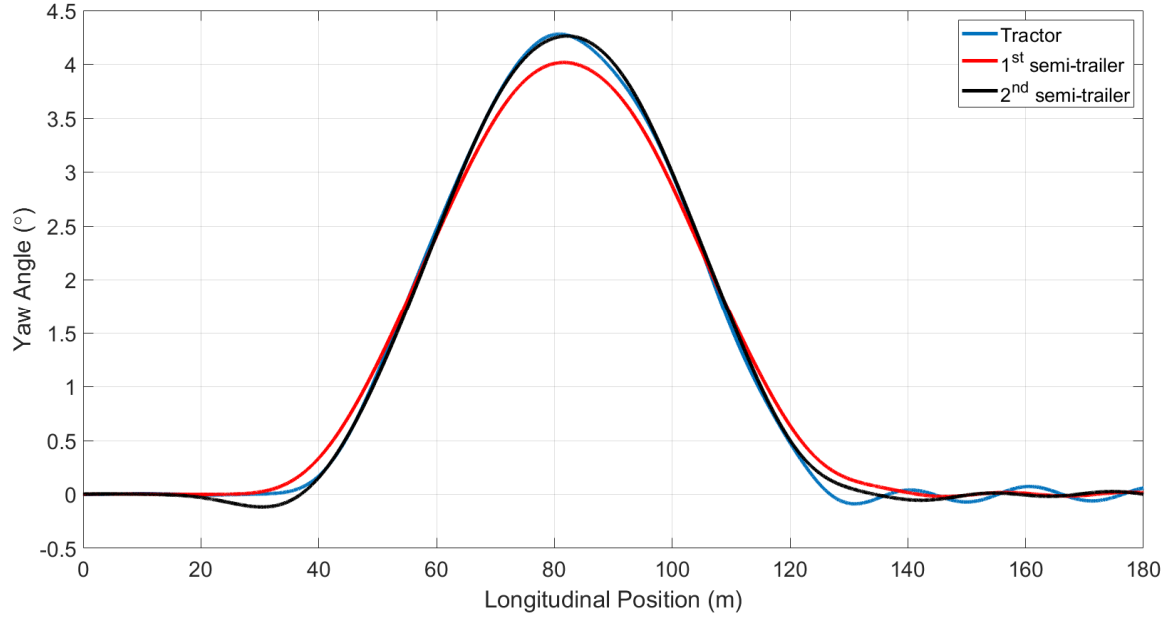


Figure 7-8: Yaw angle variation for different units during the maneuver.

In addition, the effects of varying driver and autonomous controller weights as well as driver parameters are investigated. As shown in Figure 7-9, the actual trajectories of the tractor and the second semi-trailer are shown for varying values of the Q_c and Q_d matrices, which are weights toward the NLMPC controller and the driver, respectively. As shown in the figure, increasing the weight toward one player (i.e., autonomous controller or driver) in the game (shared control) causes the final trajectory to shift toward the desired path of that player. The driver time delay and NMS dynamics are removed. It should be noted that the R_c , R_d , S_c and S_d matrices have normalized elements, each equal to 0.5.

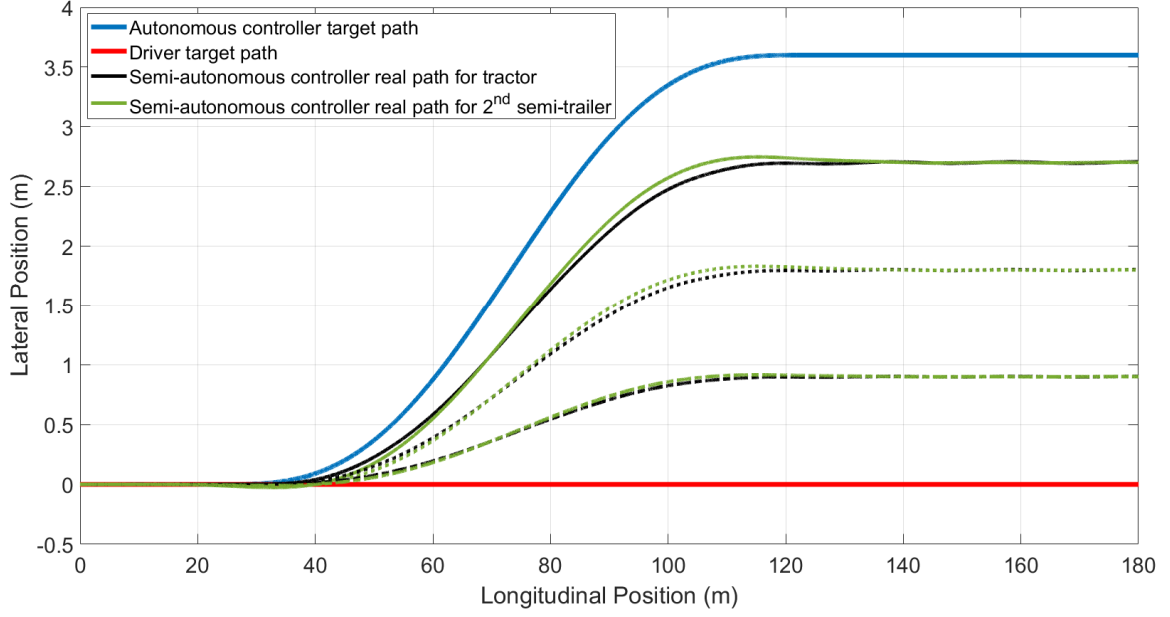
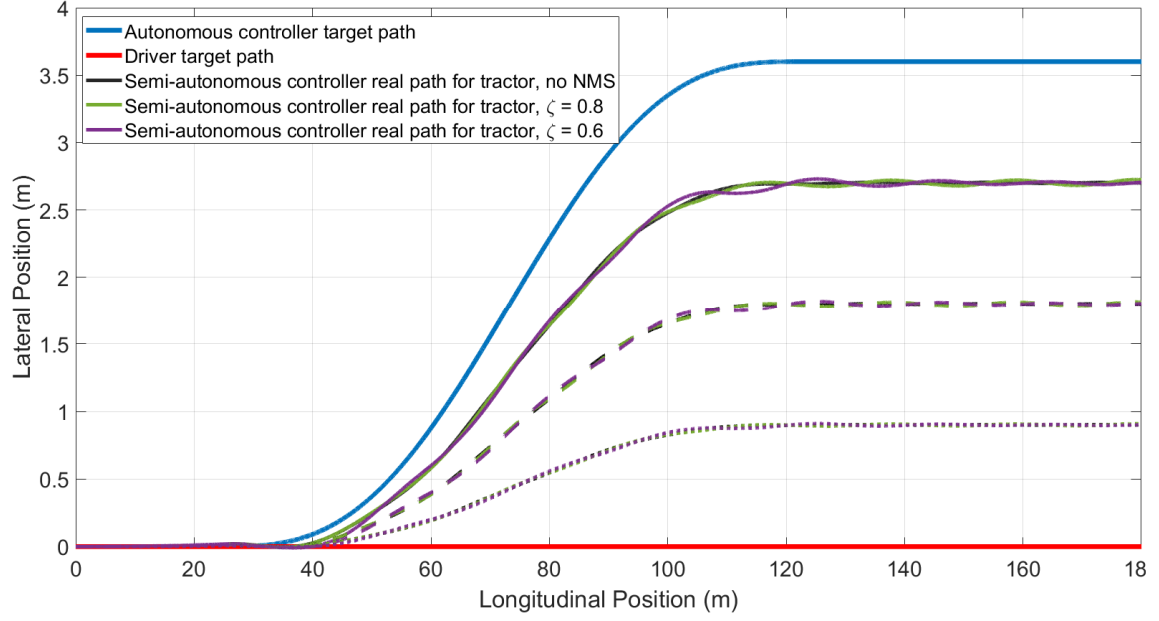


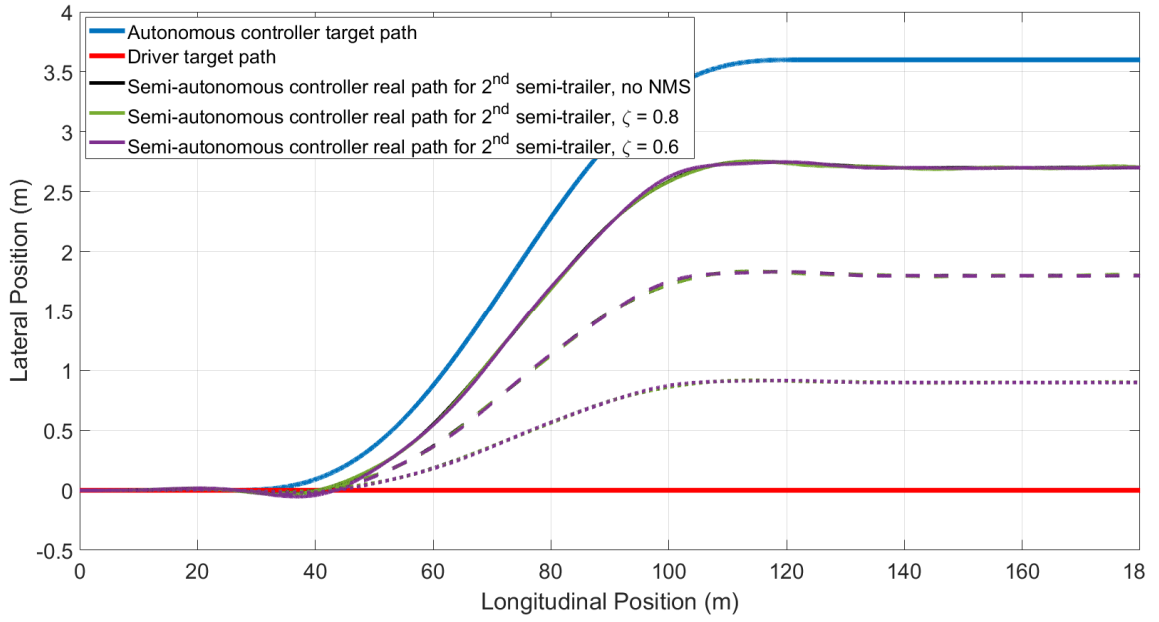
Figure 7-9: Tractor and second semi-trailer actual paths, reflecting the influence of $\mathbf{Q}_c$ and $\mathbf{Q}_d$ weighting matrices: Solid: $\mathbf{Q}_c = \text{diag}(0.66, 0.66, 0.66, 0.66)$, $\mathbf{Q}_d = \text{diag}(0.34, 0.34, 0.34)$; Dotted: $\mathbf{Q}_c = \text{diag}(0.50, 0.50, 0.50, 0.50)$, $\mathbf{Q}_d = \text{diag}(0.50, 0.50, 0.50)$; Dashed: $\mathbf{Q}_c = \text{diag}(0.34, 0.34, 0.34, 0.34)$, $\mathbf{Q}_d = \text{diag}(0.66, 0.66, 0.66)$.

Figure 7-10 illustrates the influence of the driver dynamics (arm NMS dynamics and delay block) on the semi-autonomous driving control response in presence of varying $\mathbf{Q}_c$ and $\mathbf{Q}_d$ weighting matrices. It is observed that the driver's time delay has no significant impact on the final response. This could be attributed to the fact that the NLMPC controller anticipates and compensates for the driver's delay. The figure displays the simulation results for two different values for the driver's damping ratios ζ , i.e., 0.6 and 0.8. It is observed that when the damping ratio for the driver is considered, small oscillations appear in the transient response before reaching the final value. However, the effects of these oscillations diminish as the value of the driver's weighting matrix ($\mathbf{Q}_d$) increases with respect to the autonomous controller weighting matrix ($\mathbf{Q}_c$). Figure 7-10 (a) and (b) illustrate the lateral position of the tractor and second semi-trailer, respectively. The resulting MTAHV is positioned midway between two adjacent lanes. Under normal

highway conditions, such a position is not permitted, as vehicles are required to remain within a single lane. However, within the context of this chapter, this situation represents emergency scenarios in which the driver and the NLMPC controller interact to prevent severe accidents. During these critical events, the MTAHV may intentionally occupy the space between lanes as part of a controlled evasive maneuver aimed at maintaining stability and mitigating the severity of potential collisions.



(a)



(b)

Figure 7-10: Tractor (a) and second semi-trailer (b) actual paths in the presence of driver arm NMS dynamics and delay ($\omega_n = 1.5 \times 2\pi$ rad/s, $N_d = 10$), influence of $\mathbf{Q}_c$ and $\mathbf{Q}_d$ weighting matrices: Solid: $\mathbf{Q}_c = \text{diag}(0.66, 0.66, 0.66, 0.66)$, $\mathbf{Q}_d = \text{diag}(0.34, 0.34, 0.34)$; Dotted: $\mathbf{Q}_c = \text{diag}(0.50, 0.50, 0.50, 0.50)$, $\mathbf{Q}_d = \text{diag}(0.50, 0.50, 0.50)$; Dashed: $\mathbf{Q}_c = \text{diag}(0.34, 0.34, 0.34, 0.34)$, $\mathbf{Q}_d = \text{diag}(0.66, 0.66, 0.66)$.

At this stage, with an increase in the ratio of the driver's weighting matrix ($\mathbf{Q}_d$) to the controller's weighting matrix ($\mathbf{Q}_c$), it is observed that when this ratio reaches 6.2, the

final response converges toward the driver's reference path. This phenomenon can be interpreted as a change in the behavior of a nonlinear system with the variation of one of its parameters. It appears that when the ratio of the driver's weighting matrix to the controller's weighting matrix exceeds 6.2, the region of attraction shifts, and the controller begins to focus on maintaining the current lane. This means that the driver can override the controller and take control of the MTHAV by applying a certain level of force. Figure 7-11 shows the actual paths for tractor and second semi-trailer based on the different amount of damping ratios (ζ).

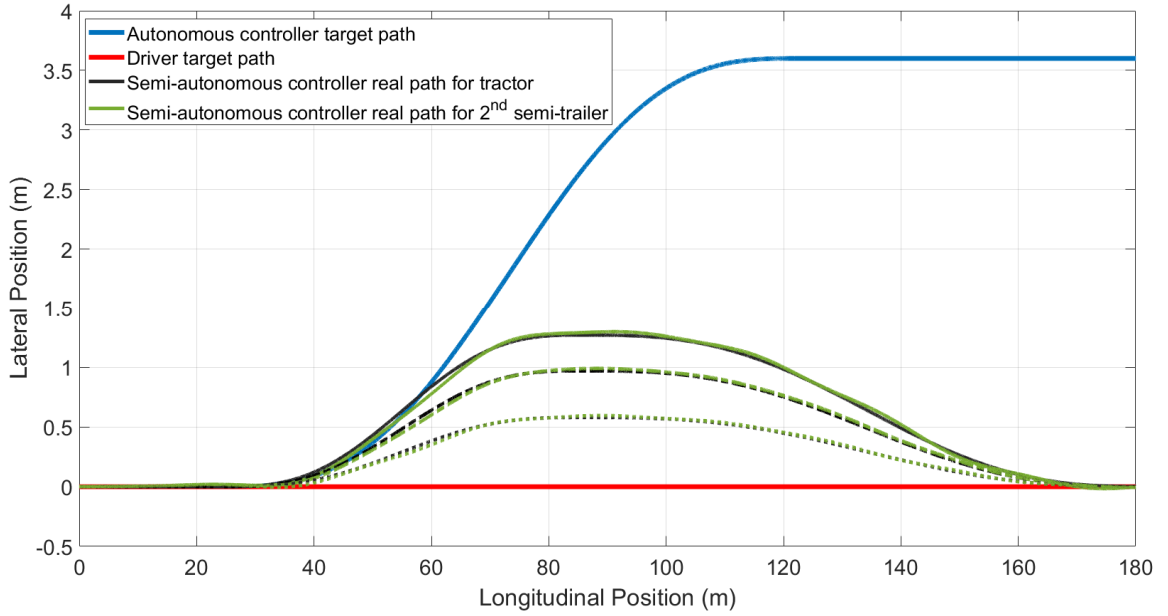


Figure 7-11: Tractor and second semi-trailer actual paths for $\mathbf{Q}_c = \text{diag}(0.125, 0.125, 0.125, 0.125)$, $\mathbf{Q}_d = \text{diag}(0.875, 0.875, 0.875)$ in the presence of driver arm NMS dynamics and delay ($\omega_n = 1.5 \times 2\pi \text{ rad/s}$, $N_d = 10$), influence of damping ratio (ζ): Solid: $\zeta = 0.6$; Dashed: $\zeta = 0.7$; Dotted: $\zeta = 0.8$.

In addition, Figure 7-12 shows the actual paths for tractor and second semi-trailer based on the different number of natural frequencies (ω_n). It is evident that as the natural frequency increases, the settling time also becomes longer.

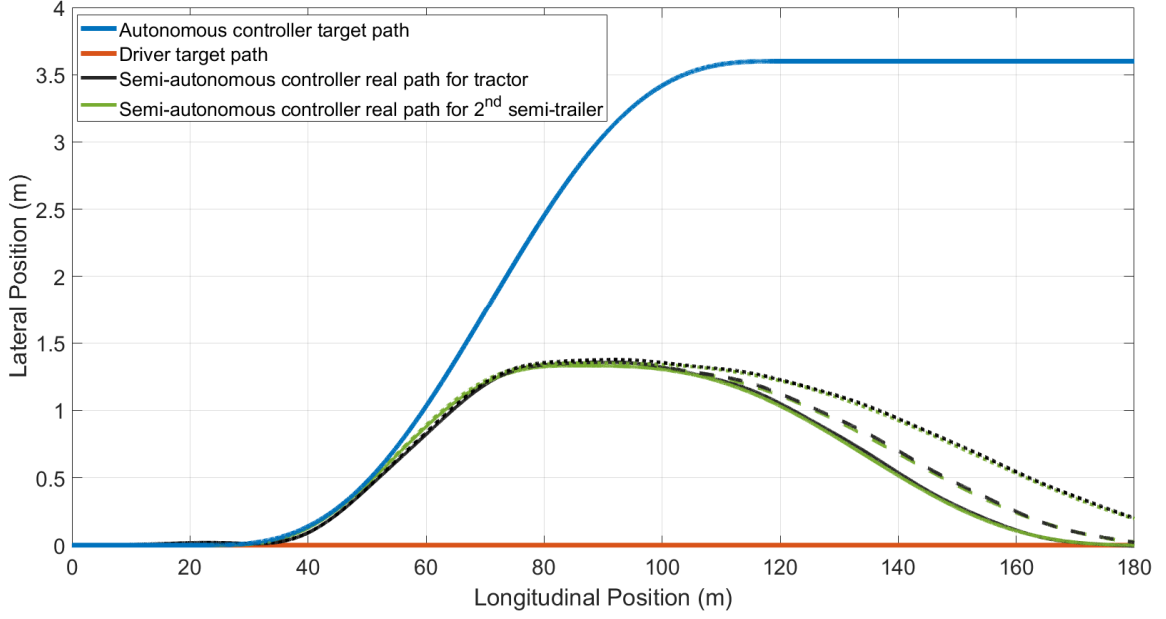


Figure 7-12: Tractor and second semi-trailer actual paths for $\mathbf{Q}_c = \text{diag}(0.125, 0.125, 0.125, 0.125)$, $\mathbf{Q}_d = \text{diag}(0.875, 0.875, 0.875)$ in the presence of driver arm NMS dynamics and delay ($\zeta = 0.6$, $N_d = 10$), influence of natural frequency (ω_n): Solid: $\omega_n = 1.5 \times 2\pi$ rad/s; Dashed: $\omega_n = 1.6 \times 2\pi$ rad/s; Dotted: $\omega_n = 1.8 \times 2\pi$ rad/s.

7.4 Summary

Conventional semi-autonomous control solutions are insufficient for MTAHVs due to their inherent nonlinearity and the complex dynamic interactions between the tractor and multiple trailers. To address this, a novel augmented system is proposed that couples an 11-DOF nonlinear MTAHV model with an Integrated MVDM driver model. This comprehensive driver model accounts for three critical human factors: preview capability based on experience, reaction time delays, and arm Neuromuscular System (NMS) dynamics. The resulting framework operates with a single steering input but manages two distinct reference trajectories—one from the driver and one from the autonomous system—providing a robust basis for analyzing lateral control under dynamic conditions.

Building on this augmented model, a NLMPC is designed to optimize steering and forward speed in real-time, ensuring the vehicle stays aligned with desired trajectories despite its nonlinear dynamics. The research concludes with a benchmark evaluation focused on non-cooperative semi-autonomous driving scenarios, where the human driver and the automated system may have conflicting objectives. This evaluation aims to resolve stability issues arising from these disagreements, offering solutions for complex, real-world situations where the driver and the controller are effectively competing for control of the vehicle.

It should be noted that, when the controller overrides the driver, it utilizes an arbitration logic that applies a torque overlay, effectively adding a counterforce to the steering wheel to physically resist the driver's unsafe inputs. This creates a conflict phase where the driver feels the wheel stiffen or pull away, but to comply with safety standards like ISO 26262, the system includes a break-through threshold; if the driver fights this resistance hard enough, the controller must instantly disengage. This sudden release of counter-torque often results in the pop-through effect, a critical instability risk where the vehicle violently lurches in the direction of the driver's exertion, potentially causing oscillations or loss of control that must be carefully accounted for in future simulations.

Chapter 8

Conclusions

8.1 Summary

This PhD thesis explores the semi-autonomous driving control of MTAHVs, addressing critical challenges in autonomous controller design, driver modeling, decision-making, and semi-autonomous driving control. The major conclusions drawn from the study are briefly summarized as follows:

- 1) A design framework is proposed for developing NLMPC-based trajectory-tracking controllers for autonomous MTAHVs. With the objectives of improving the lateral stability, enhancing path-tracking performance, achieving good ride comfort, the weighting matrices for the NLMPC-based trajectory-tracking controllers are automatically tuned off-line. Simulations demonstrate the reliability of the proposed design approach and the robustness of the NLMPC tracking-controller.
- 2) A new driver model is developed for MTAHVs. This driver model considers human drivers' visual perception mechanisms and the motion clues of trailers, and it incorporates the effects of driver time-delay and arm neuromuscular dynamics. The driver model is validated against on-road measurement data, demonstrating its strong alignment and improved accuracy compared to existing driver models.
- 3) A deep reinforcement learning based motion decision-making scheme is proposed and developed. The DRL-based motion decision-making scheme is designed for

autonomous and semi-autonomous MTAHVs operating on highways with multiple lanes. Numerical simulations demonstrate the effectiveness of the DRL-based motion decision-making scheme.

- 4) A non-cooperative game based semi-autonomous driving control strategy is proposed and designed for MTAHVs. The proposed semi-autonomous driving control strategy provides valuable guidelines and effective solutions to the problems arising from semi-autonomous driving control of MTAHVs, thereby handling complex and real-world scenarios, where the driver and automated system may have distinct driving intentions. Through an innovative study, this thesis makes original contributions to the advancement of semi-autonomous MTAHVs.

8.2 Conclusions

This research has generated critical insights into the modeling, autonomous control, decision making, and semi-autonomous control for MTAHVs highway driving. The key scientific conclusions drawn from the preceding chapters are detailed below.

- 1) The validation of the derived 11-DOF nonlinear model demonstrated that planar (2D) dynamics are sufficient for reproducing standard highway maneuvers, showing strong correlation with high-fidelity TruckSim benchmarks. However, a critical boundary to this model's validity was identified regarding dynamic vertical tire loads. The assumption of static normal tire loads creates a blind spot regarding limit-handling stability. It is concluded that while full multi-body physics are computationally prohibitive for real-time control, the integration of a quasi-static load transfer

approximation is a requisite for future prediction plants to accurately anticipate rollover risks during high-lateral-acceleration events.

- 2) This research established that although the proposed offline NLMPC weighting matrices tuning has great improvement in path-following accumulated error, future attempts should incorporate other methods such as Adaptive Model Predictive Control (AMPC) to enable real-time modulation of weighting matrices in response to evolving driving conditions. Furthermore, the validation framework requires expansion to include rigorous quantitative benchmarking against high-complexity operational scenarios, such as the double lane change.
- 3) The development of the Integrated MVDM highlighted the importance of based on the visual information and second trailer relative heading angle as well as incorporating arm neuromuscular system dynamics and reaction time delays achieved a higher degree of correlation with real-world validation data compared to idealized mathematical driver models. This finding confirms that reliable validation of semi-autonomous systems requires virtual agents that replicate human imperfections, specifically the physiological delays and stiffness limitations inherent to manual driving.
- 4) The comprehensive co-simulation framework for decision-making, controller/driver modelling and vehicle plant reveals a significant lesson that complexity does not equal safety. Throughout this thesis, two different agents was considered as the decision makers: 1- NLMPC controller, and 2- Integrated MVDM driver model. The simulations showed a fundamental trade-off. The NLMPC controller which is enriched with more surrounding vehicles data is excellent at efficiency—it drove faster and made smoother lane changes in traffic. However, the simpler model

(Integrated MVDM) was better at safety, achieving a higher collision-avoidance rate (98.2%). This says that a single super-algorithm is not the answer. Instead, the future of autonomous trucking likely requires a hybrid architecture: using the smart, efficient controller for highway cruising, but instantly switching to the robust driver during emergencies.

- 5) Finally, investigating the stability limits of a semi-autonomous A-train double during non-cooperative driving conflicts shows that the human driver and vehicle must be modeled as a coupled augmented system rather than separate entities. Simulations reveal a critical non-linear bifurcation threshold: when the ratio of the driver's control weight to the controller's weight exceeds 6.2, the system instantaneously shifts authority to the driver, confirming the physical existence of a break-through torque threshold required by ISO 26262. However, the study identifies a dangerous pop-through effect associated with this binary handover, where the sudden release of steering torque combined with the driver's tensed neuromuscular impedance causes violent lateral lurching and prolonged oscillation. Consequently, the research concludes that binary arbitration logic is operationally unsafe for articulated vehicles and recommends the implementation of variable impedance control to ensure smooth, dissipative soft handovers that prevent instability during emergency interventions.

8.3 Recommendations for future studies

In addition to all presented recommendations in previous section, for future studies in the field of semi-autonomous driving, the following directions are proposed for consideration:

- 1) **Advanced driver-vehicle interaction models** Develop more sophisticated driver models that account for human emotions, fatigue, and reaction time variability to enhance shared control systems.
- 2) **Exploration of complex road scenarios** Extend the validation of control algorithms to include urban environments, off-road conditions, and adverse weather scenarios.
- 3) **Real-time Implementation and Testing** Transition from simulation to hardware-in-the-loop (HIL) and full-scale vehicle testing to evaluate real-world performance of the control strategies.
- 4) **Integration with V2X Communication** Investigate the impact of Vehicle-to-Everything (V2X) communication on semi-autonomous control, enabling better coordination between MTAHVs and surrounding road traffic.

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Appendices

Appendix A

$$M_{11} = m_1 + m_2 + m_3 + m_4$$

$$M_{12} = -((a_3 + c_2)m_3 + (a_3 + a_4 + c_2 + c_3)m_4 + (m_2 + m_3 + m_4)a_2 + (m_2 + m_3 + m_4)c_1)$$

$$M_{14} = -((a_3 + c_2)m_3 + (a_3 + a_4 + c_2 + c_3)m_4 + (m_2 + m_3 + m_4)a_2)$$

$$M_{16} = -(a_3m_3 + (a_3 + a_4 + c_3)m_4)$$

$$M_{18} = -a_4m_4$$

$$M_{21} = -(a_2 + c_1)m_2 - (a_2 + a_3 + c_1 + c_2)m_3 - (a_2 + a_3 + a_4 + c_1 + c_2 + c_3)m_4$$

$$M_{22} = j_1 + j_2 + j_3 + j_4 + ((a_2 + c_1)^2)m_2 + ((a_2 + a_3 + c_1 + c_2)^2)m_3 \\ + ((a_2 + a_3 + a_4 + c_1 + c_2 + c_3)^2)m_4$$

$$M_{24} = j_2 + j_3 + j_4 + a_2(a_2 + c_1)m_2 + (a_2 + a_3 + c_2)(a_2 + a_3 + c_1 + c_2)m_3 + (a_2 + a_3 + a_4 \\ + c_2 + c_3)(a_2 + a_3 + a_4 + c_1 + c_2 + c_3)m_4$$

$$M_{26} = j_3 + j_4 + a_3(a_2 + a_3 + c_1 + c_2)m_3 + (a_3 + a_4 + c_3)(a_2 + a_3 + a_4 + c_1 + c_2 + c_3)m_4$$

$$M_{28} = j_4 + a_4(a_2 + a_3 + a_4 + c_1 + c_2 + c_3)m_4$$

$$M_{31} = -a_2m_2 - (a_2 + a_3 + c_2)m_3 - (a_2 + a_3 + a_4 + c_2 + c_3)m_4$$

$$M_{32} = j_2 + j_3 + j_4 + a_2(a_2 + c_1)m_2 + (a_2 + a_3 + c_2)(a_2 + a_3 + c_1 + c_2)m_3 + (a_2 + a_3 + a_4 \\ + c_2 + c_3)(a_2 + a_3 + a_4 + c_1 + c_2 + c_3)m_4$$

$$M_{34} = j_2 + j_3 + j_4 + (a_2^2)m_2 + (a_2 + a_3 + c_2)^2m_3 + (a_2 + a_3 + a_4 + c_2 + c_3)^2m_4$$

$$M_{36} = j_3 + j_4 + a_3(a_2 + a_3 + c_2)m_3 + (a_3 + a_4 + c_3)(a_2 + a_3 + a_4 + c_2 + c_3)m_4$$

$$M_{38} = j_4 + a_4(a_2 + a_3 + a_4 + c_2 + c_3)m_4$$

$$M_{41} = -a_3m_3 - (a_3 + a_4 + c_3)m_4$$

$$M_{42} = j_3 + j_4 + a_3(a_2 + a_3 + c_1 + c_2)m_3 + (a_3 + a_4 + c_3)(a_2 + a_3 + a_4 + c_1 + c_2 + c_3)m_4$$

$$M_{44} = j_3 + j_4 + a_3(a_2 + a_3 + c_2)m_3 + (a_3 + a_4 + c_3)(a_2 + a_3 + a_4 + c_2 + c_3)m_4$$

$$M_{46} = j_3 + j_4 + m_3a_3^2 + m_4(a_3 + a_4 + c_3)^2$$

$$M_{48} = j_4 + a_4(a_3 + a_4 + c_3)m_4$$

$$M_{51} = -a_4m_4$$

$$M_{52} = j_4 + a_4(a_2 + a_3 + a_4 + c_1 + c_2 + c_3)m_4$$

$$M_{54} = j_4 + a_4(a_2 + a_3 + a_4 + c_2 + c_3)m_4$$

$$M_{56} = j_4 + a_4(a_3 + a_4 + c_3)m_4$$

$$M_{58} = j_4 + a_4^2m_4$$

$$M_{63} = 1$$

$$M_{75} = 1$$

$$M_{87} = 1$$

$$a_{11} = -(C_{s1f} + C_{s1r} + C_{s2r} + C_{s3r} + C_{s4r})/V_X$$

$$a_{12} = (-a_1C_{s1f} + b_1C_{s1r} + (a_2 + b_2 + c_1)C_{s2r} + (a_2 + a_3 + b_3 + c_1 + c_2)C_{s3r}$$

$$+ (a_2 + a_3 + a_4 + b_4 + c_1 + c_2 + c_3)C_{s4r} - m_1V_X^2 - m_2V_X^2 - m_3V_X^2 - m_4V_X^2)/V_X$$

$$a_{13} = C_{s2r} + C_{s3r} + C_{s4r}$$

$$a_{14} = ((a_2 + b_2)C_{s2r} + (a_2 + a_3 + b_3 + c_2)C_{s3r} + (a_2 + a_3 + a_4 + b_4 + c_2 + c_3)C_{s4r})/V_X$$

$$a_{15} = C_{s3r} + C_{s4r}$$

$$a_{16} = ((a_3 + b_3)C_{s3r} + (a_3 + a_4 + b_4 + c_3)C_{s4r})/V_X$$

$$a_{17} = C_{s4r}$$

$$a_{18} = ((a_4 + b_4)C_{s4r})/V_X$$

$$a_{21} = (-a_1C_{s1f} + b_1C_{s1r} + (a_2 + b_2 + c_1)C_{s2r} + (a_2 + a_3 + b_3 + c_1 + c_2)C_{s3r} + (a_2 + a_3 + a_4 + b_4 + c_1 + c_2 + c_3)C_{s4r})/V_X$$

$$a_{22} = -((a_1^2C_{s1f} + b_1^2C_{s1r} + a_2^2C_{s2r} + 2a_2b_2C_{s2r} + b_2^2C_{s2r} + 2a_2c_1C_{s2r} + 2b_2c_1C_{s2r} + c_1^2C_{s2r} + a_2^2C_{s3r} + 2a_2a_3C_{s3r} + a_3^2C_{s3r} + 2a_2b_3C_{s3r} + 2a_3b_3C_{s3r} + b_3^2C_{s3r} + 2a_2c_1C_{s3r} + 2a_3c_1C_{s3r} + 2b_3c_1C_{s3r} + c_1^2C_{s3r} + 2a_2c_2C_{s3r} + 2a_3c_2C_{s3r} + 2b_3c_2C_{s3r} + 2c_1c_2C_{s3r} + c_2^2C_{s3r} + a_2^2C_{s4r} + 2a_2a_3C_{s4r} + a_3^2C_{s4r} + 2a_2a_4C_{s4r} + 2a_3a_4C_{s4r} + a_4^2C_{s4r} + 2a_2b_4C_{s4r} + 2a_3b_4C_{s4r} + 2a_4b_4C_{s4r} + b_4^2C_{s4r} + 2a_2c_1C_{s4r} + 2a_3c_1C_{s4r} + 2a_4c_1C_{s4r} + 2b_4c_1C_{s4r} + c_1^2C_{s4r} + 2a_2c_2C_{s4r} + 2a_3c_2C_{s4r} + 2a_4c_2C_{s4r} + 2b_4c_2C_{s4r} + 2c_1c_2C_{s4r} + c_2^2C_{s4r} + 2a_2c_3C_{s4r} + 2a_3c_3C_{s4r} + 2a_4c_3C_{s4r} + 2b_4c_3C_{s4r} + 2c_1c_3C_{s4r} + 2c_2c_3C_{s4r} + c_3^2C_{s4r} - a_2m_2V_X^2 - c_1m_2V_X^2 - a_2m_3V_X^2 - a_3m_3V_X^2 - c_1m_3V_X^2 - c_2m_3V_X^2 - a_2m_4V_X^2 - a_3m_4V_X^2 - a_4m_4V_X^2 - c_1m_4V_X^2 - c_2m_4V_X^2 - c_3m_4V_X^2)/V_X)$$

$$a_{23} = -b_2C_{s2r} - c_1C_{s2r} - a_3C_{s3r} - b_3C_{s3r} - c_1C_{s3r} - c_2C_{s3r} - a_3C_{s4r} - a_4C_{s4r} - b_4C_{s4r} - c_1C_{s4r} - c_2C_{s4r} - c_3C_{s4r} - a_2(C_{s2r} + C_{s3r} + C_{s4r})$$

$$\begin{aligned}
a_{24} = & -((b_2^2 C_{s2r} + b_2 c_1 C_{s2r} + a_3^2 C_{s3r} + 2a_3 b_3 C_{s3r} + b_3^2 C_{s3r} + a_3 c_1 C_{s3r} + b_3 c_1 C_{s3r} \\
& + 2a_3 c_1 C_{s3r} + 2b_3 c_2 C_{s3r} + c_1 c_2 C_{s3r} + c_2^2 C_{s3r} + a_3^2 C_{s4r} + 2a_3 a_4 C_{s4r} \\
& + a_4^2 C_{s4r} + 2a_3 b_4 C_{s4r} + 2a_4 b_4 C_{s4r} + b_4^2 C_{s4r} + a_3 c_1 C_{s4r} + a_4 c_1 C_{s4r} \\
& + b_4 c_1 C_{s4r} + 2a_3 c_2 C_{s4r} + 2a_4 c_2 C_{s4r} + 2b_4 c_2 C_{s4r} + c_1 c_2 C_{s4r} + c_2^2 C_{s4r} \\
& + 2a_3 c_3 C_{s4r} + 2a_4 c_3 C_{s4r} + 2b_4 c_3 C_{s4r} + c_1 c_3 C_{s4r} + 2c_2 c_3 C_{s4r} + c_3^2 C_{s4r} \\
& + a_2^2 (C_{s2r} + C_{s3r} + C_{s4r}) + a_2 (2b_2 C_{s2r} + c_1 (C_{s2r} + C_{s3r} + C_{s4r}) + 2(b_3 C_{s3r} \\
& + c_2 C_{s3r} + a_4 C_{s4r} + b_4 C_{s4r} + c_2 C_{s4r} + c_3 C_{s4r} + a_3 (C_{s3r} + C_{s4r}))))/V_X)
\end{aligned}$$

$$\begin{aligned}
a_{25} = & -b_3 C_{s3r} - c_1 C_{s3r} - c_2 C_{s3r} - a_4 C_{s4r} - b_4 C_{s4r} - c_1 C_{s4r} - c_2 C_{s4r} - c_3 C_{s4r} - a_2 (C_{s3r} \\
& + C_{s4r}) - a_3 (C_{s3r} + C_{s4r})
\end{aligned}$$

$$\begin{aligned}
a_{26} = & -((b_3^2 C_{s3r} + b_3 c_1 C_{s3r} + b_3 c_2 C_{s3r} + a_4^2 C_{s4r} + 2a_4 b_4 C_{s4r} + b_4^2 C_{s4r} + a_4 c_1 C_{s4r} \\
& + b_4 c_1 C_{s4r} + a_4 c_2 C_{s4r} + b_4 c_2 C_{s4r} + 2a_4 c_3 C_{s4r} + 2b_4 c_3 C_{s4r} + c_1 c_3 C_{s4r} \\
& + c_2 c_3 C_{s4r} + c_3^2 C_{s4r} + a_3^2 (C_{s3r} + C_{s4r}) + a_2 (b_3 C_{s3r} + (a_4 + b_4 + c_3) C_{s4r} \\
& + a_3 (C_{s3r} + C_{s4r})) + a_3 (2b_3 C_{s3r} + c_2 C_{s3r} + 2a_4 C_{s4r} + 2b_4 C_{s4r} + c_2 C_{s4r} \\
& + 2c_3 C_{s4r} + c_1 (C_{s3r} + C_{s4r}))))/V_X)
\end{aligned}$$

$$a_{27} = -(a_2 + a_3 + a_4 + b_4 + c_1 + c_2 + c_3) C_{s4r}$$

$$a_{28} = -((a_4 + b_4)(a_2 + a_3 + a_4 + b_4 + c_1 + c_2 + c_3) C_{s4r})/V_X$$

$$\begin{aligned}
a_{31} = & (b_2 C_{s2r} + (a_3 + b_3 + c_2) C_{s3r} + (a_3 + a_4 + b_4 + c_2 + c_3) C_{s4r} + a_2 (C_{s2r} + C_{s3r} \\
& + C_{s4r}))/V_X
\end{aligned}$$

$$\begin{aligned}
a_{32} = & -((b_2^2 C_{s2r} + b_2 c_1 C_{s2r} + (a_3 + b_3)^2 C_{s3r} + a_3 c_1 C_{s3r} + b_3 c_1 C_{s3r} + 2a_3 c_2 C_{s3r} \\
& + 2b_3 c_2 C_{s3r} + c_1 c_2 C_{s3r} + c_2^2 C_{s3r} + a_3^2 C_{s4r} + 2a_3 a_4 C_{s4r} + a_4^2 C_{s4r} \\
& + 2a_3 b_4 C_{s4r} + 2a_4 b_4 C_{s4r} + b_4^2 C_{s4r} + a_3 c_1 C_{s4r} + a_4 c_1 C_{s4r} + b_4 c_1 C_{s4r} \\
& + 2a_3 c_2 C_{s4r} + 2a_4 c_2 C_{s4r} + 2b_4 c_2 C_{s4r} + c_1 c_2 C_{s4r} + c_2^2 C_{s4r} + 2a_3 c_3 C_{s4r} \\
& + 2a_4 c_3 C_{s4r} + 2b_4 c_3 C_{s4r} + c_1 c_3 C_{s4r} + 2c_2 c_3 C_{s4r} + c_3^2 C_{s4r} + a_2^2 (C_{s2r} \\
& + C_{s3r} + C_{s4r}) - ((a_3 + c_2)m_3 + (a_3 + a_4 + c_2 + c_3)m_4)V_X^2 + a_2(2b_2 C_{s2r} \\
& + 2(a_3 + b_3 + c_2)C_{s3r} + 2(a_3 + a_4 + b_4 + c_2 + c_3)C_{s4r} + c_1(C_{s2r} + C_{s3r} \\
& + C_{s4r}) - (m_2 + m_3 + m_4)V_X^2))/V_X)
\end{aligned}$$

$$\begin{aligned}
a_{33} = & -b_2 C_{s2r} - a_3 C_{s3r} - b_3 C_{s3r} - c_2 C_{s3r} - a_3 C_{s4r} - a_4 C_{s4r} - b_4 C_{s4r} - c_2 C_{s4r} - c_3 C_{s4r} \\
& - a_2(C_{s2r} + C_{s3r} + C_{s4r})
\end{aligned}$$

$$\begin{aligned}
a_{34} = & -((b_2^2 C_{s2r} + a_3^2 C_{s3r} + 2a_3 b_3 C_{s3r} + b_3^2 C_{s3r} + 2a_3 c_2 C_{s3r} + 2b_3 c_2 C_{s3r} + c_2^2 C_{s3r} \\
& + a_3^2 C_{s4r} + 2a_3 a_4 C_{s4r} + a_4^2 C_{s4r} + 2a_3 b_4 C_{s4r} + 2a_4 b_4 C_{s4r} + b_4^2 C_{s4r} \\
& + 2a_3 c_2 C_{s4r} + 2a_4 c_2 C_{s4r} + 2b_4 c_2 C_{s4r} + c_2^2 C_{s4r} + a_3 c_3 C_{s4r} + 2a_4 c_3 C_{s4r} \\
& + 2b_4 c_3 C_{s4r} + 2c_2 c_3 C_{s4r} + c_3^2 C_{s4r} + a_2^2 (C_{s2r} + C_{s3r} + C_{s4r}) + 2a_2(b_2 C_{s2r} \\
& + b_3 C_{s3r} + c_2 C_{s3r} + a_4 C_{s4r} + b_4 C_{s4r} + c_2 C_{s4r} + c_3 C_{s4r} + a_3(C_{s3r} \\
& + C_{s4r}))))/V_X)
\end{aligned}$$

$$\begin{aligned}
a_{35} = & -b_3 C_{s3r} - c_2 C_{s3r} - a_4 C_{s4r} - b_4 C_{s4r} - c_2 C_{s4r} - c_3 C_{s4r} - a_2(C_{s3r} + C_{s4r}) - a_3(C_{s3r} \\
& + C_{s4r})
\end{aligned}$$

$$\begin{aligned}
a_{36} = & -((b_3^2 C_{s3r} + b_3 c_2 C_{s3r} + a_4^2 C_{s4r} + 2a_4 b_4 C_{s4r} + b_4^2 C_{s4r} + a_4 c_2 C_{s4r} + b_4 c_2 C_{s4r} \\
& + 2a_4 c_3 C_{s4r} + 2b_4 c_3 C_{s4r} + c_2 c_3 C_{s4r} + c_3^2 C_{s4r} + a_3^2 (C_{s3r} + C_{s4r}) \\
& + a_2(b_3 C_{s3r} + (a_4 + b_4 + c_3)C_{s4r} + a_3(C_{s3r} + C_{s4r})) + a_3(2b_3 C_{s3r} + 2(a_4 \\
& + b_4 + c_3)C_{s4r} + c_2(C_{s3r} + C_{s4r}))))/V_X)
\end{aligned}$$

$$a_{37} = -(a_3 + a_4 + b_4 + c_2 + c_3)C_{s4r}$$

$$a_{38} = -(((a_4 + b_4)(a_2 + a_3 + a_4 + b_4 + c_2 + c_3)C_{s4r})/V_X)$$

$$a_{41} = ((a_3 + b_3)C_{s3r} + (a_3 + a_4 + b_4 + c_3)C_{s4r})/V_X$$

$$\begin{aligned} a_{42} = & -((b_3^2 C_{s3r} + a_2(a_3 + b_3)C_{s3r} + b_3 c_1 C_{s3r} + b_3 c_2 C_{s3r} + (a_4 + b_4)^2 C_{s4r} + a_4 c_1 C_{s4r} \\ & + b_4 c_1 C_{s4r} + a_4 c_2 C_{s4r} + b_4 c_2 C_{s4r} + 2a_4 c_3 C_{s4r} + 2b_4 c_3 C_{s4r} + c_1 c_3 C_{s4r} \\ & + c_2 c_3 C_{s4r} + c_3^2 C_{s4r} + a_2(a_3 + a_4 + b_4 + c_3)C_{s4r} + a_3^2(C_{s3r} + C_{s4r}) - (a_4 \\ & + c_3)m_4 V_X^2 + a_3((2b_3 + c_1 + c_2)C_{s3r} + 2(a_4 + b_4)C_{s4r} + (c_1 + c_2 \\ & + 2c_3)C_{s4r} - (m_3 + m_4)V_X^2))/V_X) \end{aligned}$$

$$a_{43} = -b_3 C_{s3r} - (a_4 + b_4 + c_3)C_{s4r} - a_3(C_{s3r} + C_{s4r})$$

$$\begin{aligned} a_{44} = & -((b_3^2 C_{s3r} + b_3 c_2 C_{s3r} + a_4^2 C_{s4r} + 2a_4 b_4 C_{s4r} + b_4^2 C_{s4r} + a_4 c_2 C_{s4r} + b_4 c_2 C_{s4r} \\ & + 2a_4 c_3 C_{s4r} + 2b_4 c_3 C_{s4r} + c_2 c_3 C_{s4r} + c_3^2 C_{s4r} + a_3^2(C_{s3r} + C_{s4r}) \\ & + a_2(b_3 C_{s3r} + (a_4 + b_4 + c_3)C_{s4r} + a_3(C_{s3r} + C_{s4r})) + a_3(2b_3 C_{s3r} + 2(a_4 \\ & + b_4 + c_3)C_{s4r} + c_2(C_{s3r} + C_{s4r}))) / V_X) \end{aligned}$$

$$a_{45} = -b_3 C_{s3r} - (a_4 + b_4 + c_3)C_{s4r} - a_3(C_{s3r} + C_{s4r})$$

$$\begin{aligned} a_{46} = & -((b_3^2 C_{s3r} + (a_4 + b_4 + c_3)^2 C_{s4r} + a_3^2(C_{s3r} + C_{s4r}) + 2a_3(b_3 C_{s3r} + (a_4 + b_4 \\ & + c_3)C_{s4r}))/V_X) \end{aligned}$$

$$a_{47} = -(a_3 + a_4 + b_4 + c_3)C_{s4r}$$

$$a_{48} = -(((a_4 + b_4)(a_3 + a_4 + b_4 + c_3)C_{s4r})/V_X)$$

$$a_{51} = ((a_4 + b_4)C_{s4r})/V_X$$

$$a_{52} = -(((a_4 + b_4)(a_2 + a_3 + a_4 + b_4 + c_1 + c_2 + c_3)C_{s4r} - a_4 m_4 V_X^2)/V_X)$$

$$a_{53} = -(a_4 + b_4)C_{s4r}$$

$$a_{54} = -(((a_4 + b_4)(a_2 + a_3 + a_4 + b_4 + c_2 + c_3)C_{s4r})/V_X)$$

$$a_{55} = - (a_4 + b_4)C_{s4r}$$

$$a_{56} = - (((a_4 + b_4)(a_3 + a_4 + b_4 + c_3)C_{s4r})/V_X)$$

$$a_{57} = - (a_4 + b_4)C_{s4r}$$

$$a_{58} = - (((a_4 + b_4)^2 C_{s4r})/V_X)$$

Appendix B

$$\int_0^t \phi_1(\xi) \ddot{\tau}(\xi) d\xi = ?$$

$$\tau(\xi) = \frac{v_X(\xi) - \sqrt{v_X(\xi)^2 - 2\bar{a}(c_1 + a_2 + c_2 + a_3 + c_3 + a_4)}}{\bar{a}},$$

$$l_0 = (c_1 + a_2 + c_2 + a_3 + c_3 + a_4), \quad \bar{a} = cte$$

$$\tau(\xi) = \frac{v_X(\xi) - \sqrt{v_X(\xi)^2 - 2\bar{a}l_0}}{\bar{a}}$$

$$\dot{\tau}(\xi) = 1 - \frac{v_X(\xi)}{\sqrt{v_X(\xi)^2 - 2\bar{a}l_0}}$$

$$\ddot{\tau}(\xi) = \frac{-\bar{a}(v_X(\xi)^2 - 2\bar{a}l_0) + v_X(\xi)^2 \bar{a}}{(v_X(\xi)^2 - 2\bar{a}l_0)^{3/2}} = \frac{2\bar{a}^2 l_0}{(v_X(\xi)^2 - 2\bar{a}l_0)^{3/2}} \approx 0, \text{ for highway driving}$$

$$\int_0^t \phi_1(\xi) \ddot{\tau}(\xi) d\xi \approx 0, \text{ for highway driving}$$